

Guaranteed indirect delivery in directed networks

Extremal bounds, sparse constructions, and verified finite cases

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1 Introduction

A messenger has just obtained a message at a station s of a directed network. A pursuer occupies another station, and both positions are visible. They move alternately, the messenger first, using one directed connection or waiting. The messenger wants to deliver to a designated recipient t before interception. Receipt completes the task immediately, even if the pursuer is at t : the pursuer can intercept the messenger but cannot disable the recipient.

A direct connection $s \rightarrow t$ makes delivery immediate. We study the additional communication supported by the network: source–recipient pairs without a direct connection for which delivery is nevertheless guaranteed against every allowed initial pursuer position. Write $I(G)$ for their number. The three extremal quantities are

$$M_V(n) = \max_{|V(G)|=n} I(G), \quad M_E(m) = \max_{|E(G)|=m} I(G), \quad M(n, m) = \max_{|V(G)|=n, |E(G)|=m} I(G).$$

Isolated vertices are allowed. Removing them preserves I , so the arc-only maximum is finite: a graph with $m > 0$ arcs has at most $2m$ nonisolated vertices.

The original problem and a layered subset construction arose in Alexandra Ignatova’s 2023 project, mentored by Angel Raychev. A public 2023 abstract records that provenance (Ignatova 2023); the retained manuscript is dated February 2024 (Ignatova 2024). Ignatova and Raychev had already established the upper bound $I(G) \leq n(n-3)$ and investigated two-in/two-out-regular constructions. This account reconstructs that bound, distinguishes the original construction from its corrected analysis, and supplies new attaining families among the September 2026 extensions. The full extremal classification remains open.

1.1 Results and proof boundaries

The principal results are now an eventual classification, with separate finite exceptions and explicit proof boundaries.

1. **Vertex and even arc maxima.** For every $n \geq 12$,

$$M_V(n) = M_E(2n) = M(n, 2n) = n(n-3).$$

A six-type repeating construction with controlled vertex removals covers every $n \geq 85$; two independent exact solvers and full rank certificates supply the finite bridge $12 \leq n \leq 84$ (Theorem 40).

2. **Odd arc maxima.** For every odd $m = 2k + 1 \geq 25$,

$$M_E(m) = k(k-3) + 2.$$

Six local constructions cover all residue classes of the support size. Their strategies are independently verified and composed by a uniform ordinary argument (Theorem 45 and Corollary 45.1).

3. **The entire main interval.** For every sufficiently large n , every integer budget $2n \leq m \leq n(n-3)$ attains

$$M(n, m) = n(n-1) - m.$$

Thus Raychev's full-interval conjecture is proved. Theorem 44 gives the explicit sufficient threshold $n \geq 1800$. Exact conditional counting makes the probabilistic portion deterministic and polynomial in n ; this does not claim a small practical runtime (Theorem 43).

4. **Explicit smaller-order intervals.** A protected-difference construction attains every $3n \leq m \leq n(\lceil n/7 \rceil - 1)$ for $n \geq 25$. A second six-type certificate attains every $2n \leq m \leq 3n$ for $n \geq 1075$. Other arithmetic families give additional ranges at smaller orders (Theorems 34–39 and 52).
5. **The complete final dense band.** For every $n \geq 75$,

$$M(n, n(n-3) + r) = 2n - r - \left\lceil \frac{r}{2} \right\rceil - 1, \quad 1 \leq r \leq n.$$

At $r = 0$, the value is $2n$ for every $n \geq 52$; above $r = n$, it is zero. Explicit constructions attain every budget (Theorems 19 and 32, Corollary 33).

6. **A combined formula.** For $n \geq 1800$, these results determine every $M(n, m)$ with $m \geq 24$: the sparse arc maximum, the missing-pair bound in the main interval, the final staircase, and then zero (Corollary 47). The threshold is deliberately conservative and is not asserted to be the smallest possible.
7. **Small cases.** Complete joint maxima are known through $n = 6$, with $M_V(1), \dots, M_V(6) = 0, 0, 0, 2, 4, 7$. Two independent exhaustive checks additionally prove $M_V(7) = 21$ and $M_V(8) = 32$ (Theorem 46). The arc-only maxima through eleven arcs are $0, 0, 0, 0, 1, 1, 2, 2, 4, 5, 6, 8$ (Theorem 49). The remaining odd endpoint satisfies $M_E(23) = 83$ (Theorem 53). Other finite witnesses and nonattainment results are stated separately; no unsolved small value is filled by the eventual formulas.

The general arguments are ordinary mathematical proofs. The accompanying Lean development formalizes the actual alternating game, a certificate-to-strategy theorem, safe branching, camping obstructions, a complete universally winning 12-vertex example, complementary-block two-move delivery, and the numerical staircase deduction from explicit counting hypotheses. The graph-to-counting bridge and existence of attaining blocks remain ordinary arguments. It does not claim to formalize all the general extremal theorems.

2 The game and its elementary structure

Let $G = (V, E)$ be a finite loopless digraph. Opposite arcs are allowed. A legal move from x goes to x or to an outgoing neighbor; write

$$C(x) = \{x\} \cup N^+(x).$$

Fix $s, t \in V$ and $c \neq s$. If $s = t$, receipt has already occurred. Otherwise the messenger moves first. On its turn it chooses $v \in C(s)$. If $v = t$, delivery wins immediately. If $v \neq t$ and $v = c$, it is captured. Otherwise the pursuer chooses $c' \in C(c)$; equality $c' = v$ is capture. The process repeats. Infinite play without delivery is a messenger loss.

Let $W_G(s, t)$ mean that, for every initial $c \neq s$, there exists a messenger strategy guaranteeing delivery against every pursuer strategy. The messenger observes c ; its strategy may depend on it. Then

$$I(G) = \#\{(s, t) : s \neq t, (s, t) \notin E, W_G(s, t)\}.$$

If $w(G)$ counts all guaranteed pairs including the diagonal, then exactly

$$w(G) = n + m + I(G).$$

The terminal rule matters. In the directed diamond $s \rightarrow a \rightarrow t$, $s \rightarrow b \rightarrow t$, the messenger chooses the intermediate vertex not occupied by the pursuer and delivers on its next turn. But reaching a vertex that would be safe as a *recipient* does not justify using it as an intermediate stop. The pursuer gets an intervening move before the message can be forwarded again.

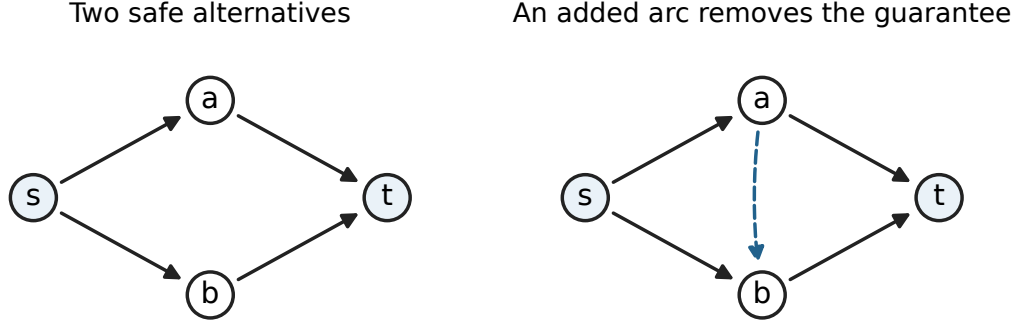


Figure 1: Two routes allow a guaranteed indirect delivery. Adding the arc from a to b lets a pursuer at a block both first moves.

2.1 Camping obstructions

Lemma 1 (source domination). If $c \neq s$ and $N^+(s) \subseteq C(c)$, there is no guaranteed indirect pair with source s .

Proof. The pursuer starts at c and waits while the messenger waits. Every nonterminal departure from s is either an immediate collision or is captured on the following pursuer turn. Waiting forever is not delivery. $\square$

Lemma 2 (predecessor domination). Put $D_t = N^-(t)$. If $D_t \subseteq C(c)$, an indirect guaranteed source for t must equal c .

Proof. For any other source the pursuer may start at c . Every indirect delivery must first enter D_t , where the pursuer can capture before the next messenger turn. $\square$

In particular, a source counted by I has outdegree at least two, and a recipient counted by I has indegree at least two. An indegree-one recipient is blocked by its unique predecessor. In an undirected network, interpreted as two opposite arcs per edge, the pursuer can start at t and catch the messenger upon its first arrival at a neighbor of t . Thus $I = 0$ for every such network.

The two degree conditions are necessary, not sufficient. Neither the absence of a one-vertex directed separator nor the existence of two internally disjoint routes characterizes this game. Adding $a \rightarrow b$ to the diamond destroys its only indirect guarantee, although both routes remain. Consequently I is not monotone under adding arcs.

Lemma 3 (two-move criterion). For an indirect pair (s, t) , put $B = N^+(s) \cap N^-(t)$. Delivery is guaranteed within two messenger moves exactly when

$$\text{for every } c \neq s, \quad B \not\subseteq C(c).$$

Proof. Choose an intermediate vertex of $B \setminus C(c)$, survive every pursuer response, and deliver.

Conversely, any successful two-move indirect delivery must use such an intermediate. Waiting on the first move cannot reach a nonadjacent recipient on the second. $\square$

2.2 Components, zero minima, and the vertex bound

Removing isolated vertices preserves I . More generally,

$$I(G \sqcup H) = I(G) + I(H).$$

There are no cross-component routes; a pursuer in the other component cannot interfere with a route in the messenger's component.

Proposition 4. At every admissible (n, m) the minimum of I is zero.

Proof. For $n \geq 2$, list all nonloop arcs row by row and retain the first m . There are full outgoing rows, at most one partial row, and empty rows. Full rows have only direct recipients; empty rows cannot move. If a full row exists, its vertex blocks every indirect attempt from the partial row by Lemma 1. Otherwise successors of the partial row have no outgoing arcs, so no indirect route exists. The empty and one-vertex cases are immediate. $\square$

Theorem 5 (original upper bound). For every $n \geq 3$,

$$I(G) \leq n(n-3).$$

Proof. A recipient of indegree zero or one has no indirect guaranteed sources. A recipient of indegree $d \geq 2$ has at most $n-1-d \leq n-3$, after excluding itself and its direct predecessors. Sum over recipients. $\square$

For $n \geq 4$, equality forces every recipient to have indegree two and every missing pair to be guaranteed. Every source must then have outdegree at least two; the degree sum forces outdegree exactly two. Conversely, a two-in/two-out-regular graph with every pair guaranteed attains equality. Thus attaining the vertex bound is exactly a sparse universal-delivery construction problem.

3 Sharp arc-budget upper bounds

Let A be the vertices of outdegree at least two and B those of indegree at least two. Write $q = |A|$, $r = |B|$, $b = |A \cap B|$, and let $e(A, B)$ count arcs from A to B . Every counted pair belongs to $A \times B$, and direct and diagonal pairs are excluded. Hence

$$I(G) \leq qr - b - e(A, B), \quad e(A, B) \geq \max\{0, 2q + 2r - m\}. \quad (1)$$

The second inequality follows by adding the total outgoing degree in A and incoming degree in B : each arc is counted at most once except those in $A \times B$.

We also need two elementary obstructions. If x has indegree zero and $x \rightarrow t$ with t of indegree two, then t has no indirect guaranteed sources. Its other predecessor blocks every indirect route: a route from elsewhere cannot reach x , while $x \rightarrow t$ itself is direct. Dually, if t is a sink, $x \rightarrow t$, and x has outdegree two, then x has no indirect guaranteed recipients. Its other successor blocks every potentially successful indirect departure. This dual argument is proved directly and does not assume invariance under reversing the graph.

3.1 Even numbers of arcs

Theorem 6. If $m = 2k$ and $k \geq 5$, then

$$I(G) \leq k(k-3).$$

Proof. We have $q, r \leq k$. Applying (1), with $b \geq 0$, gives the bound except possibly when (q, r) is one of

$$(k, k), \quad (k-1, k), \quad (k, k-1).$$

For completeness, this numerical reduction follows by maximizing the bilinear expression $qr - \max(0, 2q + 2r - 2k)$ on the two regions $q + r \leq k$ and $q + r \geq k$: on each horizontal or vertical segment its maximum occurs at an endpoint. Outside the three displayed corners the maximum is at most $k(k-3)$ for $k \geq 5$.

If $q = r = k$, every arc starts in A and ends in B , and all these outgoing and incoming degrees equal two. If $A = B$, (1) gives $k^2 - k - 2k$. Otherwise choose $x \in A \setminus B$. It has indegree zero. Its two successors are zero-contribution recipient columns. The other $k-2$ columns each receive two direct arcs from A , giving

$$I(G) \leq k(k-2) - 2(k-2) = (k-2)^2 \leq k(k-3).$$

If $q = k-1, r = k$, (1) gives $I \leq k(k-3) + 2 - b$, so $b \geq 2$ suffices. Otherwise there is an indegree-zero vertex of $A \setminus B$. Its two successor columns contribute zero. The other $k-2$ columns receive $2k-4$ arcs in total; at most two start outside A . Therefore

$$I(G) \leq (k-1)(k-2) - (2k-6) = k^2 - 5k + 8 \leq k(k-3).$$

The case $q = k, r = k-1$ uses the sink obstruction to remove two zero source rows and gives the same bound. $\square$

3.2 Odd numbers of arcs

Theorem 7. If $m = 2k+1$ and $k \geq 5$, then

$$I(G) \leq k(k-3) + 2.$$

Proof. Again $q, r \leq k$, and (1) suffices outside the same three corners. If $r = k$, the total excess incoming degree in B , together with all arcs ending outside B , is at most one. Thus at most one vertex outside B has positive indegree. An indegree-zero source in A has an indegree-two successor: either every vertex of B has indegree two and at most one arc ends outside B , or one vertex of B has indegree three and every arc ends in B . Its at least two distinct successors cannot all be exceptions. This gives a zero recipient column.

Suppose $q = k-1, r = k$. Formula (1) gives $I \leq k(k-3) + 3 - b$, so $b \geq 1$ suffices. If $b = 0$, choose an indegree-zero member of A . Remove a zero recipient column. The other $k-1$ columns receive at least $2k-2$ arcs, at most three of which start outside A , giving

$$I \leq (k-1)^2 - (2k-5) = k^2 - 4k + 6 \leq k(k-3) + 2.$$

The symmetric corner uses the zero-row argument.

Finally let $q = r = k$. Formula (1) suffices when $b \geq k-1$. If $2 \leq b \leq k-2$, there is an indegree-zero source in $A \setminus B$, hence a zero column. The remaining columns contain at least $2k-3$ direct pairs and at least $b-1 \geq 1$ diagonal pairs. Thus

$$I \leq k(k-1) - (2k-3) - 1 = k(k-3) + 2.$$

If $b \leq 1$, at least $k-2 \geq 3$ vertices of $A \setminus B$ have indegree zero. Each has an indegree-two successor, and at most two can share any such successor. There are consequently at least two zero columns. The others contain at least $2k-5$ direct pairs, so

$$I \leq k(k-2) - (2k-5) = k^2 - 4k + 5 \leq k(k-3) + 2.$$

This completes the proof. $\square$

3.3 Equality structure

Theorem 8. For $k \geq 6$, equality in Theorem 6 holds exactly when deleting isolated vertices leaves a two-in/two-out-regular graph on k vertices with every pair guaranteed.

Proof. The numerical bounds away from the three corners are strict. In the $(k-1, k)$ corner, a member of $A \setminus B$ gives the strict zero-column estimate above; if $A \subseteq B$, then $b = k-1$ gives a strict bound directly. The other corner is analogous. In the (k, k) corner, $A \neq B$ gives the strict bound $(k-2)^2$. Hence $A = B$, all $2k$ arcs lie on these k vertices, and all incoming and outgoing degrees equal two. Equality then says that every eligible missing pair is guaranteed. $\square$

Theorem 9. For $k \geq 6$, equality in Theorem 7 holds exactly for graphs whose nonisolated support has the following properties: it has $k+1$ vertices; one vertex u has outdegree one and indegree two; a different vertex v has indegree one and outdegree two; all other degrees are two; $u \rightarrow v$ is absent; and every missing nonloop pair with source of outdegree two and recipient of indegree two is guaranteed.

Proof. Strengthening the strict inequalities in Theorem 7 excludes all cases except $q = r = k$ and $b = k-1$. In the corner $(q, r) = (k-1, k)$, values $b \geq 2$ are strict directly. If $b \leq 1$, at least $k-2$ members of A lie outside B , and at most one has positive indegree. The zero-column argument gives $I \leq k^2 - 4k + 6 < k(k-3) + 2$. The dual corner uses the zero-row argument. In the remaining (k, k) corner, for $b = 2$ at least three indegree-zero sources give two zero columns; for $3 \leq b \leq k-2$ the one-column argument retains at least two diagonal exclusions. The other excluded cases already have strict bounds for $k \geq 6$.

Write $A \setminus B = \{v\}$ and $B \setminus A = \{u\}$. Equality forces $e(A, B) = 2k-1$. The outgoing degree sum in A and incoming degree sum in B are each at least $2k$, while their sum is at most $m + e(A, B) = 4k$. Thus all these degrees equal two. Exactly one arc starts outside A and one ends outside B , and they are distinct. Vertex v cannot have indegree zero: it would give a zero column containing at least $k-3 > 0$ eligible missing pairs. Hence it receives the unique arc ending outside B . Dually u emits the unique arc starting outside A . Every vertex outside $A \cup B$ is isolated, and the two exceptional arcs being distinct implies that $u \rightarrow v$ is absent. Counting gives the converse. $\square$

Adding the missing arc $u \rightarrow v$ in Theorem 9 gives a two-regular digraph. It is not asserted that this completed graph is automatically universally winning.

4 Sparse constructions with a linear deficit

4.1 A safe branching principle

Lemma 10. Suppose distinct vertices have closed move sets intersecting in at most one vertex. If a vertex r has two distinct outgoing neighbors that each guarantee delivery to t , then r guarantees delivery to t .

Proof. For every pursuer position $c \neq r$, one of the two successors lies outside $C(c)$. Move there. It survives every intervening pursuer response, after which its guaranteed strategy applies. If the chosen vertex is t , delivery is already terminal. This is a finite concatenation of actual strategies. $\square$

Theorem 11. Let $a \geq 3$, $h = 2a + 1$, $n \geq 2h + 1$, and $\gcd(n, h) = 1$. The digraph on $\mathbb{Z}/n\mathbb{Z}$ with outgoing steps $1, a, h$ has every pair guaranteed. It has $3n$ arcs and

$$I(G) = n(n - 4).$$

Proof. The positive differences in $\{0, 1, a, 2a + 1\}$ are

$$1, a - 1, a, a + 1, 2a, 2a + 1,$$

all distinct. Since $n > 2h$, no positive difference coincides modulo n with a negative one. Distinct closed move sets therefore intersect in at most one vertex.

Translate the recipient to zero. Vertices $0, -1, -a, -h$ are already delivered or have direct arcs to it. Vertex $-a - 1$ is guaranteed via its neighbors $-a, -1$. Thus all four vertices

$$Q(z) = \{z, z - 1, z - a, z - a - 1\}$$

are guaranteed for $z = 0$. If $Q(z)$ is guaranteed, apply Lemma 10 in the following order:

New source	Two guaranteed successors
$z - h$	$z - a - 1, z$
$z - h - 1$	$z - h, z - 1$
$z - h - a$	$z - h, z - a$
$z - h - a - 1$	$z - h - a, z - a - 1$

This proves $Q(z - h)$. Iterating at most $n - 1$ times covers every vertex because h is coprime to n . The induction is finite; it supplies terminating strategies, not merely perpetual evasion. Finally subtract the n diagonal pairs and $3n$ arcs from n^2 . $\square$

The choice $a = 3$ proves the theorem for every $n \geq 15$ not divisible by seven. Adding an isolated vertex gives a linear-deficit construction for the remaining large sizes as well. The next elementary lemma removes that padding and improves the constant.

Lemma 12. For every $n \geq 25$, there is an odd integer h with $7 \leq h < n/2$ and $\gcd(n, h) = 1$.

The proof and its explicit choices are included in the construction appendix below. Taking $a = (h - 1)/2$ gives, for every $n \geq 25$,

$$n(n - 4) \leq M_V(n) \leq n(n - 3), \quad 3n \leq n^2 - M_V(n) \leq 4n.$$

These estimates motivate the sharper endpoint result: Theorem 40 subsequently proves equality in the upper bound for every $n \geq 12$.

Known quartet

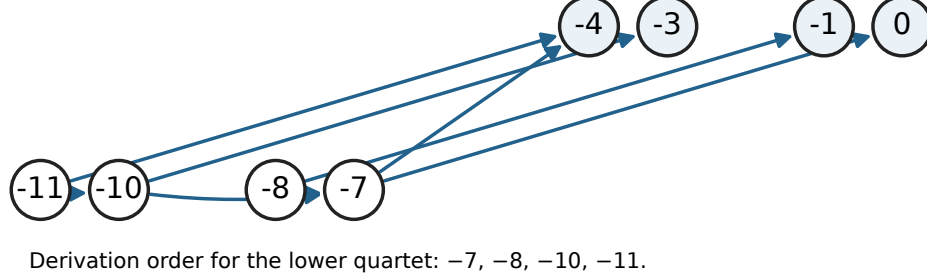


Figure 2: A guaranteed quartet produces the next quartet seven positions earlier in the step-1,3,7 construction.

5 The dense boundary

Theorem 13. For $n \geq 2$,

$$M(n, n(n-2)) = \left\lfloor \frac{n-2}{2} \right\rfloor,$$

and $M(n, m) = 0$ for every admissible $m > n(n-2)$.

Proof. If (s, t) is indirect guaranteed, row s misses $s \rightarrow t$. By Lemma 1, every other row c must miss an arc into $N^+(s)$. Thus every row misses at least one nonloop arc, and $m \leq n(n-2)$.

Now suppose $m = n(n-2)$. If $I = 0$, the upper bound is immediate. Otherwise the preceding argument shows that every row misses exactly one arc; write $f(v)$ for its endpoint. Necessarily $f(v) \neq v$. A possible indirect pair is $(s, f(s))$. Lemma 1 says

$$f^{-1}(s) = \emptyset, \quad f^{-1}(f(s)) = \{s\}. \quad (2)$$

These conditions suffice: for initial pursuer $c \neq s$, choose the intermediate $f(c)$. It is different from $s, f(s)$, is inaccessible on the pursuer's next move, and has an arc to $f(s)$. This is a two-move guarantee.

Pairs satisfying (2) have disjoint source and target vertices. If there are j pairs, the remaining vertices contain a cycle of the loopless functional graph of f and therefore number at least two. Hence $2j \leq n-2$. For attainment use pairs $s_i \rightarrow t_i$ in the missing-arc graph, send every t_i to a , put $a \leftrightarrow b$, and send any remaining vertex to a . Take all nonloop arcs except these missing arcs. $\square$

Theorem 14. A noncomplete graph with every pair guaranteed has at most $n(n-3)$ arcs.

Proof. Such a graph has an indirect pair. No row can be complete: its vertex would block every indirect source other than itself and have no indirect target of its own. If a row c missed only $c \rightarrow t$, then $N^+(t) \subseteq C(c)$. Vertex t also has an indirect recipient, contradicting Lemma 1. Every row therefore misses at least two arcs. $\square$

The next section gives a uniform construction attaining this density on an infinite set of orders. Its proof is independent of the sparse-game certificate for the complementary graph proved later.

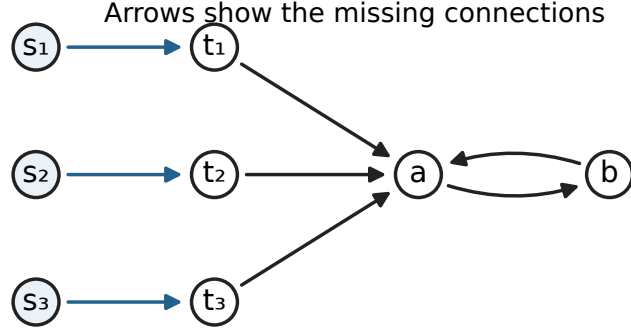


Figure 3: The missing arcs at the dense boundary form a functional graph. Each displayed two-vertex branch contributes one guaranteed indirect pair.

5.1 A periodic family with three-move delivery

Let $n = 5k$, $k \geq 4$, and write vertices as $5i + j$, with i modulo k and $0 \leq j \leq 4$. Put $\alpha(v) = v + 1$ and

$$\begin{aligned} \beta(5i) &= 5(i + 1), & \beta(5i + 1) &= 5(i - 1) + 2, \\ \beta(5i + 2) &= 5(i - 1) + 4, & \beta(5i + 3) &= 5(i - 1) + 3, \\ \beta(5i + 4) &= 5(i - 1) + 1. \end{aligned}$$

Let H have arcs $v \rightarrow \alpha(v)$, $v \rightarrow \beta(v)$, and let G be its loopless directed complement. Both maps are permutations, without fixed points or coincident images at a vertex; thus G has $n(n - 3)$ arcs.

Theorem 15. Every pair in G is guaranteed within three messenger moves. Consequently $M(5k, 5k(5k - 3)) = 10k$ for every $k \geq 4$.

Proof. For a missing pair (s, t) and initial pursuer $c \neq s$, a safe two-move intermediate is any member of

$$\{\alpha(c), \beta(c)\} \setminus F(s, t), \quad F(s, t) = \{s, t, \alpha(s), \beta(s), \alpha^{-1}(t), \beta^{-1}(t)\}.$$

Its exclusion from F ensures the two required arcs in G , while its membership in the displayed pair makes it inaccessible to the pursuer's next move.

Translate the source by a multiple of five so that $0 \leq s \leq 4$. A failure requires $c \in F - 1$, since $\alpha(c) \in F$. The complete local calculation is:

s	t	$F(s, t)$	$(c, \beta(c))$, with $c \in F - 1$, $c \neq s$
0	1	$\{0, 1, 5, 9\}$	$(-1, -9), (4, -4), (8, 3)$
0	5	$\{0, 1, 4, 5\}$	$(-1, -9), (3, -2), (4, -4)$
1	2	$\{-3, 1, 2, 6\}$	$(-4, -8), (0, 5), (5, 10)$
1	-3	$\{-4, -3, 1, 2\}$	$(-5, 0), (-4, -8), (0, 5)$
2	3	$\{-1, 2, 3, 8\}$	$(-2, -7), (1, -3), (7, 4)$
2	-1	$\{-2, -1, 2, 3\}$	$(-3, -6), (-2, -7), (1, -3)$
3	4	$\{-2, 3, 4, 7\}$	$(-3, -6), (2, -1), (6, 2)$
3	-2	$\{-3, -2, 3, 4\}$	$(-4, -8), (-3, -6), (2, -1)$
4	5	$\{-4, 0, 4, 5\}$	$(-5, 0), (-1, -9), (3, -2)$
4	-4	$\{-5, -4, 4, 5\}$	$(-6, -14), (-5, 0), (3, -2)$

The only candidate with $\beta(c) \in F$ is $(s, t, c) = (4, 5, -5)$. Every difference being tested has absolute value at most 19, so no extra equality appears modulo any $5k \geq 20$. In that exceptional state move from 4 to 0. This is legal and safe: the pursuer at -5 has no arc to 0. Afterwards the pair $(0, 5)$ is covered by the nonexceptional second row for every surviving pursuer position, and delivery takes at most two more moves. $\square$

6 Correcting the original layered construction

The original graph remains a useful construction, but its printed count is not correct. We give its exact count and separate safe intermediate movement from terminal delivery.

Let $q = 2x$, $a = \binom{2x}{x}/2$, and choose a family $\mathcal{F}$ containing one of each complementary pair of x -subsets of $[2x]$, with every coordinate occurring in exactly $a/2$ members. Such a family exists precisely when positive x is not a power of two; a proof follows below.

There are row vertices A_X, A'_X for $X \in \mathcal{F}$, and coordinate vertices b_i, b'_i, c_i, c'_i for $i \in [q]$. The only arcs are

- $A_X \rightarrow b_i, b'_i$ when $i \in X$;
- $b_i \rightarrow A'_Y$ when $i \in Y$, and $b'_i \rightarrow A'_Y$ when $i \notin Y$;
- $A'_Y \rightarrow c_i, c'_i$ when $i \in Y$;
- $c_i \rightarrow A_X$ when $i \in X$, and $c'_i \rightarrow A_X$ when $i \notin X$.

The order is $N = 2a + 4q$. The four directed layers are $A, B \cup B', A', C \cup C'$.

Theorem 16. For the balanced original family,

$$I(G) = 2a(N - 1 - q) + 4q.$$

At $x = 3$, the 44-vertex graph has exactly 764 indirect guaranteed pairs, rather than the draft's claimed lower bound of 1252.

Proof. The rows form an antichain. The $2q$ literal columns, indicating membership or nonmembership of each coordinate, also form an antichain, with size $a/2 \geq 5$. They cannot coincide: equality on the selected rows would remain true on their complements and hence on every x -subset, which is impossible for distinct literals. Every vertex has at least two successors.

From any noncollision state the messenger can safely advance one layer. If both players are in the same layer, use incomparable outgoing neighborhoods to choose a successor inaccessible to the

pursuer. If the pursuer is in the next layer, avoid its occupied vertex; it has no within-layer arc. In the other two cases it cannot reach the next layer in one move. This includes waiting responses.

From A_X to a specified A'_Y there is a two-move guarantee. Against a pursuer A_Z , choose a coordinate in $X \setminus Z$, then choose its primed or unprimed intermediate according to membership in Y . Against a pursuer in the intermediate layer, choose another coordinate of X ; the other layers cannot interfere. The reverse direction is identical. Safely advancing to the appropriate row layer first therefore gives delivery to any row recipient in at most five moves. There are $2a$ such recipients, each with $N - 1 - q$ indirect sources.

For target b_i or b'_i , its predecessor set is exactly $N^+(c_i)$. Lemma 2 excludes every indirect source except c_i . That exceptional source wins in two moves: literal-column incomparability handles a pursuer in its own layer, and the other layer cases are as above. Thus (c_i, b_i) and (c_i, b'_i) both count. Symmetrically (b_i, c_i) and (b_i, c'_i) count, and no other indirect sources do. These give exactly $4q$ further pairs. $\square$

The correction preserves the original $N^2 - O(N \log N)$ leading estimate along its constructed orders. The new circulant family supplies the stronger linear-deficit result at every sufficiently large order. The original construction's regular incidence design and the later extremal construction are distinct contributions.

6.1 Balanced complementary halves

Lemma 17. For positive x , the x -subsets of $[2x]$ can be partitioned into two coordinate-balanced families with complementary subsets in opposite families if and only if x is not a power of two.

Proof. Put $C = \binom{2x}{x}$. Each coordinate occurs in $C/2$ subsets, so balance requires $4 \mid C$. The factorial valuation formula gives

$$v_2 \binom{2x}{x} = s_2(x),$$

where $s_2(x)$ is the number of ones in the binary expansion. Thus divisibility by four is equivalent to x not being a power of two.

For sufficiency, distinguish coordinate $2x$ and cyclically rotate the other $L = 2x - 1$ coordinates. Subsets containing $2x$ correspond to binary circular words with $x - 1$ ones and x zeros. Every orbit has length L : a proper repetition would have a repetition count dividing both consecutive integers. The number $C/(2L)$ of orbits is even, since L is odd and $4 \mid C$.

Choose half of those orbits. In the first family put their subsets, and put the complements of the subsets in the unchosen orbits. Exactly one of each complementary pair is selected. The distinguished coordinate occurs $C/4$ times. Rotation invariance gives a common incidence count r for every other coordinate; counting all incidences yields

$$(2x - 1)r + C/4 = xC/2,$$

so $r = C/4$. The other family is balanced as well. $\square$

This proof replaces the draft's incorrect stronger divisibility assertion $4x \mid \binom{2x}{x}$, which already fails at $x = 3$.

7 High-girth complements and exact dense intervals

The preceding periodic construction attains the largest possible density of a noncomplete all-delivery network on an arithmetic progression. A different construction handles every sufficiently large order and an interval of arc counts. Here the sparse graph specifies the connections to omit from an otherwise complete directed network.

Lemma 18 (high-girth complement). Let F be an orientation of a finite simple undirected graph of girth at least six. Suppose every vertex has outdegree at least two in F . Let G contain exactly the nonloop arcs absent from F . Then every source–recipient pair in G is guaranteed within two messenger moves.

Proof. Diagonal and direct pairs need no argument. For a missing pair (s, t) of G , the arc $s \rightarrow t$ belongs to F . Set

$$B = N_F^+(s) \cup N_F^-(t).$$

Both s and t belong to B . In the underlying undirected graph, the vertices of B form an induced double star: its centers are s, t , the other outneighbors of s are leaves at s , and the other predecessors of t are leaves at t . The two sets of leaves are disjoint, since an overlap would make a triangle. An additional edge inside B would make a triangle or quadrilateral. Thus the distance between any two vertices in this induced tree is at most three.

Fix an initial pursuer vertex $c \neq s$. If $c \notin B$, two neighbors of c in B would form an undirected cycle of length at most five, so c has at most one outgoing F -neighbor in B . The same bound holds when $c \in B \setminus \{s\}$: a leaf at s has its only incident edge in the tree directed toward it; a leaf at t has only its edge toward t ; and all tree edges incident with t point into t .

Since c has at least two outgoing F -neighbors, choose

$$u \in N_F^+(c) \setminus B.$$

Then $u \neq s, t$, and $s \rightarrow u$ and $u \rightarrow t$ are arcs of G . Also $u \neq c$ and $c \rightarrow u$ is absent from G . The messenger moves to u , survives every pursuer response, and delivers to t on its next turn. $\square$

The lemma only requires a lower bound on outgoing degrees. In particular, edges can later be added to the underlying graph and oriented arbitrarily, provided its girth remains at least six.

Theorem 19 (the densest noncomplete all-delivery networks). For every $n \geq 52$,

$$M(n, n(n-3)) = 2n.$$

An attaining graph guarantees every pair within two messenger moves. The construction for $n \geq 130$ has the ordinary proof below; the extension to $52 \leq n < 130$ uses an explicit finite structural certificate.

Proof. We first construct a four-regular undirected graph of girth at least six at each required order. Orient an Euler tour in each of its components. Every vertex then has two incoming and two outgoing arcs. Such tours exist by the elementary closed-trail construction: even degrees prevent an unused-edge trail from stopping away from its starting vertex, and further closed trails can be spliced into the first until all edges of the component have been used. Lemma 18 applies to the resulting orientation F . It has $2n$ arcs, so its directed complement has $n(n-3)$ arcs and exactly $2n$ missing pairs, all guaranteed. The missing-pair count is also an upper bound for I .

For a building block, take point vertices $p \in \mathbb{Z}/13\mathbb{Z}$ and line vertices $\ell \in \mathbb{Z}/13\mathbb{Z}$, joining them when

$$p - \ell \in \{0, 1, 3, 9\}.$$

This graph has 26 vertices and is four-regular. The twelve ordered differences between distinct members of $\{0, 1, 3, 9\}$ are precisely the twelve nonzero residues modulo 13. Consequently two distinct points cannot have two common line neighbors: the two common neighbors would give two representations of the same nonzero difference. There is no four-cycle. Bipartiteness excludes odd cycles, so the girth is at least six.

The following operation increases the order by one. Choose edges ab and cd such that every endpoint of the second is at distance at least four from every endpoint of the first. Delete these two edges, add a new vertex x , and join x to a, b, c, d . The endpoints are distinct and all degrees remain four. Any new cycle uses two of these new edges. If their old endpoints belonged to the same removed edge, the path between them after its removal has length at least five, giving a new cycle of length at least seven. Otherwise their old distance was at least four, giving a new cycle of length at least six. Thus the operation preserves the required girth.

Such an edge pair exists in every four-regular graph with more than 120 vertices. Fix an edge ab and let S consist of the vertices within distance three of a or b . The branching bound from the two ends of the edge gives

$$|S| \leq 2(1 + 3 + 9 + 27) = 80.$$

The induced graph on S is connected, hence has at least $|S| - 1$ edges. If there were no edge with both endpoints outside S , four-regularity would give

$$4|V \setminus S| = e(S, V \setminus S) = 4|S| - 2e(S) \leq 2|S| + 2.$$

It would follow that

$$n \leq \left\lfloor \frac{3|S| + 1}{2} \right\rfloor \leq 120.$$

Therefore an outside edge exists and can serve as cd .

Start with five disjoint copies of the 26-vertex building block. Repeated insertion now gives every order $n \geq 130$. For the smaller stated orders, two disjoint building blocks give order 52. The supplied certificate lists 78 admissible insertions, producing every order from 52 through 130. Each insertion satisfies the same distance condition and therefore has the proof of correctness just given. Its finite verification is described below. $\square$

The density $n(n - 3)$ is maximal among noncomplete all-delivery graphs, by the earlier dense-boundary theorem. This construction therefore settles the joint extremum at that density for every $n \geq 52$; it is not a claim about the arc-only maximum at the same number of arcs.

Finite-certificate reproduction. The file `research/extremal-bounds-girth-growth.json` records the two removed edges at every insertion, starting from the explicit 52-vertex graph above. The generator and replayer `src/extremal_bounds_girth_family.py` checks the endpoint distances, all degrees, and the absence of short cycles. A separate implementation, `src/extremal_bounds_girth_verify.py`, rebuilds the incidence graph directly and checks girth by enumerating simple paths and testing whether they close into cycles of lengths three, four, or five. It verifies all 79 graph orders independently of the generator's edge-deleted breadth-first checks. The receipt `research/extremal-bounds-girth-verification.json` records the certificate and verifier hashes. These are finite graph certificates attached to the proved insertion rule, not extrapolation

from successful pursuit-game searches. They are executable checks rather than a Lean formalization; omitting this finite certificate leaves the ordinary theorem for every $n \geq 130$.

Theorem 20 (an interval of exact dense extrema). Fix integers $n \geq 52$ and $5 \leq D \leq n - 1$, and put

$$\begin{aligned} B_D &= 1 + D(1 + (D - 1) + (D - 1)^2 + (D - 1)^3) \\ &= D^4 - 2D^3 + 2D^2 + 1, \\ L_D(n) &= \left\lfloor \frac{Dn - (D - 4)B_D}{2} \right\rfloor. \end{aligned}$$

For every integer h satisfying $2n \leq h \leq L_D(n)$,

$$M(n, n(n - 1) - h) = h.$$

Each equality is attained by an all-delivery graph with a two-move strategy. An empty interval makes no assertion.

Proof. Begin with the four-regular girth-six graph from Theorem 19. Repeatedly add an undirected edge between two vertices of current degree less than D whose distance is at least five. Vertices in different components have infinite distance. Each addition preserves simplicity, the degree cap D , and girth at least six.

When no further addition is possible, let U be the vertices of degree less than D . If U is nonempty, all its vertices lie within distance four of any fixed member of U . A graph of maximum degree D has at most B_D vertices in such a ball, so $|U| \leq B_D$. This inequality also holds when U is empty. The minimum degree remains at least four. The final edge count is consequently at least

$$\frac{D(n - |U|) + 4|U|}{2} \geq \frac{Dn - (D - 4)B_D}{2}.$$

Starting from $2n$ edges and adding one edge at a time produces a graph with every edge count through $L_D(n)$.

Orient the initial four-regular graph by Euler tours, and orient each added edge arbitrarily. Each vertex retains its original two outgoing arcs. Applying Lemma 18 at the stage with h edges gives a directed complement with $n(n - 1) - h$ arcs and all its h missing pairs guaranteed. No graph with that arc count can have more than h indirect pairs. $\square$

The statement has both order and density as parameters. For example, $B_D \leq D^4$ for $D \geq 2$. When $n \geq 2D^4$, the displayed bound gives

$$L_D(n) \geq n(D/4 + 1) - 1.$$

Thus choosing $D = \lfloor (n/2)^{1/4} \rfloor$ for sufficiently large n yields exact joint extrema for an interval of missing-arc counts extending to a positive constant times $n^{5/4}$. As with Theorem 19, the range $n \geq 130$ has an ordinary construction, while the extension down to 52 uses the explicit finite certificate.

8 An infinite family attaining the sparse bounds

The five-type graph used in the dense construction also attains the sparse vertex and arc-budget bounds. This requires a different strategy: its messenger may need to travel around a substantial

part of the ring. The proof combines fixed local certificates with a simulation valid at every sufficiently large circumference, followed by six finite base cases.

For clarity, define the sparse graph H_k again. Its vertices are

$$(i, j) \in (\mathbb{Z}/k\mathbb{Z}) \times \{0, 1, 2, 3, 4\}.$$

The first outgoing arc follows the cycle

$$(i, 0) \rightarrow (i, 1) \rightarrow (i, 2) \rightarrow (i, 3) \rightarrow (i, 4) \rightarrow (i + 1, 0).$$

The second outgoing arc is specified by

$$\begin{aligned} (i, 0) &\rightarrow (i + 1, 0), & (i, 1) &\rightarrow (i - 1, 2), \\ (i, 2) &\rightarrow (i - 1, 4), & (i, 3) &\rightarrow (i - 1, 3), \\ (i, 4) &\rightarrow (i - 1, 1). \end{aligned}$$

Both arc maps are permutations, are loopless, and have distinct images at every vertex when $k \geq 4$. Hence H_k has $5k$ vertices and $10k$ arcs, with indegree and outdegree two everywhere. Waiting is an additional legal action, as throughout the paper.

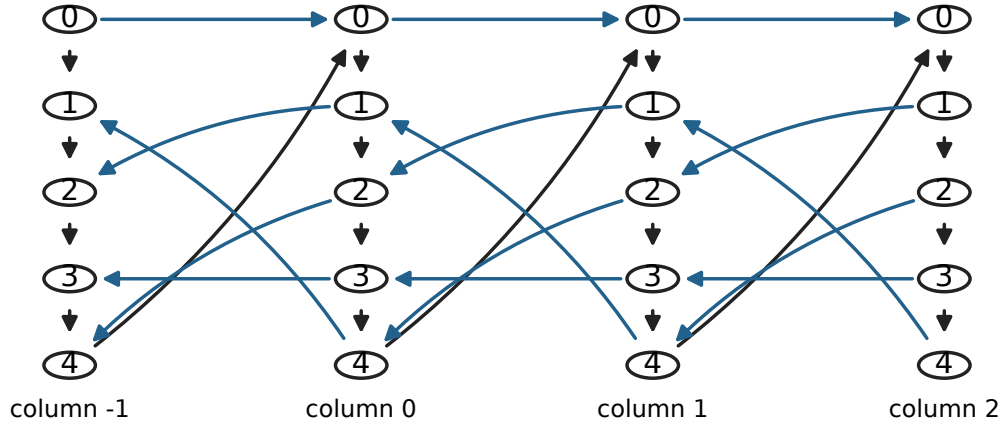


Figure 4: The five station types in consecutive columns. Black arrows follow the long cycle and blue arrows give the second permutation. Identifying columns modulo k closes the strip into the graph H_k .

Theorem 21 (uniform sparse attainment). For every integer $k \geq 4$, every source–recipient pair in H_k is guaranteed. Consequently

$$M_V(5k) = 5k(5k - 3), \quad M_E(10k) = 5k(5k - 3).$$

For $k \geq 10$, a sufficient delivery time is $8k + 16$ messenger moves. The theorem is computer-assisted: its finite ingredients are the two local objectives specified below and the six rings $4 \leq k \leq 9$.

8.1 The stronger local arena

Unwrap the column coordinate to the integers, retaining the same two arc rules. Confine the messenger to columns $-3, \dots, 3$, disallowing moves out of this interval. Represent the pursuer by its exact vertex in columns $-4, \dots, 4$, together with an exterior state Ω . Thus there are 35 messenger positions and 46 pursuer positions.

Within its nine-column window the pursuer has all its usual moves and its wait; a move leaving the window goes to Ω . From Ω it may wait or enter any vertex of either boundary column -4 or 4 . This grants more freedom than a particular physical pursuer outside the window. The messenger and Ω can never coincide.

The two objectives have different stopping rules:

- A **safe transfer** ends in column zero at the start of a messenger turn, with the positions distinct. Entering column zero on a messenger move is insufficient: the intervening pursuer response must also be survived.
- A **delivery** ends immediately when the messenger enters a specified vertex $(0, t)$, before collision is checked there, exactly as in the original game.

For each objective, the certificate assigns nonnegative integer ranks to winning states and selects a legal move at every positive-rank messenger state. Every selected move decreases rank, and every legal response at a positive-rank pursuer state also decreases rank. The zero-rank states are precisely the applicable terminal successes. These conditions prove termination by induction on rank. For safe transfer, only noncollision **messenger-turn** states in column zero are seeded as successes; pursuer-turn states there still require all their responses to be safe.

The checked finite bounds are:

Local objective	Initial column	Initial noncollision states	Maximum messenger moves
Safe transfer to column zero	1	225	8
Delivery to $(0, 0)$	2	225	24
Delivery to $(0, 1)$	2	225	16
Delivery to $(0, 2)$	2	225	16
Delivery to $(0, 3)$	2	225	17
Delivery to $(0, 4)$	2	225	18

Each row covers all five source types and all 46 pursuer states other than the source itself. In particular, it includes an initially exterior pursuer. Only the bounds 8 and 24 are needed below; optimality of these local times is not claimed as a separate result.

8.2 Projection to a ring

Fix $k \geq 10$ and any reference column. Its nine offsets $-4, \dots, 4$ represent distinct columns of the ring. Map a physical pursuer in this window to its exact local position and every other pursuer position to Ω .

Every physical pursuer move has a permitted image. A move within the window is unchanged, one leaving it goes to Ω , and a move remaining outside is represented by waiting at Ω . A move

from outside into the window can enter only a boundary column, because each arc changes the column by at most one; every such entry is permitted from Ω . At $k = 10$ the exterior consists of a single column adjacent to both boundaries, and both entries are already allowed. At $k = 9$ the two boundary columns themselves have wraparound adjacencies, so this argument would fail; those rings are deliberately covered by finite checks instead.

Every messenger move permitted by the local certificate is a genuine ring move. Its seven represented columns lie strictly inside the pursuer window. Consequently a physical collision with the messenger is exactly a local collision; an exterior pursuer cannot coincide with it. Following a local winning strategy therefore succeeds against every physical pursuer strategy. This is a simulation of legal moves and collisions, not an assumption that winning strategies pass automatically to graph covers.

8.3 Composition and termination

Proof of Theorem 21. Translate the recipient's column to zero, and let the messenger's current column be d . Translate the safe-transfer window so that its local column zero is the physical column $d - 1$; the messenger then starts in local column one. The certificate reduces its physical column by one modulo k and finishes only after the pursuer's response, so another transfer can begin with the correct turn order. Recenter the window between transfers; each new local certificate is valid for every resulting noncollision pursuer position.

After the least nonnegative residue of $d - 2$ modulo k such transfers, at most $k - 1$, the messenger is in column two. Apply the delivery certificate for the recipient's type. This takes at most

$$8(k - 1) + 24 = 8k + 16$$

messenger moves. If a transfer encounters the actual recipient earlier, terminal receipt wins immediately and only shortens the play. Thus the argument guarantees finite delivery, not merely indefinite safe movement.

For $4 \leq k \leq 9$, the explicit finite-ring certificates verify every source and pursuer position for each of the five recipient types in column zero. Translating columns gives every recipient. These six checks, together with the uniform argument for $k \geq 10$, establish universality for all $k \geq 4$.

Finally, a universal graph with $n = 5k$ vertices and $2n$ arcs has

$$I(H_k) = n^2 - n - 2n = n(n - 3).$$

The vertex upper bound and the even arc-budget upper bound both equal this number. Hence both asserted extremal equalities follow. $\square$

Certificate reproduction and scope. The file `research/five-type-certificates.json` contains the fixed local arena, ranks, and chosen messenger moves, generated and directly checked by `src/five_type_certificate.py`. The independent implementation `src/five_type_local_verify.py` constructs the explicit alternating-state graph from the arc rules and checks every winning rank condition, as well as closure of the losing region. Its full ranks and selected moves are saved in `research/five-type-independent-local-certificates.json`, together with the implementation hash. Both implementations agree on the six local bounds above. The six finite bases are stored in `research/five-type-small-ring-certificates.json`, including complete arc lists, messenger-move layers, and explicit alternating-state rank certificates for the five representative recipient types. The script `src/five_type_small_certificates.py` reproduces those checks using

both game solvers and verifies every certificate transition. The ordinary projection and composition proof explains why these finite arrays certify infinitely many rings. This theorem is not currently claimed as a Lean formalization.

Corollary 22 (sharp arc-budget leading term). As $m \rightarrow \infty$,

$$M_E(m) = \frac{m^2}{4} - O(m).$$

More explicitly, for every $m \geq 40$,

$$\frac{m^2}{4} - 6m \leq M_E(m) \leq \left\lfloor \frac{m}{2} \right\rfloor^2.$$

Proof. Set $k = \lfloor m/10 \rfloor \geq 4$ and $r = m - 10k$, so $0 \leq r \leq 9$. Take H_k together with r disjoint one-way arcs. Each added component has no indirect guaranteed pair, and additivity over components preserves $I(H_k)$. This graph has exactly m arcs and

$$I = 25k^2 - 15k = \frac{(m-r)^2}{4} - \frac{3(m-r)}{2} \geq \frac{m^2}{4} - 6m.$$

The upper bound follows because there are at most $\lfloor m/2 \rfloor$ possible sources and at most that many possible recipients of indirect guaranteed pairs. Thus the quadratic coefficient $1/4$ is sharp for arbitrary arc counts, not only the progression $m = 10k$. $\square$

Exact equality at arbitrary orders and arc counts remains a separate question. The padding argument establishes the leading term; it does not fill the gaps between the sharp values supplied by Theorem 21.

9 Exact extrema with at most seven arcs

Proposition 23. Allowing arbitrarily many vertices, including isolated vertices,

$$(M_E(0), M_E(1), \dots, M_E(6)) = (0, 0, 0, 0, 1, 1, 2).$$

Proof. Write $A = \{v : d^+(v) \geq 2\}$, $B = \{v : d^-(v) \geq 2\}$, $q = |A|$, $r = |B|$, and $b = |A \cap B|$. Every counted pair belongs to $A \times B$, excludes the diagonal, and excludes arcs. Consequently

$$I(G) \leq qr - b - e(A, B), \quad e(A, B) \geq \max\{0, 2q + 2r - m\}, \quad q, r \leq \lfloor m/2 \rfloor. \quad (23.1)$$

For completeness, the second inequality follows by counting the arcs leaving A and those entering B : their respective totals are at least $2q$ and $2r$, and their union has at most m arcs.

A counted pair (s, t) requires two arcs leaving s and two entering t . These four arcs are distinct because $s \rightarrow t$ is absent. Thus $I(G) = 0$ when $m \leq 3$.

We will use two elementary camping observations. If $d^-(z) = 0$, $z \rightarrow t$, and $d^-(t) = 2$, then no indirect pair ending at t is guaranteed. Indeed, let p be its other predecessor. An indirect source is neither z nor p . A pursuer starting at p can wait there: the messenger cannot reach z , and cannot enter p without being captured before reaching t . Dually, if $d^+(z) = 0$, $s \rightarrow z$, and $d^+(s) = 2$, then no indirect pair starting at s is guaranteed. A pursuer waits at the other successor of s ; the messenger's only alternative departure reaches a sink different from the indirect recipient.

Suppose first that $m = 4$ or $m = 5$. Then $q, r \leq 2$. If either is zero there is nothing to count, and if both are one the bound is immediate. When $q = r = 2$, (23.1) gives $I(G) \leq m - 4 - b \leq 1$. When $q = 1$ and $r = 2$, two counted pairs would have the same source and distinct recipients. The source's two outgoing arcs are then disjoint from the at least four arcs entering those recipients, which would require $m \geq 6$. The case $q = 2, r = 1$ follows by the same count with sources and recipients interchanged: at least four arcs leave the two counted sources, and the recipient's two incoming arcs are disjoint from them. Hence $I(G) \leq 1$.

Now let $m = 6$, so that $q, r \leq 3$. If $q = 1$, three counted pairs would require two outgoing arcs from their common source and six disjoint arcs entering their three recipients; this is impossible. Thus $I(G) \leq 2$. The same argument applies when $r = 1$; cases with a zero coordinate are again immediate. If $q = r = 2$, (23.1) gives $I(G) \leq 2 - b \leq 2$.

Consider $q = 2, r = 3$. Every arc ends in B , and every vertex of B has indegree exactly two. If $A \subseteq B$, then $b = 2$ and (23.1) gives $I(G) \leq 6 - 2 - 4 = 0$. Otherwise choose $z \in A \setminus B$. It has indegree zero and at least two distinct successors in B . The first camping observation eliminates both corresponding recipient columns. At most one column remains, containing at most the two sources in A . Thus $I(G) \leq 2$.

For $q = 3, r = 2$, every arc starts in A and every vertex of A has outdegree exactly two. If $B \subseteq A$, the same count gives zero. Otherwise a vertex $z \in B \setminus A$ has outdegree zero and at least two predecessors in A . The second camping observation eliminates their two source rows, leaving at most one row and two recipients. Again $I(G) \leq 2$.

Finally, if $q = r = 3$, all six arcs go from A to B , and all relevant indegrees and outdegrees equal two. If $A = B$, (23.1) gives $I(G) \leq 9 - 3 - 6 = 0$. Otherwise a vertex of $A \setminus B$ has indegree zero and eliminates two recipient columns by the first camping observation. The remaining column has two distinct predecessors in A , so at most one of the three sources can contribute an indirect pair. This completes the upper bounds in all cases.

For the lower bounds, the four-arc diamond

$$E = \{(0, 1), (0, 2), (1, 3), (2, 3)\}$$

guarantees the indirect pair $(0, 3)$: the messenger chooses the branch not occupied by the pursuer and delivers on the next move. Adding a disjoint arc gives the five-arc example. The six-arc graph

$$E = \{(0, 1), (0, 2), (1, 3), (1, 4), (2, 3), (2, 4)\}$$

guarantees both $(0, 3)$ and $(0, 4)$. If the pursuer starts on one branch, choose the other; if it starts at either sink, choose either branch. In each case it cannot reach the selected branch on its next move, and receipt then occurs on the messenger's second move. Hence $I(G) = 2$, as required. $\square$

These arguments establish the arc-only maximum over every finite vertex set; no restriction on the number of nonisolated vertices is used.

Proposition 24 (with a finite exhaustive computation). We have

$$M_E(7) = 2.$$

Proof. First consider a weakly connected digraph with at most seven arcs and at least seven vertices. Forget the directions but retain one undirected edge for each arc, so that opposite arcs

become parallel edges. The resulting connected multigraph has at most as many edges as vertices. It is therefore a tree or has exactly one cycle; a parallel pair is allowed as a cycle of length two.

In a tree, any reachable indirect source–recipient pair has an internal vertex separating its endpoints. A pursuer can start and wait there, preventing delivery. Thus no indirect pair is guaranteed. In the unicyclic case, a pair with either endpoint outside the unique cycle likewise has an internal separating vertex, unless its endpoints are adjacent in the underlying graph. For adjacent endpoints off the cycle, a directed route exists only if their connecting arc has the required direction, making the pair direct. Thus every counted pair must have both endpoints on the cycle.

There are exactly two simple undirected paths along that cycle between two distinct cycle vertices. If only one is a directed path from the source to the recipient, it is either a direct arc or has an internal vertex where the pursuer can wait. If neither is directed, the recipient is unreachable. Excursions into attached trees cannot bypass an internal cycle vertex. Consequently a counted pair requires both cycle paths to be directed from its source to its recipient. This forces the cycle’s unique divergence at that source and unique convergence at that recipient; every other cycle vertex has one entering and one leaving cycle edge. A fixed orientation can therefore support at most one such ordered pair. A cycle of length two consists of opposite arcs and supports no indirect pair. We have proved that every weakly connected digraph with at most seven arcs and at least seven vertices has $I(G) \leq 1$.

For a general digraph with seven arcs, distinct weak components contribute additively to $I(G)$. Each component with a positive contribution uses at least four arcs by Proposition 23, so at most one component contributes. If that component uses at most six arcs, Proposition 23 bounds its contribution by two. If it uses seven arcs and has at least seven vertices, the preceding argument bounds its contribution by one. In the only remaining case it has at most six vertices. Adding isolated vertices to make six does not change its indirect guarantee count, and the complete six-vertex enumeration gives a maximum of two at seven arcs.

This last finite assertion is the computational dependency of the proposition. It is reproduced by `src/enumerate_small_n6.cpp`; the result is recorded with `complete: true` in `research/solver-enumeration-n6.json`. The enumeration covers every six-vertex digraph up to the degree-ordering reduction described in the computational section; the witness is also checked by the separate explicit-turn solver. The structural reduction above is what extends that finite calculation to unrestricted vertex counts.

Finally, add one disjoint arc to the six-arc, two-recipient construction in Proposition 23. This preserves its two guarantees and supplies seven arcs, proving the matching lower bound. $\square$

Theorem 49 below settles the next four budgets, eight through eleven.

10 Attainment across densities and losses near completeness

Angel Raychev proposed that, for every sufficiently large n , the missing-pair bound should be attained at **every** integer budget

$$2n \leq m \leq n(n-3).$$

Theorems 38–44 below now prove this conjecture, including a deterministic construction and an explicit sufficient threshold. The results in this section provide the probabilistic and dense ingredients of that proof. Raychev also proposed separating existence from explicit constructions and studying how domination destroys missing pairs as the graph approaches completeness. These

distinctions motivate the statements below; their proofs were developed in the September 2026 follow-up investigation.

Write $L = n(n - 1)$ and $h = L - m$. Thus h is the number of missing nonloop arcs, and always $I(G) \leq h$.

10.1 An exact-budget existence theorem

Theorem 25. Let $n \geq 4$, let $0 < m < L$ be an integer, and put $p = m/L$. If

$$(n - 3)p^2(1 - p) > 5 \log n,$$

then some graph with exactly n vertices and m arcs guarantees every source–recipient pair within two messenger moves. In particular,

$$M(n, m) = L - m.$$

Proof. Choose each nonloop arc independently with probability p . Fix distinct vertices s, t and a legal initial pursuer position $c \neq s$. Each $u \notin \{s, t, c\}$ is a safe intermediate vertex when

$$s \rightarrow u, \quad u \rightarrow t \quad \text{are present, and} \quad c \rightarrow u \quad \text{is absent.}$$

These three distinct arcs give probability $p^2(1 - p)$, including when $c = t$. Different candidates use disjoint sets of arcs, so their success events are independent. There are at least $n - 3$ candidates. The probability that this triple has no safe intermediate is at most

$$\exp(-(n - 3)p^2(1 - p)).$$

There are fewer than n^3 triples. The probability that any triple fails is therefore less than n^{-2} under the hypothesis.

It remains to enforce the exact arc count. That count has the binomial distribution with parameters L, p , and m is a mode because $p = m/L$. Indeed, the ratio of the probabilities at $j + 1$ and j is

$$\frac{(L - j)p}{(j + 1)(1 - p)},$$

which decreases with j , is greater than one at $j = m - 1$, and is less than one at $j = m$. Hence the probability of exactly m arcs is at least $1/(L + 1) > n^{-2}$. It exceeds the entire failure probability, so at least one graph with exactly m arcs satisfies every safe-intermediate condition.

In that graph the messenger moves to its chosen intermediate vertex. The pursuer neither occupies that vertex nor can reach it on its response. The messenger then reaches the recipient, and receipt precedes any further interception. All $L - m$ missing pairs are consequently guaranteed. $\square$

This is an exact integer-budget existence theorem, rather than a statement about an expected number of arcs. It also gives a finite randomized search: sample graphs with exactly m arcs and check the safe-intermediate condition. No practical running-time bound near the threshold is asserted.

Corollary 26. There is an absolute N such that, for every $n \geq N$ and every integer budget

$$\left\lceil 10n^{3/2}\sqrt{\log n} \right\rceil \leq m \leq n(n - 3),$$

we have $M(n, m) = n(n-1) - m$. Attainment can always be achieved with delivery in at most two moves.

Proof. Let $p = m/L$ and $h = L - m$. For $n \geq 6$, we have $n-3 \geq n/2$. If $p \leq 1/2$, the lower bound on m gives $p \geq 10\sqrt{\log n/n}$ and therefore

$$(n-3)p^2(1-p) \geq 25 \log n > 5 \log n.$$

If $p \geq 1/2$ and $h \geq 100n \log n$, then $1-p = h/L \geq 100 \log n/n$, giving

$$(n-3)p^2(1-p) \geq \frac{100}{8} \log n > 5 \log n.$$

Theorem 25 covers both cases.

For the remaining dense budgets use Theorem 20 with

$$D = \lfloor (n/2)^{1/4} \rfloor.$$

For all sufficiently large n its hypotheses hold, and its upper endpoint satisfies

$$L_D(n) \geq n(D/4 + 1) - 1.$$

Since D grows as a positive constant times $n^{1/4}$, this endpoint eventually exceeds $100n \log n$. Theorem 20 supplies every integer missing-arc count from $2n$ to $L_D(n)$, and the probabilistic range supplies the rest. The intervals overlap, so no admissible integer budget is omitted. Both proofs give two-move delivery. $\square$

The absolute threshold is not optimized or asserted to be moderate. In particular, this corollary does not establish the full interval at $n = 52$. The general sparse interval between $2n$ and order $n^{3/2}\sqrt{\log n}$ remains outside its conclusion.

10.2 A growing loss in the final dense band

Let F be the loopless directed complement of G , and write $q(v) = d_F^+(v)$. The number

$$\Delta = h - I(G)$$

counts missing pairs that fail to guarantee delivery.

Theorem 27. For $n \geq 2$ and every admissible integer $n \leq h < 2n$,

$$I(G) \leq h - \left\lceil \frac{2n-h}{2} \right\rceil - 1.$$

Equivalently, writing $r = 2n - h$, we have

$$M(n, n(n-3) + r) \leq 2n - r - \lceil r/2 \rceil - 1 \quad (1 \leq r \leq n).$$

Proof. If $I(G) = 0$, the conclusion holds: the right-hand side is increasing with h and at $h = n$ equals $\lfloor (n-2)/2 \rfloor \geq 0$. Assume henceforth that $I(G) > 0$.

Every $q(v)$ is positive. Otherwise a vertex c with $q(c) = 0$ can intercept every nonterminal departure from any other source, while its own source row has no missing pair. Put

$$S = \{c : q(c) = 1\}, \quad \ell = |S|.$$

Since $h \geq \ell + 2(n - \ell)$, we have $\ell \geq 2n - h \geq 1$. For $c \in S$ let $f(c)$ be its unique missing neighbor; thus $f(c) \neq c$. Set

$$U = f(S), \quad a = |U \cap S|, \quad b = |U \setminus S|.$$

Every source $x \in U$ loses all of its $q(x)$ missing pairs. To see this, choose $c \in S$ with $f(c) = x$. The pursuer's closed outgoing neighborhood in G is $V \setminus \{x\}$. It can wait at c while the messenger waits at x and intercept every nonterminal departure. An indirect recipient cannot be reached by a direct departure.

A second obstruction applies when two distinct vertices $c, c' \in S$ have the same image x . The missing pair (c, x) loses against a pursuer starting at c' , whose closed outgoing neighborhood contains every possible nonterminal intermediate vertex.

Let R be the set of $c \in S \setminus U$ for which $f(c)$ has exactly one preimage in S , and put $\rho = |R|$. Every missing edge with tail in $S \setminus R$ loses by one of the two obstructions. The killed rows in $U \setminus S$ have at least two missing edges each, with tails outside S . These losses are disjoint, so

$$\Delta \geq \ell - \rho + 2b.$$

If $b \geq 1$, then $\rho \leq \ell - a$ because $R \subseteq S \setminus U$, and $\rho \leq a + b$ because its members have distinct images in U . Consequently

$$\Delta \geq \max\{a + 2b, \ell - a + b\} \geq \left\lceil \frac{\ell + 3b}{2} \right\rceil \geq \lceil \ell/2 \rceil + 1.$$

If $b = 0$, the function f maps S into itself without fixed points. Its finite functional graph has a directed cycle of length at least two. Every cycle vertex belongs to U and already has a preimage in U , so it cannot also be the image of a member of R : that member is outside U and would give a second preimage. Thus $\rho \leq a - 2$, as well as $\rho \leq \ell - a$. It follows that

$$\Delta \geq \max\{a, \ell - a + 2\} \geq \lceil \ell/2 \rceil + 1.$$

Using $\ell \geq 2n - h$ proves the bound. $\square$

Thus the forced loss increases as 2, 2, 3, 3, 4, 4, ... as the arc budget moves one, two, three, four, five, six, and further steps above $n(n - 3)$. At the other endpoint, $h = n$, the bound agrees with Theorem 13.

10.3 Combining two-move constructions

Lemma 28. Suppose a missing-edge graph F is a disjoint union of graphs F_i , each with positive outgoing degree at every vertex. Let G_i be the complement on its own block. If an indirect pair in G_i can be guaranteed within two messenger moves, then it remains guaranteed in the complement G of the entire F .

Proof. Let the source and recipient lie in block i . If the initial pursuer also lies there, the original two-move strategy chooses an intermediate vertex outside its closed outgoing neighborhood. The same is true in G . The pursuer may respond by moving outside the block, but this cannot prevent receipt on the messenger's next move.

If the pursuer starts at c in another block, choose any outgoing F -neighbor u of c in that block. The arcs $s \rightarrow u$ and $u \rightarrow t$ are present in G because they cross blocks, whereas $c \rightarrow u$ is absent. Also $u \neq c, s, t$. This gives the required safe two-move delivery. $\square$

The lemma concerns two-move strategies. No preservation theorem for arbitrary longer strategies under this join is assumed.

Theorem 29. For every integer n and every $2 \leq r \leq n - 52$,

$$M(n, n(n-3) + r) = 2n - r - \lceil r/2 \rceil - 1.$$

Equivalently, equality in Theorem 27 holds throughout

$$n + 52 \leq h \leq 2n - 2.$$

There is an explicit attaining construction at every such integer budget.

Proof. Put $k = n - r \geq 52$. On a block of r vertices use the functional missing-edge extremizer from Theorem 13. Explicitly, put $j = \lfloor (r-2)/2 \rfloor$ and use vertices a, b, s_i, t_i for $1 \leq i \leq j$, with an additional vertex z when r is odd. Its missing arcs are

$$s_i \rightarrow t_i, \quad t_i \rightarrow a, \quad a \rightarrow b, \quad b \rightarrow a,$$

together with $z \rightarrow a$ when needed. Every vertex has exactly one missing outgoing arc. Its complement guarantees the j pairs (s_i, t_i) within two moves, by the proof of Theorem 13.

On a disjoint block of k vertices, orient Euler circuits in the four-regular girth-six graph of Theorem 19. This missing-edge graph has two incoming and two outgoing arcs at every vertex. Lemma 18 guarantees all $2k$ missing pairs in its complement within two moves.

Take the disjoint union of these missing-edge graphs and complement it. It has $n = r + k$ vertices and $h = r + 2k = 2n - r$ missing arcs. Lemma 28 preserves at least

$$j + 2k = \lfloor (r-2)/2 \rfloor + 2(n-r) = 2n - r - \lceil r/2 \rceil - 1$$

guaranteed missing pairs. Theorem 27 supplies the matching upper bound. $\square$

The attaining construction uses the explicit finite girth-growth certificate when $52 \leq k < 130$ and the ordinary all-order growth proof thereafter. The interval is nonempty for $n \geq 54$. It leaves $r = 1$ and the bounded-offset range $n - 51 \leq r \leq n - 1$ outside its uniform conclusion; $r = n$ is already covered by Theorem 13. The existing 26-vertex incidence block also gives equality at $r = n - 26$ whenever $r \geq 2$, by the same join.

10.4 A restriction on permutation constructions

Proposition 30. Suppose the two outgoing neighbors of each vertex are $\alpha(v)$ and $\beta(v)$, where α, β are commuting permutations, both images differ from v , and $\alpha(v) \neq \beta(v)$. Then $I(G) \leq n$. In fact all indirect guarantees are decided by a two-move criterion.

Proof. Fix a recipient t and put $s_t = \alpha^{-1}\beta^{-1}(t)$. Commutation gives

$$N^+(s_t) = \{\beta^{-1}(t), \alpha^{-1}(t)\} = N^-(t).$$

The predecessor camping lemma excludes every indirect source except possibly s_t : a pursuer starting at s_t can intercept the messenger when it first reaches a predecessor of t . Thus each recipient contributes at most one pair.

When $s_t \neq t$, the pair (s_t, t) is missing because neither predecessor of a loopless recipient equals that recipient. This pair is guaranteed exactly when no $c \neq s_t$ has $N^-(t) \subseteq C(c)$. Necessity is the same

camping lemma. For sufficiency, from each legal initial pursuer position choose $u \in N^-(t) \setminus C(c)$. The messenger moves from s_t to this safe predecessor and then delivers. $\square$

The bound is sharp: on $\mathbb{Z}/7\mathbb{Z}$, take arcs $x \rightarrow x+1$ and $x \rightarrow x+3$. For recipient zero the predecessors are 6, 4, and the only closed outgoing neighborhood containing both is that of 3. The criterion gives $(3, 0)$; translation gives exactly the seven pairs $(s, s+4)$. Since $n(n-3) > n$ for $n \geq 5$, commuting permutations cannot supply a universal vertex-extremal graph in that range. Noncommutation is necessary, not asserted sufficient.

10.5 An extension that fills the small dense offsets

Proposition 31. Let $r \geq 2$, $k \geq 1$, and $n = r + k$. Put $j = \lfloor (r-2)/2 \rfloor$. If

$$k \leq \binom{j}{2},$$

then an explicit graph attains

$$M(n, n(n-3) + r) = 2n - r - \lceil r/2 \rceil - 1.$$

In particular, for every $n \geq 75$ this holds at each offset $k = n - r$ with $1 \leq k \leq 51$.

Proof. Begin with the functional missing-edge graph on r vertices from Theorem 29: its vertices are a, b, s_i, t_i for $1 \leq i \leq j$, with an extra z if r is odd, and its arcs are

$$a \rightarrow b, \quad b \rightarrow a, \quad s_i \rightarrow t_i, \quad t_i \rightarrow a,$$

together with $z \rightarrow a$ when needed. Add a set H of k vertices. For each $v \in H$ choose a distinct two-element subset $A_v \subseteq \{s_1, \dots, s_j\}$ and give v precisely those two outgoing missing arcs. Add no other missing arcs. These choices are possible by the assumed inequality. The resulting missing-edge graph F has n vertices and $h = r + 2k$ arcs.

We show that its complement guarantees each pair (v, t) with $v \in H$ and $t \in A_v$. If the pursuer starts at a functional vertex c , let u be its unique outgoing missing neighbor. This vertex lies among $a, b, t_1, \dots, t_j$, so $v \rightarrow u \rightarrow t$ are present arcs of the complement while $c \rightarrow u$ is absent. If the pursuer starts at another vertex $c \in H \setminus \{v\}$, distinctness and equal cardinality give an element $u \in A_c \setminus A_v$. Then $v \rightarrow u$ is present, and $u \rightarrow t$ is present because $u = s_\ell$ has only t_ℓ as a missing neighbor, whereas t is one of the s_i . Also $u \neq t$ because $t \in A_v$. In each case u is distinct from the pursuer and from both endpoints, and cannot be reached by the pursuer on its next move. This proves all $2k$ of these guarantees in two moves.

Each original pair (s_i, t_i) remains guaranteed as well. Against a functional pursuer $c \neq s_i$, its unique missing neighbor u is neither endpoint: no functional missing arc enters s_i , and the only one entering t_i starts at s_i . Hence $s_i \rightarrow u \rightarrow t_i$ are present and safe. Against a pursuer $c \in H$, choose $s_\ell \in A_c$ with $\ell \neq i$, which is possible because A_c has two members. The route $s_i \rightarrow s_\ell \rightarrow t_i$ is present and safe. Thus at least $j + 2k$ missing pairs are guaranteed.

As in Theorem 29,

$$j + 2k = 2n - r - \lceil r/2 \rceil - 1,$$

and Theorem 27 supplies the upper bound. Finally, if $n \geq 75$ and $1 \leq k \leq 51$, then $r = n - k \geq 24$, so $j \geq 11$ and $\binom{j}{2} \geq 55 \geq k$. $\square$

The construction is generated by `src/dense_functional_extension.py`. Its direct two-move check covers all 51 offsets at $n = 75$; five small examples are also checked by both full game solvers and the independent rank-certificate verifier. The receipt is `research/dense-functional-extension-verification.json`. These checks support the ordinary proof; no finite testing is needed to justify the general parameter range. Combining Proposition 31 with Theorem 29 and the functional endpoint proves sharpness for every $2 \leq r \leq n$ when $n \geq 75$.

All proofs in this follow-up section are ordinary mathematical arguments. Theorem 25 is probabilistic existence; Theorem 29 is an explicit construction with the previously stated finite structural dependency; Proposition 31 is a direct explicit construction. No new Lean coverage is claimed. The full-density conjecture is proved by combining these ingredients with the subsequent sparse constructions in Theorems 38–44.

11 Completing the final dense band

Theorem 32. For every $n \geq 68$,

$$M(n, n(n-3) + 1) = 2n - 3.$$

An explicit attaining graph guarantees all but two of its missing pairs within two messenger moves.

Proof. Theorem 27 gives the upper bound. For attainment put $k = n - 4 \geq 64$, and take a four-regular undirected graph H of girth at least six on k vertices, supplied by Theorem 19. Choose four distinct vertices a, b, d, e with

$$\text{dist}(a, b), \text{dist}(d, e) \geq 4, \quad \text{dist}(u, v) \geq 2 \quad (u \in \{a, b\}, v \in \{d, e\}).$$

Distances between components are infinite. A radius-three ball in a graph of maximum degree four has at most $1 + 4 + 12 + 36 = 53$ vertices. After choosing a , choose b outside that ball. Choose d outside the closed neighborhoods of a, b , whose union has size at most ten. Choose e outside this union and the radius-three ball of d . At most 63 vertices are excluded, so all choices exist when $k \geq 64$.

Orient Euler tours of H . Add four vertices c, x, y, z and the seven missing arcs

$$c \rightarrow x, \quad x \rightarrow y, z, \quad y \rightarrow a, b, \quad z \rightarrow d, e.$$

Let F be this missing-edge graph and G its directed complement. Every vertex has outdegree two in F except c , whose only outgoing arc is $c \rightarrow x$. The unique incoming neighbors of x, y, z in F are respectively c, x, x . We have

$$|V(F)| = k + 4 = n, \quad |E(F)| = 2k + 7 = 2n - 1,$$

so G has $n(n-3) + 1$ arcs.

The underlying graph of F still has girth at least six. The new edges form a tree attached to the core at a, b, d, e , with c a pendant vertex. A new simple cycle with one segment in this tree has either a same-branch segment of length two and a core path of length at least four, or a cross-branch segment of length four and a core path of length at least two. A cycle with two tree segments must use the vertex-disjoint paths $a - y - b$ and $d - z - e$: a cross-branch tree segment would meet any other segment. Its two core paths both join opposite branches and have length at least two, giving

total length at least eight. With four attachment vertices there can be no third tree segment. Thus no cycle of length below six is introduced.

For a missing pair (s, t) , set

$$B = N_F^+(s) \cup N_F^-(t).$$

The local argument in Lemma 18 shows that every initial cop vertex $v \neq s$ with $d_F^+(v) \geq 2$ has an outgoing F -neighbor outside B . Indeed, the induced double star on B has diameter at most three, and such a cop has at most one outgoing neighbor in it. This is a safe intermediary for a two-move delivery in G . The argument only requires the degree bound at that cop vertex; the other degrees need not all satisfy it.

Only initial cop position c remains. Consider a missing pair with $s \neq c, x$. Then $x \notin N_F^+(s)$, since the sole predecessor of x is c . Also $x \notin N_F^-(t)$: otherwise t would be y or z , whose sole predecessor is x , forcing $s = x$. Consequently $x \notin B$, and x gives a safe two-move delivery. When $s = c$, that cop position is not legal initially, so the preceding degree-two argument already covers every legal initial position.

The only remaining missing pairs are (x, y) and (x, z) . Neither is guaranteed: a pursuer at c has closed outgoing neighborhood $V \setminus \{x\}$ in G , and can intercept every nonterminal departure from source x . Waiting never delivers, and neither recipient is directly accessible. Exactly these two pairs fail, leaving $2n - 3$ guarantees. $\square$

The generator `src/dense_first_step.py` implements the choices above and verifies the full graph's girth, the relevant degrees, and every safe-intermediate condition. Its receipt `research/dense-first-step-checks.json` records instances at orders 68, 75, 80, 100, and 164, covering 105,634 missing-pair/cop triples with precisely the two asserted exceptions. These checks corroborate the ordinary proof. The construction at $n \geq 134$ uses the ordinary high-girth construction starting at core order 130; its extension to $68 \leq n < 134$ uses the already stated finite structural insertion certificate.

Corollary 33 (the complete final dense band). For every $n \geq 75$ and every integer $1 \leq r \leq n$,

$$M(n, n(n-3) + r) = 2n - r - \left\lceil \frac{r}{2} \right\rceil - 1.$$

Equivalently, for every integer $n \leq h \leq 2n - 1$,

$$M(n, n(n-1) - h) = h - \left\lceil \frac{2n - h}{2} \right\rceil - 1.$$

There is an explicit attaining graph at every such budget, with a two-move strategy for every counted pair.

Proof. The upper bound is Theorem 27. The first step $r = 1$ is Theorem 32. For $r \geq 2$, put $k = n - r$. If $k \geq 52$, use Theorem 29. If $1 \leq k \leq 51$, then $r = n - k \geq 24$, and Proposition 31 applies because

$$j = \left\lceil \frac{r-2}{2} \right\rceil \geq 11, \quad \binom{j}{2} \geq 55 \geq k.$$

Finally, $k = 0$ means $r = n$, which is the functional construction of Theorem 13. These cases exhaust all integer budgets in the band. $\square$

Together with Theorem 19 at $m = n(n-3)$ and Theorem 13 beyond $m = n(n-2)$, this determines the entire joint extremum for $n \geq 75$ and $m \geq n(n-3)$. At the boundary $m = n(n-3)$ the value is $2n$; the next step has value $2n-3$, and the subsequent values follow the displayed staircase down to $\lfloor (n-2)/2 \rfloor$, after which the value is zero. This result does not resolve the general sparse interval below $n(n-3)$, nor does it assert that every smaller order has the same formula. No new Lean formalization is claimed here.

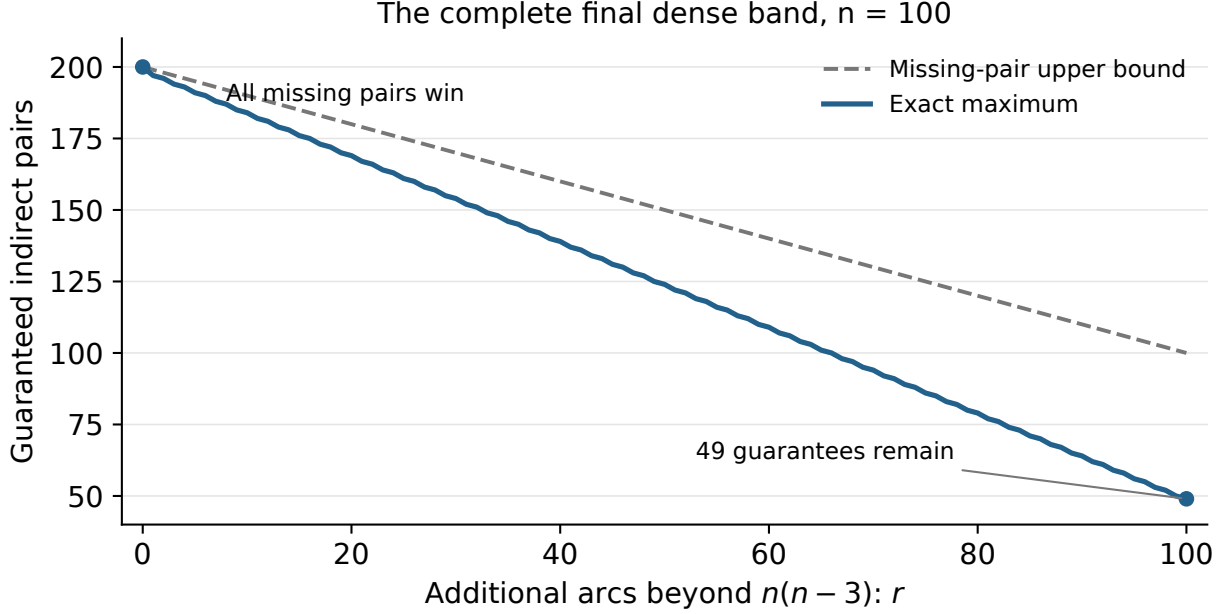


Figure 5: The complete final dense band at $n=100$. The solid curve is the proved exact maximum at every integer budget; the dashed line counts all missing pairs and does not account for forced failures.

12 Explicit attainment at every budget in a sparse interval

The circulant construction gives an exact joint extremum, not only a lower bound on the vertex maximum: for every $n \geq 25$, Theorem 11 and Lemma 12 already show that $M(n, 3n) = n(n-4)$. The next result fills consecutive budgets above $3n$. Its proof controls the extra pursuer moves; arbitrary additions of arcs would not justify that conclusion.

Theorem 34. For every $n \geq 25$,

$$M(n, 3n) = n(n-4).$$

Moreover, if $d \geq 4$ and

$$n > d^3 + 2d,$$

then at every integer budget $3n \leq m \leq dn$ there is an explicit graph attaining

$$M(n, m) = n(n-1) - m.$$

In particular, this holds throughout $3n \leq m \leq 4n$ for every $n \geq 73$. More generally the explicit interval reaches arc counts of order $n^{4/3}$.

Proof. Say that a set $A \subseteq \mathbb{Z}/n\mathbb{Z}$ has distinct nonzero differences when

$$a - b = a' - b' \neq 0, \quad a, b, a', b' \in A,$$

implies $(a, b) = (a', b')$. For such a set, two distinct translates $x + A$ and $y + A$ intersect in at most one point: two intersections would give two representations of the nonzero difference $y - x$.

Choose h as in Lemma 12 and put $a = (h - 1)/2$. The set

$$A_0 = \{0, 1, a, h\}$$

has distinct nonzero differences by the proof of Theorem 11. The base graph B has steps $1, a, h$. For any larger set A with distinct nonzero differences, let H_A have the nonzero steps in A ; we will keep $0 \in A$ throughout. Every graph G satisfying

$$B \subseteq G \subseteq H_A$$

has distinct closed move sets intersecting in at most one vertex, because $C_G(x) \subseteq x + A$.

The proof of Theorem 11 therefore works inside every such G . To make this preservation explicit, fix the recipient zero. Its direct predecessors $-1, -a, -h$ and zero are guaranteed, and $-a - 1$ is guaranteed using the two successors $-a, -1$. If

$$Q(z) = \{z, z - 1, z - a, z - a - 1\}$$

is guaranteed, the following four applications of Lemma 10 establish $Q(z - h)$:

New source	Two guaranteed successors
$z - h$	$z - a - 1, z$
$z - h - 1$	$z - h, z - 1$
$z - h - a$	$z - h, z - a$
$z - h - a - 1$	$z - h - a, z - a - 1$

All displayed moves belong to B . Coprimality of h and n makes finitely many repetitions cover all sources. The same argument with all coordinates shifted applies to every recipient, even when G itself is not translation invariant. Thus every intermediate graph is universally winning.

It remains to enlarge A_0 . Suppose $|A| = k$ and its nonzero differences are distinct. To adjoin x , exclude the following residues:

1. The k members of A itself.
2. Residues $x = a + b - c$ with $a, b, c \in A$ and $b \neq c$. These exclude every equality between an old nonzero difference and either sign of a new difference. There are at most $k^2(k - 1)$ such residues.
3. Solutions of $2x = a + b$ with $a, b \in A$. These exclude an equality $x - a = b - x$ between the two signs of new differences. There are $k(k + 1)/2$ unordered pairs with repetition, and each congruence has at most two solutions, hence at most $k(k + 1)$ excluded residues.

Within each sign the new differences are already distinct. The three exclusions are therefore sufficient, and together rule out at most

$$k + k^2(k - 1) + k(k + 1) = k^3 + 2k$$

residues. Whenever $n > k^3 + 2k$, at least one admissible residue remains; choose the smallest representative. Under the theorem's hypothesis this process continues from size four to size $d + 1$.

The resulting H_A has dn arcs and contains the $3n$ arcs of B . For a prescribed integer m in the interval, add exactly $m - 3n$ of the remaining arcs in lexicographic order. The intermediate graph is universally winning by the preservation argument, so all $n(n - 1) - m$ missing pairs are guaranteed. This matches the elementary upper bound.

For $d = 4$ the hypothesis is $n > 72$. Taking the largest admissible d gives d of order $n^{1/3}$, and therefore an upper interval endpoint of order $n^{4/3}$. The endpoint $m = 3n$ only needs the original $n \geq 25$ construction, with no extension. $\square$

This is a deterministic construction at each stated integer budget. The underlying delivery argument may require many moves; it is not a two-move theorem. Its ordinary proof is independent of the finite tests in `src/sidon_main_interval.py`, and no new Lean coverage is asserted. It provides an initial consecutive part of Raychev's full-density interval. The stronger constructions and exact-budget argument in Theorems 38–48 subsequently cover the entire interval at every sufficiently large order.

13 An explicit algebraic family at intermediate density

Theorem 35. Let $p \geq 5$ be a prime. For every integer

$$4p^2 \leq m \leq p^2(p - 1),$$

there is an explicit graph attaining

$$M(p^2, m) = p^2(p^2 - 1) - m.$$

In particular, the interval extends to $m = n^{3/2} - n$ at orders $n = p^2$. For primes $p \geq 11$, Theorem 34 extends its lower endpoint to $3p^2$.

Proof. Put $A = \{(t, t^2) : t \in \mathbb{F}_p\}$, and let H_A be the directed graph with all nonzero moves in A . Let B_0 be its four-move subgraph with parameters

$$t \in \{1, -1, -2, -3\}.$$

These are distinct and nonzero when $p \geq 5$. We prove universal delivery in every graph G satisfying $B_0 \subseteq G \subseteq H_A$, including graphs without translation symmetry.

Every closed outgoing neighborhood in G is a subset of the corresponding translate of A . Distinct translates of A intersect in at most one point. Indeed, a nonzero difference

$$(t, t^2) - (s, s^2) = (t - s, (t - s)(t + s))$$

determines t, s uniquely: if the first coordinate is nonzero, it determines their difference and sum, and division by two is valid; if the first coordinate is zero, the entire difference is zero. Thus every nonzero difference has at most one ordered representation.

This intersection property gives the following safe-branching rule. Fix a recipient. If two distinct outgoing neighbors of a vertex are already universally winning positions for that recipient, the vertex is universally winning as well. Against an initial cop at another vertex, at most one of

the two successors lies in the cop's closed outgoing neighborhood. Move to the other, survive the cop's response, and use its winning strategy. The rule concerns the actual game, including the intervening pursuer move.

We first use recipient 0 and only moves of the uniform core B_0 . Put

$$u = (1, 1), \quad v = (-2, 4), \quad h = (-3, 9) = u + 2v,$$

and define

$$Q(z) = \{z, z - u, z - v, z - u - v\}.$$

The vertices $0, -u, -v$ are initially winning, by immediate receipt or a direct final move. The remaining vertex $-u - v$ has winning successors $-u, -v$, so all of $Q(0)$ is winning.

If every vertex of $Q(z)$ is winning, so is every vertex of $Q(z - h)$. Apply the safe-branching rule in the following order; each row lists the new vertex and two already winning successors:

$$\begin{array}{c|cc} z - h & z - u - v & z \\ z - h - u & z - h & z - u \\ z - h - v & z - h & z - v \\ z - h - u - v & z - h - v & z - u - v \end{array}$$

The corresponding pairs of moves are $(v, h), (u, h), (v, h), (u, h)$, respectively. They are distinct allowed moves.

Repeated propagation makes the entire line $L = \{\lambda h : \lambda \in \mathbb{F}_p\}$ winning: the nonzero vector h has additive order p , so integer multiples cover all the displayed scalars.

Use the fourth core move $w = (-1, 1)$. We have

$$w - v = (1, -3) = -h/3 \in L,$$

while $v \notin L$ because $\det(h, v) = 6 \neq 0$. If every vertex of a coset $L + z$ is winning, then every vertex of $L + z - v$ has two distinct winning successors, obtained by adding v and w . The safe-branching rule therefore fills that entire coset as well. Iterating gives all cosets $L - jv$, which are the p distinct cosets because $v \notin L$. Thus every vertex is winning.

The same proof shifted by the recipient uses only moves of B_0 and the global closed-neighborhood intersection bound. It applies to every recipient even if the extra arcs of G are not translation invariant. Hence every intermediate graph is universally winning.

The core has $4p^2$ arcs, and H_A has $p^2(p-1)$ arcs. Add the remaining arcs one at a time in any fixed order until the prescribed integer budget m is reached. All $p^2(p^2-1) - m$ missing nonloop pairs are guaranteed, attaining the elementary upper bound. Finally, when $p \geq 11$ we have $p^2 \geq 73$, so Theorem 34 covers the adjacent interval $3p^2 \leq m \leq 4p^2$. $\square$

The proof is finite and constructive: initializing a quartet takes at most two messenger moves, and each quartet propagation increases a bound by at most four. At most $p-1$ such propagations and $p-1$ further coset steps therefore give the uniform, unoptimized bound $5p-3$ messenger moves.

The prime hypothesis is intentional. In an extension field, repeated addition only covers scalars from its prime subfield, so this four-move propagation does not justify the same conclusion for arbitrary prime powers. The construction is checked at $p = 5, 7, 11$ by `src/parabola_cayley.py`; it verifies the full-parabola intersection condition, the four-move core's complete safe-branching closure, and

the core/upper-graph inclusions for all 827 consecutive integer budgets in these examples. At $p = 5$, both full game solvers and their rank certificates independently verify every recipient. These finite checks supplement the ordinary proof and are recorded in `research/parabola-cayley-checks.json`. No new Lean formalization is claimed.

14 A consecutive interval above the two-regular endpoint

The five-type construction can tolerate a controlled set of additional arcs. The proof gives the pursuer every additional move while restricting the messenger to the original moves. This verifies all intermediate graphs at once, without assuming that winning is monotone under adding arcs.

Theorem 36 (with finite local certificates). Let $n = 5k$ with $k \geq 10$. For every integer

$$2n \leq m \leq \frac{11n}{5},$$

there is an explicit graph with n vertices and m arcs satisfying

$$M(n, m) = n(n - 1) - m.$$

Every pair can be guaranteed within $8k + 18$ messenger moves.

Proof. Use the five-type graph H_k of Theorem 21 on vertices (i, j) , where i is modulo k and $0 \leq j \leq 4$. Its two outgoing arcs are

$$\begin{aligned} \alpha(i, j) &= (i, j + 1) & (j < 4), & & \alpha(i, 4) &= (i + 1, 0), \\ \beta(i, 0) &= (i + 1, 0), & \beta(i, 1) &= (i - 1, 2), \\ \beta(i, 2) &= (i - 1, 4), & \beta(i, 3) &= (i - 1, 3), & \beta(i, 4) &= (i - 1, 1). \end{aligned}$$

There are $2n$ arcs. Add any chosen subset of the k further arcs

$$(i, 0) \longrightarrow (i - 1, 4).$$

They are distinct from the original arcs and from each other, so every integer budget in the stated interval is obtained.

We verify the stronger asymmetric game in which the messenger may use only α, β and waiting, while the pursuer may also use all k additional arcs. Construct the same local arena as in Theorem 21: messenger columns $-3, \dots, 3$, pursuer columns $-4, \dots, 4$, and a pursuer state Ω representing the outside. The messenger is constrained to its local columns. The pursuer may leave into Ω , wait there, and re-enter at any vertex in either boundary column -4 or 4 . Within its columns it has the two original arcs and the additional arc above whenever its type is zero.

Two independently implemented finite attractor computations verify the following objectives against every legal initial pursuer state, including Ω :

Objective	Initial messenger column	Maximum messenger moves
Reach column zero safely at the beginning of a messenger turn	1	8
Deliver to (0, 0)	2	26
Deliver to (0, 1)	2	19
Deliver to (0, 2)	2	17
Deliver to (0, 3)	2	17
Deliver to (0, 4)	2	18

Each row covers all five initial messenger types and all 45 legal pursuer positions per type, for 225 required initial states. The transfer goal is seeded only at a messenger turn after the pursuer response, with unequal positions. Delivery goals use immediate receipt before collision, as required by the game. The rank checks verify every selected messenger move and every permitted pursuer response; these are complete finite certificates for the displayed statements.

For $k \geq 10$, the nine represented pursuer columns are distinct on the ring. All moves, including the additional ones, change the column by at most one. Therefore an actual pursuer outside those columns can first enter only a boundary column; its behavior projects into the adversarial Ω state. The actual messenger can follow every certified original move. The local certificates consequently apply at every translated column of every such ring, independently of which additional arcs the actual graph contains.

Fix a recipient in column zero. Repeatedly translate and apply the safe-transfer certificate to move the messenger one column backward until it reaches column two. This needs at most $k - 1$ transfers, each leaving a legal messenger-turn state for the next certificate. Then apply the appropriate delivery certificate. The total is at most

$$8(k - 1) + 26 = 8k + 18.$$

Thus every pair is guaranteed, and all $n(n - 1) - m$ missing pairs are counted. The missing-pair upper bound proves equality. $\square$

The primary generator and full ranks are `src/five_type_sparse_interval.py` and `research/five-type-sparse-interval-certificates.json`. The independent explicit alternating-state implementation is `src/five_type_augmented_verify.py`, with its full certificate in `research/five-type-augmented-independent-certificates.json`. The ring projection and concatenation are ordinary proofs; the bounded local facts depend on these finite certificates. This theorem is not presently formalized in Lean, and it makes no claim for the small rings $k = 4, \dots, 9$ or for budgets beyond $11n/5$.

Theorem 37 (with finite local certificates). Let $n = 5k$ with $k \geq 16$. At every integer budget $2n \leq m \leq 3n$, there is an explicit graph attaining

$$M(n, m) = n(n - 1) - m.$$

Every pair can be guaranteed within $8k + 16$ messenger moves.

Proof. Keep the original five-type graph, and add any subset of the n arcs

$$(i, j) \longrightarrow (i + 3, 0), \quad 0 \leq j \leq 4.$$

They are pairwise distinct and are not original arcs. It suffices to win with the messenger restricted to the original graph and the pursuer allowed all the additional arcs.

Use messenger columns $[-3, 3]$, pursuer columns $[-6, 6]$, and the outside state Ω . Give the represented pursuer its original and added moves; moves leaving the interval enter Ω . From Ω permit waiting and re-entry at every vertex in the six columns $-6, -5, -4, 4, 5, 6$. Independent synchronous and explicit alternating-state certificates verify safe transfer from column one to zero in at most eight moves, and delivery from column two to the five recipient types at zero in at most 24, 16, 16, 17, 18 moves. Each objective covers all 325 legal initial states: five messenger types and 65 pursuer positions per type. The transfer ends on a safe messenger turn, and delivery has immediate terminal priority, exactly as in Theorem 36.

We check the larger-jump projection explicitly. The thirteen represented pursuer columns are distinct when $k \geq 16$, and at least three consecutive columns lie outside them. No move, whose column displacement is at most three, can jump over that entire outside gap and land at the opposite represented boundary. Thus a represented move projects either to its represented destination or to Ω . An outside pursuer can enter the represented interval only in one of the six allowed boundary columns. Every actual pursuer behavior is therefore included in the finite arena. This gap condition is why the proof uses $k \geq 16$, not merely enough columns to make the represented labels distinct.

At most $k - 1$ safe backward transfers reach column two relative to the recipient. The final delivery costs at most 24 further moves, giving $8(k - 1) + 24 = 8k + 16$. The strategy is available in every intermediate graph, and its pursuer has no more moves than the certified adversary. Hence all pairs are guaranteed at each of the $n + 1$ integer budgets. Counting missing pairs proves equality. $\square$

The full primary ranks are `research/five-type-full-sparse-interval-certificates.json`, generated by `src/five_type_sparse_interval.py`. The independent 4620-state implementation and certificate are `src/five_type_long_augmented_verify.py` and `research/five-type-long-augmented-independent-certificates.json`. The statement is a uniform theorem with a finite local proof ingredient, not an extrapolation from tested circumferences. No new Lean coverage is claimed. For orders divisible by five with $n \geq 80$, Theorems 34 and 37 combine into a consecutive explicit interval starting at $2n$ and reaching order $n^{4/3}$.

15 Protecting the moves used by the strategy

The distinct-difference condition in Theorem 34 is stronger than the delivery strategy needs. Its branching steps only compare pairs of the three original outgoing moves. Preserving those comparisons permits a much larger set of additional moves.

Theorem 38. For every $n \geq 25$ and every integer

$$3n \leq m \leq n \left(\left\lceil \frac{n}{7} \right\rceil - 1 \right),$$

there is an explicit graph attaining

$$M(n, m) = n(n - 1) - m.$$

The strategy uses only three outgoing moves per vertex; the remaining arcs may be added individually up to the stated budget.

Proof. Choose h using Lemma 12, put $a = (h - 1)/2$, and set

$$S = \{1, a, h\}, \quad A_0 = \{0, 1, a, h\} \subseteq \mathbb{Z}/n\mathbb{Z}.$$

The proof of Theorem 11 shows that A_0 has distinct nonzero differences. In particular, each of the six ordered differences

$$D = \{u - v : u, v \in S, u \neq v\} = \{\pm(a - 1), \pm(a + 1), \pm 2a\}$$

has just its original ordered representation in A_0 . We only preserve uniqueness for these six differences.

Starting from $A = A_0$, adjoin any residue outside

$$A \cup (A + D), \quad A + D = \{x + d : x \in A, d \in D\}.$$

No new ordered pair involving the adjoined residue has a difference in D , because $D = -D$. Thus the six protected differences retain their original unique representations. When $|A| = k$, at most $7k$ residues are excluded. If $k < \lceil n/7 \rceil$, then $7k < n$, so the extension remains possible. Choosing the smallest available representative constructs a set of size at least $\lceil n/7 \rceil$. For $25 \leq n \leq 28$, the initial set already has that size.

Let B be the uniform three-move graph with steps in S , and let H_A have all nonzero steps in A . Consider any graph

$$B \subseteq G \subseteq H_A.$$

For a source vertex r and distinct core moves $u, v \in S$, no different cop vertex c can have both $r + u$ and $r + v$ in its closed outgoing neighborhood. Indeed, that neighborhood is contained in $c + A$. If both successors belonged to it, their corresponding elements $x, y \in A$ would satisfy $x - y = u - v \in D$. Protected uniqueness forces $(x, y) = (u, v)$ and therefore $c = r$, a contradiction.

Consequently, whenever two successors obtained using core moves are universally winning for a fixed recipient, their source is universally winning: at least one of those successors is inaccessible to the initial cop and survives its response. This argument does not assert that arbitrary pairs of actual outgoing successors have the same property.

The entire proof of Theorem 11 uses exactly these protected pairs. For recipient zero, the direct predecessors $-1, -a, -h$ are winning, and $-a - 1$ uses the core moves $1, a$. If

$$Q(z) = \{z, z - 1, z - a, z - a - 1\}$$

is winning, the four steps establishing $Q(z - h)$ use the move pairs

$$(a, h), \quad (1, h), \quad (a, h), \quad (1, h),$$

respectively, in the order given in Theorem 34. Hence that quartet propagates in G . Since $\gcd(h, n) = 1$, its successive translates cover every source. Shifting the proof by the recipient covers every target, regardless of whether the added arcs are translation invariant.

Thus every graph between B and H_A is universally winning. The first has $3n$ arcs and the latter has at least $n(\lceil n/7 \rceil - 1)$. Add the extra arcs in lexicographic order until the prescribed integer budget is reached. All $n(n - 1) - m$ missing pairs are guaranteed, meeting the elementary upper bound. $\square$

The improvement from order $n^{4/3}$ to order n^2 comes from preserving only the distinctions actually used by the strategy. Other differences may repeat many times. The graph may also have larger closed-neighborhood intersections; the proof protects its particular pairs of core successors instead of imposing a global intersection bound.

For example, at $n = 29$ the construction gives core moves $1, 3, 7$ and the envelope $A = \{0, 1, 3, 7, 8\}$. Its protected differences are $\pm 2, \pm 4, \pm 6$, each still uniquely represented. The unprotected difference -1 occurs twice, as $0 - 1 = 7 - 8$. Thus this envelope genuinely fails the full distinct-difference condition, yet every intermediate graph with $87 \leq m \leq 116$ is universally winning.

Corollary 39. There is an absolute N such that, for every $n \geq N$ and every integer

$$3n \leq m \leq n(n-3),$$

we have

$$M(n, m) = n(n-1) - m.$$

For every sufficiently large multiple of five n , the same formula holds at every integer budget $2n \leq m \leq n(n-3)$.

Proof. The upper endpoint in Theorem 38 satisfies

$$n(\lceil n/7 \rceil - 1) \geq n^2/7 - n.$$

Since $\sqrt{n/\log n} \rightarrow \infty$, this endpoint eventually exceeds $\lceil 10n^{3/2}\sqrt{\log n} \rceil$, the lower endpoint in Corollary 26. The two integer intervals overlap. Theorem 38 covers the lower one and Corollary 26 covers all remaining budgets through $n(n-3)$, including its dense end via the high-girth complement construction. Taking a common sufficiently large threshold establishes the first statement.

For multiples of five, Theorem 37 supplies every budget from $2n$ through $3n$ once $n \geq 80$. Combining the intervals proves the second statement. $\square$

This settles the large-order joint problem across the main interval except for the remaining sparse range $2n \leq m < 3n$ at general orders. The proof supplies explicit constructions through the quadratic endpoint of Theorem 38 and probabilistic existence over the remaining part supplied by Corollary 26. It does not assert a practical numerical value of N or a short explicit construction at every budget in the latter part. The statement for multiples of five also retains the finite local certificates used by Theorem 37. No new Lean verification is claimed.

The generator `src/main_interval_resume.py` checks protected differences at eight orders from 25 through 1000. It checks every protected source-pair/cop triple at the seven listed orders through 200, and verifies 24 selected recipient games with both independent solvers and the explicit rank-certificate checker. Those tests include intermediate graphs lacking translation symmetry. The receipt `research/main-interval-resume-checks.json` records their exact scope; the parameterized proof above is independent of these finite checks.

16 Sparse endpoint attainment at every order from twelve

A second periodic construction admits controlled removal of vertices. Unlike a finite modification checked at one circumference, the removal is certified in a fixed local game that applies at every translated position. Separating the removals then provides all residue classes of the vertex count.

Theorem 40 (with finite certificates). For every integer $n \geq 12$, there is an explicit two-in/two-out-regular universally winning digraph on n vertices. Consequently

$$M_V(n) = M_E(2n) = M(n, 2n) = n(n - 3).$$

Every source–recipient pair in the construction can be guaranteed within $9\lceil n/6 \rceil + 18$ messenger moves.

Proof. First construct a uniform family for $n \geq 85$. Begin with k columns, indexed modulo k , containing vertices (i, j) of types $0 \leq j \leq 5$. One permutation follows the Hamilton cycle

$$0_i \rightarrow 1_i \rightarrow 2_i \rightarrow 3_i \rightarrow 4_i \rightarrow 5_i \rightarrow 0_{i+1}.$$

The other permutation has arcs

$$\begin{aligned} 0_i &\rightarrow 5_i, & 1_i &\rightarrow 4_{i+1}, & 2_i &\rightarrow 1_{i+1}, \\ 3_i &\rightarrow 0_{i-1}, & 4_i &\rightarrow 2_{i+1}, & 5_i &\rightarrow 3_{i-1}. \end{aligned}$$

Choose columns S whose consecutive cyclic gaps are at least three. At each $h \in S$, remove 4_h and smooth its path separately in the two permutations:

$$3_h \rightarrow 4_h \rightarrow 5_h \quad \mapsto \quad 3_h \rightarrow 5_h,$$

$$1_{h-1} \rightarrow 4_h \rightarrow 2_{h+1} \quad \mapsto \quad 1_{h-1} \rightarrow 2_{h+1}.$$

All remaining arcs stay unchanged. The resulting graph has $6k - |S|$ vertices. Smoothing preserves each permutation’s bijectivity. There are no loops or coincident outgoing arcs: the modified first-permutation arc leaves type three for type five, while its other arc ends at type zero; the modified second-permutation arc leaves type one for type two two columns ahead, while its first-permutation arc ends at type two in the same column. Thus the graph remains simple and two-in/two-out regular.

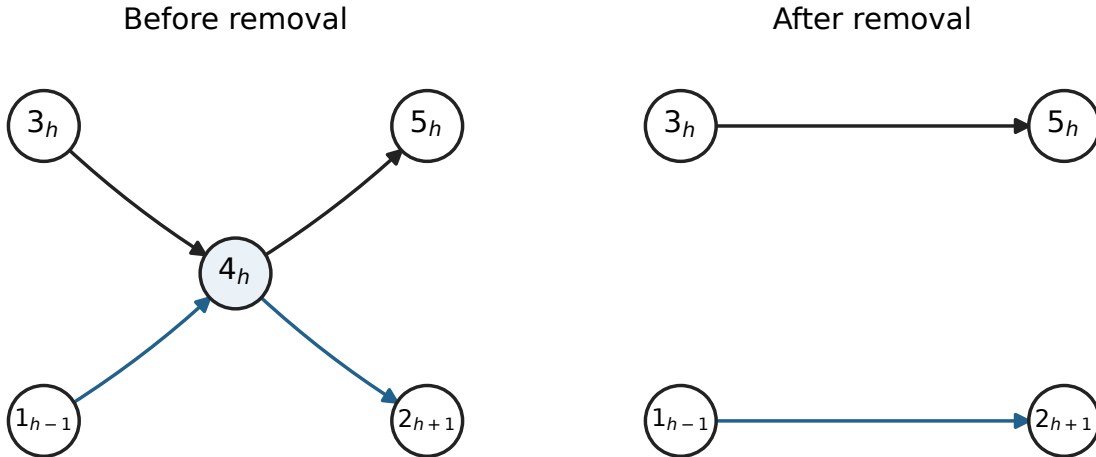


Figure 6: Remove one type-four vertex and reconnect each color separately. All undisplayed arcs stay unchanged. Spaced removals give every residue class of the total number of vertices; the local certificates prove that delivery remains possible.

We establish a uniform strategy when $k \geq 15$. Around an arbitrary reference column zero, restrict the messenger to columns $[-4, 4]$ and represent the pursuer in columns $[-6, 6]$, together with a state Ω for all outside positions. Retain both players' waits and graph arcs. Messenger moves leaving its window are disallowed; pursuer moves leaving its window lead to Ω . From Ω allow waiting and entry at any existing vertex in columns $-6, -5, 5, 6$.

The removals inside the represented pursuer window form a subset of $\{-6, \dots, 6\}$ with consecutive gaps at least three. There are exactly 189 such subsets. For every one of these patterns, the following finite local claims have complete rank certificates:

Objective	Initial messenger column	Maximum messenger moves
Reach column zero safely at the start of a messenger turn	-1	9
Deliver to any specified existing vertex in column zero	-2	27

Each claim covers every existing initial messenger type and every noncolliding initial pursuer position, including Ω . Safe-transfer goals are seeded only at messenger turns, after the preceding pursuer response. Delivery goals instead use immediate receipt before collision. There are 1,287 certificates: seven objectives for each of the 189 patterns, with delivery to 4_0 omitted in the 36 patterns where that vertex is removed. An independent explicit alternating-state implementation verifies all 560,885 required initial states and all 10,271,740 states of these finite games, including winning rank descent and losing-region closure.

We check that these local games represent every actual play on the ring. The thirteen pursuer columns are distinct and leave at least two exterior columns. Every graph arc changes the column by at most two, so a represented pursuer cannot jump across that exterior gap and land at the opposite represented boundary in one move. Every move from outside into the window lands in one of its two boundary layers, where Ω permits entry.

A removal outside the represented window does not change any represented-to-represented arc. The only change to an outward arc occurs when removing 4_7 redirects $1_6 \rightarrow 4_7$ to $1_6 \rightarrow 2_8$; both heads represent Ω . Outside-to-inside changes are already included among its permitted boundary entries. At the messenger boundary, any analogous changed outgoing arc still leaves the allowed messenger window. The local model therefore depends only on the removals inside the pursuer window. Their consecutive gaps are at least three, so it always has one of the 189 certified patterns.

Fix a recipient in column zero. Recenter and apply safe transfer repeatedly: each phase increases the messenger's column by one and ends on an uncaught messenger turn, ready for the next phase. After at most $k - 1$ phases it reaches column -2 , where the delivery certificate applies. The total is at most

$$9(k - 1) + 27 = 9k + 18.$$

If a transfer encounters the actual recipient earlier, play simply ends with delivery. This proves universal winning for every permitted set S .

Finally, given $n \geq 85$, put $k = \lceil n/6 \rceil$ and $r = 6k - n$. Then $k \geq 15$ and $0 \leq r \leq 5$. Choose

$$S = \{0, 3, \dots, 3(r - 1)\}$$

when $r > 0$, and take S empty when $r = 0$. The displayed gaps are three, and the wraparound gap is at least $15 - 12 = 3$. The graph has exactly n vertices and $2n$ arcs. Universal winning gives $I = n(n - 1) - 2n = n(n - 3)$, and the vertex and even arc-budget upper bounds give the three stated equalities for $n \geq 85$.

The complete primary certificates and their direct checker are `research/sparse-composition-multi-certificates.json.gz` and `src/sparse_composition_multi_certificate.py`. The independently written verifier uses direct arc formulas and an explicit alternating graph: `src/six-type-independent-spacing.py`, with full ranks in `research/six-type-independent-spacing-3-certificates.json.gz` and a summary in `research/six-type-independent-spacing-3-summary.json`. The graph definition, spacing argument and simulation are ordinary proofs; the two bounded local claims depend on the finite certificates. No Lean formalization of this theorem is asserted.

For completeness, the finite range $12 \leq n \leq 84$ is supplied by 73 explicit graphs. Each graph has $2n$ distinct nonloop arcs and both incoming and outgoing degrees two. Two independent exact game algorithms agree, and every actual-game rank obligation is checked for every recipient and every allowed initial pursuer. The finite witnesses also satisfy the displayed delivery-time bound. Their arc lists are in `research/sparse-finite-bridge-witnesses.json`, the complete ranks in `research/sparse-finite-bridge-certificates.jsonl.gz`, and the verification receipt in `research/sparse-finite-bridge-verification.json`. The unified generator `src/sparse_finite_bridge_construct.py` returns the corresponding finite witness below 85 and the uniform construction above it. This proves the remaining finite cases of the theorem. $\square$

Together with the small-order results of Theorem 46, only orders nine through eleven remain unclassified in the vertex problem. The separate Theorem 45 addresses the odd arc-budget bound. Both its finite bridge and uniform local certificates remain outside the current Lean coverage.

17 Bridging the entire sparse interval at every large order

The six-type construction can also support optional arcs. We keep those arcs away from deleted vertices, allowing two separately verified local strategies to cover every part of the graph.

Theorem 41 (with finite local certificates). For every integer $n \geq 2419$ and every integer $2n \leq m \leq 3n$, there is an explicit graph with

$$M(n, m) = n(n - 1) - m.$$

Every pair can be delivered within $9\lceil n/6 \rceil + 26$ messenger moves.

Proof. Put $k = \lceil n/6 \rceil \geq 404$ and $r = 6k - n \in \{0, \dots, 5\}$. Use the six-type graph of Theorem 40, with deletions of type 4 in columns

$$S = \{0, 17, \dots, 17(r - 1)\}$$

when $r > 0$, and no deletions when $r = 0$. The cyclic gaps are at least 17. Smooth both permutation paths as in that theorem, obtaining exactly n vertices and $2n$ arcs.

The following eight triples (j, d, t) specify optional arcs $(i, j) \rightarrow (i + d, t)$:

$$(1, -3, 5), (3, -3, 5), (4, -3, 0), (4, 3, 1), (2, 3, 0), (3, 3, 0), (3, -3, 0), (5, -2, 0).$$

Allow them only at columns i whose cyclic distance from every member of S exceeds 16. Their target types are never 4, and their source columns contain no deleted vertex. They are distinct nonloop arcs, absent from the smoothed base graph: all but the last have column displacement ± 3 ; the last has displacement -2 , whereas the base displacements are $0, \pm 1$ and the sole smoothed displacement is $+2$.

For $r \geq 1$, the forbidden source columns are contained in the cyclic image of the integer interval

$$[-16, 17(r-1) + 16],$$

which has $17r + 16 \leq 101$ elements. At least $k - 101$ columns therefore permit all eight optional arcs, giving at least

$$8(k - 101) = 8k - 808 \geq 6k \geq n$$

optional arcs. With no deletions there are $8k \geq n$. Thus any prescribed $m - 2n \leq n$ of these arcs may be selected. It remains to prove that every such selection preserves universal delivery.

Consider local messenger columns $[-4, 4]$ and pursuer columns $[-8, 8]$, with an outside state Ω . The messenger uses only base arcs and may not leave its window. A pursuer move leaving its window enters Ω ; from Ω it may wait or enter any existing vertex in columns $-8, -7, -6, 6, 7, 8$. These three boundary layers cover every possible three-column re-entry.

There are two kinds of certified arenas:

1. There is no deletion inside the pursuer window. The messenger has only the original six-type moves, while the pursuer has all eight optional arc types at every source column in its window.
2. There is one deletion, at some offset $h \in [-8, 8]$. Both players use the smoothed base moves; the pursuer still has the same three-layer outside re-entry rule.

The finite certificates prove safe transfer from any existing type in column -1 to column zero in at most nine messenger moves, and delivery from any existing type in column -2 to any existing recipient in column zero in at most 35 moves. The transfer completes only on a messenger turn after the pursuer response, with distinct positions. Delivery has immediate receipt priority. There are 125 certificates: seven for the augmented unmodified arena, and seven for each of the 17 possible deletion offsets, except for the absent recipient at $(0, 4)$ when $h = 0$.

Every actual local window fits one of these cases. Its 17 represented pursuer columns have diameter 16, so cannot contain two deletions. If it contains a deletion h , each represented pursuer source column is at cyclic distance at most 16 from h , and hence none of its optional arcs was allowed. If it contains no deletion, the augmented arena contains every possible actual pursuer move and every messenger move used by the strategy. An exterior deletion changes no represented-to-represented base arc: the only affected boundary tail has both its original and smoothed heads outside. Discarded messenger exits are likewise unaffected.

All actual moves have column displacement at most three. Since $k \geq 404$, the exterior gap between the two ends of the represented pursuer window has at least three columns. No single move can cross that entire gap between two represented vertices. An exterior return lands in one of the three allowed boundary layers. Thus mapping every exterior pursuer to Ω correctly includes all possible responses. The actual pursuer has no options beyond the relevant certified arena.

Repeatedly use safe transfer to advance the messenger by one column, ending each phase after the pursuer response. At most $k - 1$ transfers place it two columns before the recipient. Apply the

appropriate delivery certificate. The total is at most

$$9(k - 1) + 35 = 9k + 26.$$

This proves universal delivery for every selected subset of optional arcs. All $n(n - 1) - m$ missing pairs are consequently guaranteed. $\square$

The primary generator and saved-rank checker are `src/six_type_budget_certificate.py` and `research/six-type-budget-certificates.json.gz`. The independent explicit alternating-game implementation checks all 125 modules, 75,085 required initial states, and 1,365,108 complete states, including winning rank decrease and closure of losing states. Its sources and receipts are `src/six_type_independent_augmented.py` and `research/six-type-independent-augmented-certificates.json.gz`. These are finite proof certificates coupled with the ordinary uniform projection above. They are not presently Lean formalizations.

Corollary 42 (the complete main interval at all sufficiently large orders). There is an absolute integer N such that, for every $n \geq N$ and every integer $2n \leq m \leq n(n - 3)$,

$$M(n, m) = n(n - 1) - m.$$

Proof. Increase the threshold from Corollary 39, if necessary, to be at least 2419. Theorem 41 supplies every integer budget from $2n$ to $3n$; Corollary 39 supplies every remaining budget through $n(n - 3)$. $\square$

This proves Raychev’s main-interval conjecture with a sufficiently-large order threshold. The threshold N is not claimed to be 2419: the latter is sufficient for Theorem 41 only, while the combined existence argument retains the asymptotic threshold from Corollaries 26 and 39. The construction is explicit in the intervals furnished by Theorems 38 and 41; the middle-density existence step is probabilistic. The result does not settle odd arc-only extrema, the exact small-order threshold, or all joint extrema below $2n$.

18 Deterministically constructing the probabilistic examples

The probabilistic existence argument can be made algorithmic without changing its density range. We give the conditional counts explicitly, since conditioning on an exact arc budget destroys independence between different candidate intermediaries.

Theorem 43. Put $L = n(n - 1)$ and $p = m/L$. Under the hypotheses of Theorem 25, there is a deterministic algorithm, with running time polynomial in n , which constructs an n -vertex, exactly m -arc graph in which every pair is guaranteed within two messenger moves.

Proof. For each triple (s, t, c) with $s \neq t$ and $c \neq s$, call the triple bad if there is no vertex $u \notin \{s, t, c\}$ for which

$$s \rightarrow u \text{ and } u \rightarrow t \text{ are present, while } c \rightarrow u \text{ is absent.}$$

Let Z count the bad triples. The calculation in Theorem 25 bounds the expectation, not just the probability of some failure:

$$\mathbb{E}_p Z \leq n^3 \exp(-(n - 3)p^2(1 - p)) < n^{-2}.$$

For the independently sampled graph let T be its number of arcs. The same binomial-mode bound gives $\Pr(T = m) \geq 1/(L + 1) > n^{-2}$. Therefore

$$\mathbb{E}_p[Z \mid T = m] \leq \frac{\mathbb{E}_p Z}{\Pr(T = m)} < 1.$$

Conditioning on $T = m$ produces the uniform distribution on all exactly m -arc graphs.

We next show that its conditional expectation remains exactly computable after any partial assignment of present and absent arcs. Let F_1, F_0 be the assigned present and absent sets, with $\ell = L - |F_1| - |F_0|$ undecided arcs and $b = m - |F_1|$ present arcs still required. There are

$$C = \binom{\ell}{b}$$

completions, with the convention that an infeasible binomial coefficient is zero.

Fix one triple (s, t, c) . For distinct candidate vertices u , the three arc variables $(s, u), (u, t), (c, u)$ are disjoint. This also holds when $c = t$. A candidate fails to witness the triple unless its three variables take the one specified pattern: present, present, absent. If the partial assignment already contradicts that pattern, and q of its variables remain undecided, all their assignments are allowed in the bad event; its generating factor is $(1 + x)^q$. Otherwise let a of the undecided variables require presence in the witness pattern. The bad-event factor is

$$P_u(x) = (1 + x)^q - x^a.$$

Here powers of x count newly present arcs. This formula includes a fully assigned witness: $q = a = 0$ gives the zero factor, so the triple cannot be bad in any completion.

If q_0 undecided arc variables belong to none of these candidate triples, the exact number of bad completions for (s, t, c) is

$$[x^b](1 + x)^{q_0} \prod_{u \notin \{s, t, c\}} P_u(x).$$

Sum these coefficients over all (s, t, c) to obtain an integer B . It counts bad triples with multiplicity over all completions, and hence the exact conditional expectation is B/C . Independence is used only to describe disjoint arc variables before imposing the budget; extracting one coefficient imposes that budget jointly.

Initially $B < C$. Expose the next undecided arc and compute (B_0, C_0) and (B_1, C_1) for its two values. The completions partition, so

$$B = B_0 + B_1, \quad C = C_0 + C_1.$$

At least one branch satisfies $B_j < C_j$. Choose the first such branch and continue. Infeasible branches have $C_j = 0$ and cannot qualify. At the end there is exactly one completion, so $C = 1$ and the nonnegative integer $B < 1$ is zero. The resulting graph has exactly m arcs and no bad triples. Each missing pair therefore has a safe two-move intermediary against every legal initial pursuer.

For complexity, there are $O(n^2)$ decisions and $O(n^3)$ triples per decision. Each coefficient can be computed by multiplying $O(n)$ polynomials of degree at most three, followed by a binomial-coefficient sum for the remaining variables. Polynomial-length integer arrays suffice. All coefficients and counts have $O(n^2 + \log n)$ bits, since there are at most 2^L assignments and fewer than n^3 triples.

Straightforward exact arithmetic therefore takes polynomial time in n . This is a polynomial-time statement, not a claim of a small practical running time. $\square$

Together with the explicit sparse and high-girth constructions, this makes the full main-interval theorem constructive by a deterministic algorithm for every sufficiently large n . Use Theorems 38 and 41 in their sparse ranges, the algorithm above where Theorem 25 applies, and the high-girth complement construction in the final dense portion. The proofs of Corollaries 26, 39, and 42 show that these ranges overlap. The unknown moderate-order onset remains a separate issue; algorithmic construction does not lower that threshold automatically.

`src/exact_budget_derandomize.py` implements the exact coefficient count and branch selection. Its tests compare 120 conditional counts with complete small enumerations, check branch additivity, and run a complete nontrivial fixed-budget decision sequence for a selected objective. The receipt `research/exact-budget-derandomization-checks.json` records the precise finite checks. No large asymptotic instance was run, and no new Lean formalization is asserted.

18.1 An explicit threshold for the entire main interval

Theorem 44. For every integer $n \geq 1800$ and every integer

$$2n \leq m \leq n(n-3),$$

a deterministic algorithm constructs a graph with exactly n vertices and m arcs in which every pair is guaranteed. Consequently $M(n, m) = n(n-1) - m$ throughout this interval.

Proof. Write $L = n(n-1)$ and $h = L - m$. The packed-hole construction of Theorem 52 covers $2n \leq m \leq 3n$ at every $n \geq 1075$. Theorem 38 continues the interval through

$$T = n(\lceil n/7 \rceil - 1).$$

At the other end, the incidence construction of Theorem 50 covers $2n \leq h \leq 3n$ for every $n \geq 166$. The protected dense construction of Theorem 48 then covers

$$3n \leq h \leq H = n \left\lceil \frac{n-1}{13} \right\rceil.$$

In particular $H \geq L/13$.

A budget not yet covered must satisfy $m > T$ and $h > H$. Its density $p = m/L$ therefore satisfies

$$\frac{n-7}{7(n-1)} < p < \frac{12}{13}.$$

For $n \geq 1800$ the lower endpoint exceeds $71/500$. The function $p^2(1-p)$ has its minimum on this interval at an endpoint, and both endpoint values exceed $173/10000$. The overlap calculation following Proposition 51 proves, using exact rational logarithm bounds, that

$$\frac{173(n-3)}{10000} > 4 \log n + \log(8/3) \quad (n \geq 1800).$$

That proposition sharpens the conditioning estimate in Theorem 43, so the same deterministic exact-budget algorithm supplies every remaining graph. The sparse, middle, and dense ranges cover every stated integer. The missing-pair upper bound proves equality. $\square$

The sufficient threshold 1800 is not asserted optimal. The endpoint already holds at every $n \geq 12$; the first sparse interval starts at 1075 and the dense bridge at 166. Other families give many additional smaller-order ranges. Earlier overlaps gave the sufficient bounds 10^{12} , 5442, and 2419. The protected dense envelope, incidence blocks, packed deletions and sharper exact-budget estimate remove those limitations; the research log retains their derivations.

19 The odd endpoint and completion of its degree defect

The odd arc-budget equality theorem prescribes a particularly small defect: on $n = k + 1$ vertices there is one vertex u of outdegree one, a different vertex v of indegree one, all other indicated degrees are two, and the arc $u \rightarrow v$ is absent. The candidate extremal value is

$$M_E(2n - 1) = (n - 2)(n - 3).$$

The equality theorem reduces attainment to guaranteeing every missing pair whose source differs from u and whose recipient differs from v . It does not make arbitrary deletion from, or completion to, a two-regular graph preserve the relevant guarantees.

19.1 Two necessary structural properties

Lemma. Every odd-extremal graph with the preceding degree pattern is strongly connected. Moreover, for each vertex c , the graph with c deleted has a directed path from every $s \notin \{u, c\}$ to every $t \notin \{v, c\}$.

Proof. Each such ordered pair with $s \neq t$ is either a direct arc or a guaranteed missing pair. Against a pursuer initially at c and waiting there, any successful delivery path avoids c internally. Since its endpoints differ from c , it gives the asserted path in the deleted graph.

To prove strong connectivity without deleting a vertex, all vertices different from u already reach all vertices different from v . The unique predecessor b of v is different from u , because $u \rightarrow v$ is absent. Thus all vertices other than u reach v by first reaching b . The unique successor a of u differs from u , and u reaches every vertex by first going to a . $\square$

These are necessary conditions, not substitutes for the full game: the pursuer can move, so ordinary connectivity and stationary separators do not establish an attaining strategy.

19.2 Completing an odd extremizer can fail

Proposition (finite certified example). There is a graph G on 19 vertices with 37 arcs attaining $I(G) = 272 = M_E(37)$ such that adding the unique degree-completing arc yields a two-in/two-out graph which is not universally winning. Its completed indirect count is 276, whereas a universally winning two-regular graph on 19 vertices would have 304.

The example has vertex set $\{0, \dots, 18\}$ and the following outgoing neighbors:

Vertex	Outgoing neighbors	Vertex	Outgoing neighbors
0	5, 12	10	2, 17
1	15, 10	11	1, 6
2	7, 0	12	3, 8
3	1, 18	13	4, 5
4	15	14	4, 12
5	11, 9	15	0, 16
6	8, 13	16	7, 9
7	13, 17	17	14, 11
8	10, 16	18	2, 6
9	3, 14		

Its exceptional vertices are $u = 4$ and $v = 18$. The finite actual-game certificate verifies every guarantee required by odd equality and the count 272. The upper bound follows from Theorem 7 with $k = 18$.

There is also a short ordinary obstruction after adding $4 \rightarrow 18$. Fix recipient zero and start the messenger at 1, the pursuer at 4. Define

$$S_4 = \{1, 3, 5, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17\}, \quad S_{13} = \{6, 11, 12, 14\}.$$

The following four classes of messenger-turn states form a pursuer trap:

$$(s, 4) \ (s \in S_4), \quad (10, 18), \quad (17, 6), \quad (s, 13) \ (s \in S_{13}).$$

They contain the initial state and exclude every state where the messenger has reached zero. The pursuer acts as follows, in terms of the messenger's new vertex x :

Current pursuer	Response
4	If $x = 10$, move to 18. If $x = 15$ or 18, capture at x . Otherwise wait at 4.
18	If $x = 2$, capture there. If $x = 17$, move to 6. If $x = 10$, wait.
6	If $x = 11$ or 14, move to 13. If $x = 17$, wait.
13	If $x \in \{1, 3, 4, 8\}$, move to 4, capturing if $x = 4$. Otherwise wait at 13.

An immediate collision with the current pursuer already loses. Direct inspection of the outgoing-neighbor table shows that every other possible move either permits the stated capture or returns to one of the four trap classes. Waiting is included. Since infinite play without receipt loses, the messenger cannot deliver to zero. This explicitly identifies how the new arc helps the pursuer: the transition from cop position 4 to 18 after the messenger moves from 1 to 10 was unavailable before completion.

`src/odd_endpoint_completion.py` contains the fixed graph and runs both independent game solvers, checking every winning rank and every losing closure obligation. `research/odd-endpoint-completion-certificate.json` stores the complete finite certificates for the original and completed graphs. The ordinary trap proves nonuniversality without computation; the original graph's

attainment and the completed count 276 use the verified finite certificates. This example is not presently embedded in Lean. The subsequent construction settles the odd endpoint at every sufficiently large budget.

19.3 A useful degree-preserving search operation

Given a two-in/two-out graph, split one vertex w into vertices v, u . Assign one old incoming and one old outgoing arc to each new vertex, then add $v \rightarrow u$. This adds one vertex and one arc, giving exactly the odd equality degree pattern: v has indegree one and outdegree two, while u has indegree two and outdegree one. The forbidden opposite arc $u \rightarrow v$ is absent because the original graph was loopless.

This operation supplies a structured family to investigate, but it is not a preservation theorem. A split of the 20-vertex five-type graph attains $M_E(41) = 342$, while all twenty canonical splits of its 30-vertex member fail to attain the corresponding odd bound. Likewise, changing the two horizontal lane speeds to three produces attaining single-arc deletions at 40 and 50 vertices but failures at neighboring and larger orders. These counterchecks prevent promoting isolated success to a uniform construction. A different six-type family supplies the following uniform positive result.

19.4 A proved infinite family attaining the odd bound

Theorem (with finite local certificates). For every integer $q \geq 15$,

$$M_E(12q - 1) = (6q - 2)(6q - 3).$$

An explicit attaining graph has $n = 6q$ vertices and $2n - 1$ arcs. Every eligible pair can be guaranteed within $14q + 12$ messenger moves.

Proof. Use vertices (i, j) , with i modulo q and $0 \leq j \leq 5$. The first permutation follows the six types in order and then advances one column:

$$\alpha(i, j) = (i, j + 1) \ (j < 5), \quad \alpha(i, 5) = (i + 1, 0).$$

The second permutation is

$$\begin{aligned} \beta(i, 0) &= (i - 1, 0), & \beta(i, 1) &= (i - 1, 2), \\ \beta(i, 2) &= (i + 1, 5), & \beta(i, 3) &= (i + 1, 1), \\ \beta(i, 4) &= (i - 1, 4), & \beta(i, 5) &= (i + 1, 3). \end{aligned}$$

These permutations have distinct nonloop images at every vertex. Delete the single arc $u \rightarrow v$, where $u = (0, 0)$ and $v = (0, 1)$. The graph now has the exact odd equality degree pattern. We must guarantee every pair whose source differs from u and whose recipient differs from v .

Construct a local arena with messenger columns $[-4, 4]$, pursuer columns $[-6, 6]$, and a pursuer outside state Ω . The messenger may use only moves remaining in its represented columns. The pursuer may leave to Ω , wait there, and re-enter at any vertex in columns $-6, -5, 5, 6$. Include both the unchanged arena and, for each $h \in \{-6, \dots, 6\}$, the arena with the translated arc $(h, 0) \rightarrow (h, 1)$ deleted.

In every such arena the certificates prove two kinds of statement:

- From each vertex of column -1 other than the degree-one source, reach some vertex of column zero other than that source safely, at the beginning of a messenger turn, in at most 14 moves.

- From each vertex of column -2 other than the degree-one source, deliver to each recipient in column zero other than the degree-one recipient, in at most 26 moves.

All legal initial pursuer positions, including Ω , are quantified. Safe transfer goals exclude collision and are seeded only on messenger turns; delivery goals retain immediate terminal receipt. Two independent implementations check the full 97 objectives, all 44,850 required initial states, and every winning-rank and losing-closure condition. These are finite proof certificates for the displayed local statements.

For $q \geq 15$, the thirteen pursuer columns are distinct and at least two columns remain outside. Every actual move changes its column by at most one, so an outside pursuer can first re-enter only at a permitted boundary column. No represented move can jump across the outside gap. The actual game therefore projects into this stronger local arena. A deletion in the represented columns gives one of the listed translated modules; otherwise the module is unchanged.

Fix an eligible recipient. Repeated safe transfers advance the messenger one column at a time until it reaches the column two places before the recipient. At most $q - 1$ transfers are needed. Each transfer avoids the exceptional source even at its final state, so the next module's initial requirement continues to hold. The delivery module then applies. This proves finite delivery in at most $14(q - 1) + 26 = 14q + 12$ moves.

Finally, there are $(n - 1)^2$ pairs with an eligible source and recipient. Exactly $n - 2$ are diagonal. The unique arc leaving u and the unique arc entering v are distinct because $u \rightarrow v$ is absent, so precisely $(2n - 1) - 2 = 2n - 3$ arcs belong to this rectangle. Hence the number of eligible missing pairs is

$$(n - 1)^2 - (n - 2) - (2n - 3) = (n - 2)(n - 3).$$

All are guaranteed, and the odd arc-budget upper bound proves equality. $\square$

The primary proof data are generated by `src/odd_endpoint_certificates.py` and stored in `research/odd-endpoint-uniform-certificates.json.gz`. The independent checker `src/six_type_independent_odd.py` constructs its own graph and explicit alternating arena; its full ranks are in `research/six-type-independent-odd-certificates.json.gz`, with a summary and independent review in the adjacent files. The projection and finite composition argument are ordinary proofs. This result is not currently formalized in Lean. The next theorem removes the congruence restriction.

19.5 Every sufficiently large odd budget

Theorem 45 (the odd endpoint). For every integer $n \geq 145$,

$$M_E(2n - 1) = (n - 2)(n - 3).$$

Equivalently, for every odd $m \geq 289$,

$$M_E(m) = \frac{(m - 3)(m - 5)}{4}.$$

An explicit attaining graph has n vertices. Every pair required for attainment has a delivery strategy using at most $34\lceil n/6 \rceil + 27$ messenger moves. The construction and its composition proof are ordinary finite arguments; the bounded local strategy lemmas are supplied by independently verified finite certificates.

Proof. Put $q = \lceil n/6 \rceil$ and $r = 6q - n$, so $q \geq 25$ and $0 \leq r \leq 5$. We first describe a graph on the surviving vertices of $(\mathbb{Z}/q\mathbb{Z}) \times \{0, \dots, 5\}$. Its two initial permutations are

$$\alpha(i, j) = \begin{cases} (i, j+1) & j < 5, \\ (i+1, 0) & j = 5, \end{cases} \quad \beta(i, j) = (i + e_j, \sigma(j)).$$

Use the following four choices of (σ, e) , whose entries are in type order $0, 1, \dots, 5$:

Pattern	σ	e
<i>A</i>	(0, 2, 5, 1, 4, 3)	(-1, -1, 1, 1, -1, 1)
<i>B</i>	(0, 5, 1, 4, 3, 2)	(-1, 1, 1, -1, -1, 1)
<i>C</i>	(3, 5, 0, 2, 4, 1)	(-1, 1, -1, -1, -1, 0)
<i>D</i>	(4, 1, 5, 2, 0, 3)	(-1, -1, 1, 1, -1, 1)

For a set H of deleted vertices, smooth each permutation separately: from a surviving vertex, follow that permutation until first reaching another surviving vertex, and use the resulting arc. Finally delete one specified arc of the smoothed graph. The complete choices are:

r	Pattern	Tail u of deleted arc	Which arc	Direction d
0	<i>A</i>	(0, 0)	α	1
1	<i>B</i>	(1, 4)	β	1
2	<i>C</i>	(0, 5)	β	-1
3	<i>D</i>	(0, 1)	α	1
4	<i>D</i>	(0, 1)	α	1
5	<i>C</i>	(0, 5)	β	-1

The corresponding deleted vertex sets are:

- $r = 0$: $H = \emptyset$.
- $r = 1$: $H = \{(0, 3)\}$.
- $r = 2$: $H = \{(0, 0), (0, 1)\}$.
- $r = 3$: $H = \{(0, 2), (0, 3), (0, 4)\}$.
- $r = 4$: $H = \{(-1, 4), (0, 2), (0, 3), (0, 4)\}$.
- $r = 5$: $H = \{(0, 0), (0, 1), (1, 0), (1, 1), (1, 5)\}$.

Write v for the head of the final deleted arc. Every type cycle of each displayed σ has a nonzero total displacement e with $|e|$ no greater than its length ℓ . Its cyclic lift has length $\ell q / \gcd(q, e) \geq q$, whereas α is a cycle of length $6q$. At most five vertices are removed, so no permutation cycle disappears. The smoothing operation therefore preserves the permutation property on surviving vertices. Inspection of these six fixed patterns shows that the two smoothed images at each surviving vertex are distinct, are not the vertex itself, and change the column by at most two. The changes to outgoing arcs are confined to tails in columns $[-2, 2]$; removed vertices lie in columns $[-1, 1]$. These are finite checks on the displayed tables, independent of q . Because $q \geq 25$, reducing these local coordinates modulo q introduces neither a coincidence nor a loop. Thus the final graph has $n = 6q - r$ vertices, $2n - 1$ arcs, precisely one outdegree-one vertex u , precisely one indegree-one vertex v , and all other corresponding degrees two. Moreover $u \neq v$ and $u \rightarrow v$ is absent.

We now specify the finite strategy lemmas and explain exactly how they apply to arbitrary q . Work first in the infinite column strip. In a local arena the messenger is restricted to columns $[-5, 5]$; the pursuer has columns $[-7, 7]$ together with an outside state Ω . Inside these windows both players have their actual graph moves and may wait. A pursuer move leaving its window goes to Ω , from which the pursuer may wait or re-enter at any surviving vertex in columns $-7, -6, 6, 7$. Include the unchanged arena and every translation of the finite defect whose anchor belongs to $\{-10, \dots, 10\}$.

For $r \leq 4$, the certificates prove the following statements for every such arena, every admissible messenger vertex in the stated source column, and every initial pursuer position different from that vertex, including Ω :

- **Safe transfer:** from column $-d$, reach a surviving vertex of column zero different from u , at the start of a messenger turn.
- **Delivery:** from column $-2d$, deliver to any surviving recipient in column zero different from v .

In both statements the starting messenger vertex must differ from u . For $r = 5$, where $d = -1$, safe transfer instead permits either column zero or column -1 as its final column, and delivery is guaranteed from either column 2 or column 3. The exclusion of u at the starting and final transfer states remains unchanged. Safe transfer goals exclude collision and are terminal only at the start of a messenger turn. Delivery goals give receipt priority over collision, exactly as in the original game.

The maximum certified move counts are:

r	Safe transfer	Delivery
0	14	26
1	9	22
2	17	26
3	34	23
4	28	27
5	9	26

The primary checker verifies all 903 local objectives with the same windows for all six rows. A separately written checker verifies the five nonempty deletion patterns in 750 objectives, covering 425,176 required initial states and 7,590,092 explicit alternating states. The unchanged-vertex pattern has its earlier independent 97-objective certificate. Every certificate includes the full attractor ranks and checks each existential messenger move and every universal pursuer response. Thus the displayed local statements are finite lemmas, not inferences from testing increasing ring lengths.

For the projection to a cyclic graph, represent the fifteen pursuer columns by their consecutive integer lifts. All actual moves have column jump at most two. An outside pursuer can therefore first enter only one of the four boundary columns allowed from Ω . Since $q \geq 25$, the outside gap has at least ten columns, so no move can cross that gap between the two represented ends. Messenger positions remain in their eleven distinct represented columns, and Ω cannot conceal a collision there. Consequently each actual pursuer move has a permitted projected response in the stronger local arena.

The only defect anchors which can affect a represented outgoing move are within nine columns of the arena center, since changed tails lie within two columns of their anchor. Vertex deletions

and the two degree exceptions are covered by the same bound. Testing anchors from -10 to 10 therefore covers every relevant configuration. This interval has length 21 , less than q , so at most one periodic copy of the defect can occur in it. If no copy occurs, the unchanged arena applies. These observations prove the projection for every cyclic position of the local window.

Fix a source other than u and a recipient other than v . For $r \leq 4$, repeatedly apply safe transfer in direction d until the messenger is two columns before the recipient. At most $q - 1$ transfers are needed, after which the delivery module applies. Each transfer ends at a messenger turn away from u , so it satisfies its next module's initial requirements regardless of the pursuer's history.

For $r = 5$, measure the messenger's column distance to the recipient in the leftward direction by an integer $a \in \{0, \dots, q - 1\}$. Delivery applies as soon as $a \in \{2, 3\}$. Otherwise a transfer decreases this distance by one or two modulo q . If $a \geq 4$, this cannot pass below 2 without first reaching 2 or 3 . If initially $a = 0$ or 1 , at most two transfers either reach those distances or wrap to $q - 1$ or $q - 2$, after which the preceding decreasing argument applies. In every case at most q transfers suffice. This proves finite delivery, uniformly bounded by $34q + 27$ moves.

Every missing pair with source different from u and recipient different from v is therefore guaranteed. Their number is $(n - 1)^2 - (n - 2) - (2n - 3) = (n - 2)(n - 3)$, as in the preceding theorem. The odd arc-budget upper bound proves equality. $\square$

The generator `src/odd_endpoint_all_residues.py` supplies the six graph families and the complete primary proof data in `research/odd-endpoint-all-residues-certificates.json.gz`. `src/six_type_independent_odd_residues.py` independently constructs the smoothed graphs and explicit alternating games; its full proof data and summary are `research/six-type-independent-odd-residues-certificates.json.gz` and `research/six-type-independent-odd-residues-summary.json`. The original six-type pattern is additionally checked by `src/six_type_independent_odd.py`. These certificates and the ordinary projection argument establish an explicit all-order result; this new endpoint theorem has not yet been formalized in Lean.

Corollary 45.1 (the odd endpoint from 25 arcs). For every $n \geq 13$,

$$M(n, 2n - 1) = M_E(2n - 1) = (n - 2)(n - 3).$$

Consequently the exact odd arc-only extremum is

$$M_E(m) = \frac{(m - 3)(m - 5)}{4} \quad (m \geq 25 \text{ odd}).$$

Proof. Theorem 45 covers $n \geq 145$. For every one of the remaining 132 integers $13 \leq n \leq 144$, the supplementary data give a fixed simple graph with $2n - 1$ arcs, one outdegree-one vertex u , one different indegree-one vertex v , all other corresponding degrees two, and $u \rightarrow v$ absent. The displayed six-type residue families supply most of these graphs. Deleting one arc from independently verified finite two-regular examples supplies the remaining orders from 19 upward. Six further fixed witnesses cover orders 13 through 18.

For every recipient other than v , the finite certificates specify an integer rank for every explicit messenger-turn and pursuer-turn state. Terminal receipt has rank zero, nonterminal collision has rank -1 , and every other nonnegative messenger rank has a legal successor of smaller nonnegative rank. Every legal successor of a nonnegative pursuer rank has smaller nonnegative rank. The direct checker verifies these conditions against the original graph, together with the entire complementary losing closure. Every initial messenger state whose source differs from u and whose pursuer position

differs from the source has nonnegative rank. Strict rank descent therefore forces finite delivery from each required state against every pursuer policy. This proves all required guarantees for each listed graph.

The full check contains 10,236 recipient objectives and 215,967,810 explicit alternating states. The same degree count as in Theorem 45 then gives $(n - 2)(n - 3)$ guaranteed missing pairs. The arc-only upper bound applies to all support sizes, and hence also to the fixed support size n . This proves both equalities. $\square$

The 126 graphs at orders 19 through 144 are in `research/odd-endpoint-finite-bridge-graphs.json`. Their complete ranks are in the streamable archive `research/odd-endpoint-finite-bridge-certificates.jsonl.gz`, with source hashes and a full numerical summary in `research/odd-endpoint-finite-bridge-verification.json`. The producer `src/odd_endpoint_finite_certify.cpp` uses an explicit alternating-game attractor. The separate direct verifier `src/odd_endpoint_finite_verify.py` checks the original legal moves; it does not rerun or trust the producer’s queue logic. Its accelerated array checks have an equivalent standard-library scalar mode. Scalar replay of all objectives at orders 19 through 24 also passes, and the earlier two independent game algorithms agree with selected saved certificates at orders 19, 28 and 144. The complete direct rank checks, rather than these additional sample comparisons, prove the finite bridge. The six smaller witnesses are fully checked by both original game algorithms and the direct transition verifier in `src/odd_endpoint_small_verify.py`; their complete proof data and summary are `research/odd-endpoint-small-certificates.json.gz` and `research/odd-endpoint-small-verification.json`. These finite certificates have not yet been imported into Lean.

Proposition (small odd exceptions; exhaustive finite check). For $6 \leq k \leq 11$, the general odd upper bound is not attained:

$$M_E(2k + 1) \leq k(k - 3) + 1.$$

In particular, $M_E(23) \leq 89$.

Proof. Equality in the upper bound $k(k - 3) + 2$ would, by the equality characterization, have precisely $n = k + 1$ nonisolated vertices, one outdegree-one vertex u , one distinct indegree-one vertex v , all other corresponding degrees two, and $u \rightarrow v$ absent. Add this missing arc. The resulting two-in/two-out digraph is a bipartite two-regular graph when its sources and recipients are represented in separate parts. Each of its bipartite cycles has even length, so alternating edge colors decompose it into two permutation factors A, B . Choose A to contain $u \rightarrow v$ and label $u = 0, v = 1$.

Up to relabeling fixing these two vertices, the A -cycle through zero is $(0, 1, \dots, \ell - 1)$ for one $\ell \geq 2$. Its other cycles are represented by an integer partition of $n - \ell$ into parts at least two. For each such choice, enumerating all permutations B with $B(i) \neq i, A(i)$ covers every candidate graph; remove $0 \rightarrow 1$ before applying any game test.

Two necessary obstructions safely prune this enumeration. Once the outgoing neighborhood of an eligible source s is complete, if some $c \neq s$ has $N^+(s) \subseteq N^+[c]$ and there is an eligible missing recipient for s , the pursuer can wait at c and capture the first nonwaiting messenger move. Once the incoming neighborhood of an eligible recipient t is complete, if $N^-(t) \subseteq N^+[c]$ and there is an eligible source outside $N^-(t) \cup \{t, c\}$, waiting at c until the messenger first enters $N^-(t)$ likewise permits capture. In a partial assignment the cop’s outgoing row may be incomplete: coverage already present persists under later additions. The source or recipient neighborhood used in the

inclusion is always complete. The code checks explicitly that the required eligible missing partner exists.

All remaining completed graphs are evaluated by the exact finite reachability game, with receipt priority and finite termination. Enumeration finishes at every $7 \leq n \leq 12$ and finds no attaining graph. Its counts are:

n	Distinguished cycle partitions	Partial nodes	Completed graphs surviving local obstructions
7	7	756	0
8	11	6,338	21
9	15	59,375	2,092
10	22	800,507	64,992
11	30	10,079,292	767,210
12	42	164,276,910	16,902,091

These finite exclusions and the equality characterization prove the claimed strict bounds. $\square$

The complete enumerator is `src/odd_endpoint_exhaustive.cpp`; its finished receipts are `research/odd-endpoint-small-exhaustive.json` and `research/odd-endpoint-n12-exhaustive.json`. A known attaining 13-vertex graph passes every partial pruning test as a positive control. The independent review `research/odd-endpoint-twelve-independent-audit.md` checks the permutation coverage, both pruning arguments, and the final game semantics. It is an independent source and proof audit, not a separately implemented repetition of the enumeration. Neither this result nor the preceding odd endpoint certificates currently has Lean coverage. The proposition does not determine the exact small odd maxima.

Lemma (first reduction at 23 arcs). Every graph with 23 arcs satisfies $I(G) \leq 87$.

Proof. Write A, B, q, r, b, e as in the general arc-bound proof. Then $q, r \leq 11$, $e \geq 2q + 2r - 23$, and $I \leq qr - b - e$. We first show that $I \geq 88$ would force one of three degree patterns on 12 nonisolated vertices.

If $q, r \leq 10$ the elementary bound is at most 83; if either is at most 9 it is at most 82. For $(q, r) = (10, 11)$, the bound is $91 - b$, so only $b \leq 3$ needs consideration. There are then at least six indegree-zero vertices in $A \setminus B$: all indegree outside B totals at most one. They emit at least twelve arcs. At most one recipient in B can have indegree three, and then no arc ends outside B ; alternatively at most one arc ends outside B . At least nine of the twelve arcs therefore enter indegree-two recipients in B , producing at least five zero-contribution columns. This gives $I \leq 10(11 - 5) = 60$. The case $(11, 10)$ uses the corresponding zero-row argument. Hence $q = r = 11$.

If $b \leq 7$, at least three members of $A \setminus B$ have indegree zero. Their at least six outgoing arcs produce at least two indegree-two zero columns by the same argument. Deleting those columns removes at most four of the at least 21 direct pairs, so $I \leq 99 - 17 = 82$. If $b = 8$ or 9, there is at least one such zero column. Indeed, the two distinct successors of any indegree-zero member cannot both avoid indegree-two vertices of B : an indegree-three recipient already uses the whole available unit of indegree surplus. Removing the column gives $I \leq 110 - 19 - (b - 1) = 92 - b \leq 84$.

Suppose $b = 10$, and write $A \setminus B = \{v\}$, $B \setminus A = \{u\}$. If v has indegree zero or u has outdegree zero, a zero column or row gives $I \leq 82$. Otherwise their respective degrees equal one and exhaust all indegree outside B and all outdegree outside A . All degrees in A and B equal two, and all other vertices are isolated. This gives the first two patterns below, according as $u \rightarrow v$ is absent or present.

Finally suppose $b = 11$, so $A = B = C$ has 11 vertices. At most one arc starts outside C and at most one ends outside C . If all arcs lie in C , there are only $110 - 23 = 87$ missing pairs. An arc with both endpoints outside C leaves a two-regular 11-vertex core; such a core is not universal by the completed small regular exclusion, so again $I \leq 87$. If there is exactly one crossing arc, either deleting it leaves a two-regular core, or its active endpoint has degree two and a zero-degree external neighbor. In the former case a losing core strategy still applies: the messenger cannot use an inaccessible external source or enter an external sink on a successful route to an eligible recipient. In the latter case an entire eligible row or column, containing at least nine missing pairs, contributes zero. Neither permits $I \geq 88$.

If there are two crossing arcs, they are $C \rightarrow x$ and $y \rightarrow C$. When $x \neq y$, the same sink or indegree-zero-source obstruction applies. When $x = y$, this is the third pattern. Thus the only possibilities under $I \geq 88$ are:

Pattern on 12 vertices	Eligible missing pairs
Distinct degree-one source u and recipient v , all other corresponding degrees two, $u \rightarrow v$ absent	90
The same distinct exceptions, with $u \rightarrow v$ present	89
One vertex of both indegree and outdegree one, every other vertex two-in/two-out	89

All three finite classes can be covered by permutation factorization. For the first, add the missing $0 \rightarrow 1$ as before. For the second, add a second copy of $0 \rightarrow 1$ and factor the resulting bipartite multigraph; both permutation factors then contain $0 \rightarrow 1$. For the third, add the dummy loop $0 \rightarrow 0$ and choose the first factor to contain it. All other cycles of that factor have length at least two. Remove the added copy or loop before all game tests. These conventions cover every simple graph in the displayed classes.

The same camping prunes remain sound even for a search allowing a few failed pairs: a source obstruction destroys at least eight eligible missing pairs, and a recipient obstruction destroys at least seven. Since each class has at most 90 eligible missing pairs, no such graph can have $I \geq 87$. For every surviving complete graph, the exact game now counts unsuccessful eligible source-recipient pairs; evaluation may stop only once the remaining possible count falls below 87.

All three enumerations complete and find no graph with $I \geq 87$. The respective numbers of complete graphs surviving local obstructions are 16,902,091, 1,125,588, and 3,558,672. In particular none reaches 88, contradicting the proposed $I \geq 88$. Therefore $I(G) \leq 87$. $\square$

The detailed degree reduction is `research/odd-endpoint-near23-reduction.md`. The exact enumerator `src/odd_endpoint_near23_exhaustive.cpp` and its three finished receipts `research/odd-endpoint-near23-absent.json`, `research/odd-endpoint-near23-present.json`, and `research/odd-endpoint-near23-same.json` specify the finite checks. An independent source

and mathematical audit is recorded in `research/odd-endpoint-near23-independent-audit.md`; it does not claim a separately implemented repetition of the enumeration. The next lemma strengthens the treatment of eleven-vertex cores and will reduce the final upper bound by one further pair.

For comparison, a fixed 12-vertex graph with 23 arcs has 83 guaranteed indirect pairs. Both original game solvers and every direct rank condition verify this in `src/odd_endpoint_m23_lower.py`, with complete proof data in `research/odd-endpoint-m23-lower-certificate.json.gz`. This witness supplies the lower bound in the final 23-arc corollary below.

Lemma (eleven-vertex cores). Every graph on 11 vertices with 22 or 23 arcs satisfies $I(G) \leq 81$.

Proof. A vertex of outdegree at most one leaves at most ten contributing source rows, each of size at most eight, giving $I \leq 80$. A low-indegree vertex gives the same column bound. We may therefore assume minimum indegree and outdegree two. At 22 arcs all degrees equal two. At 23 arcs exactly one vertex has outdegree three and exactly one has indegree three.

If a source camping obstruction occurs, at least eight missing pairs fail at 22 arcs, and at least seven fail at 23 arcs. If a recipient camping obstruction occurs at t with pursuer position c , every source outside $N^-(t) \cup \{t, c\}$ fails. This gives at least seven failed pairs at 22 arcs and at least six at 23 arcs. The total numbers of missing pairs are 88 and 87, respectively, so every graph with one of these obstructions has $I \leq 81$.

It remains to enumerate the graphs with neither obstruction. The 22-arc case uses the two-permutation decomposition already described. For 23 arcs, the bipartite graph still has a perfect matching: a set S of source vertices emits at least $2|S|$ edges, whereas a set of possible heads T can receive at most $2|T| + 1$. Thus $|N(S)| \geq |S|$, which implies a perfect matching by the augmenting-path criterion. Its removal leaves residual degree one at every source except the unique degree-three source, where two choices remain; recipient capacities are analogous.

Fix the matching permutation to each cycle partition with no singleton. Its cycle rotations and permutations of equal-length cycles allow the degree-three source to be placed at one representative vertex for each distinct cycle length. Enumerate all positions of the degree-three recipient, then every residual row subject to the prescribed column capacities. This covers every graph in question. Only complete source rows or recipient columns are used in camping prunes; a partial cop row suffices because any coverage already present persists under later additions.

The complete 22-arc enumeration has 120,302 graphs surviving these prunes, with maximum $I = 66$. The complete 23-arc enumeration has 792,080 such graphs, with maximum $I = 74$. Both use the exact finite game recurrence and were freshly recompiled and replayed with all counts and witnesses matching. Thus the graphs surviving the prunes also have $I \leq 81$, proving the lemma. $\square$

The source and mathematical audit, including the matching argument, symmetry reduction and rollback checks, is `research/odd-endpoint-eleven-enumeration-audit.md`. The enumerators are `src/small_exceptional_regular_max.cpp` and `src/small_exceptional_one_excess.cpp`; matching replay receipts and hashes are indexed by `research/odd-endpoint-eleven-enumeration-audit.json`. The saved maximizing witnesses additionally pass both original game solvers and all direct rank checks. This is independent review and re-execution of the same enumeration sources, rather than a second independently implemented enumeration.

Corollary (intermediate 23-arc bounds).

$$83 \leq M_E(23) \leq 86.$$

Proof. Suppose $I \geq 87$. Repeat the preceding active-set reduction. The bounds for $q, r \leq 10$, for $\min(q, r) \leq 9$, and for $q = r = 11$ with $b \leq 9$ are already below 87. In the corner $(q, r) = (10, 11)$, $I \leq 91 - b$ suffices when $b \geq 5$. If $b \leq 4$, at least five vertices of $A \setminus B$ have indegree zero. Their at least ten outgoing arcs produce at least four indegree-two zero columns, giving $I \leq 10(11 - 4) = 70$. The dual treats $(11, 10)$.

Hence $q = r = 11$ and $b = 10$ or 11 . For $b = 10$, the preceding argument leaves only the two support-12 patterns with distinct exceptional vertices. For $b = 11$, the all-internal graph and every surviving 22-arc core now have at most 81 guaranteed pairs by the lemma; the external sink or indegree-zero-source cases also lie below 87. The only remaining possibility is the support-12 pattern with one vertex of both indegree and outdegree one. All three support-12 searches were completed at threshold 87 and found no graph reaching it. Therefore $I \leq 86$. The previously verified 83-pair witness supplies the lower bound. $\square$

20 Small vertex orders

The smallest order attaining the vertex bound is twelve. This statement is about attaining the bound, not about the first noncomplete network in which all pairs are guaranteed: the latter already occurs at seven vertices.

Theorem 46 (with finite exhaustive certificates). The seven- and eight-vertex maxima are

$$M_V(7) = 21, \quad M_V(8) = 32.$$

At the next three orders the following bounds hold:

n	Lower bound for $M_V(n)$	Upper bound for $M_V(n)$
9	48	53
10	62	69
11	84	87

There is no universally winning two-in/two-out regular digraph on any of 7, 8, 9, 10, 11 vertices. Together with the previous exact results through six vertices and Theorem 40, this implies that twelve is the smallest order $n \geq 4$ for which $M_V(n) = n(n - 3)$.

Proof of the lower bounds. The finite witnesses at orders seven through eleven have respectively 21, 24, 24, 28, 26 arcs and guarantee every pair. Their complete graph and winning-strategy certificates are provided in `research/small-exceptional-general-certificates.json`, with the improved nine-vertex witness in `research/root-small-structured-9-certificate.json` and the eleven-vertex witness in `research/small-exceptional-structured-11-certificate.json`. Two independent attractor algorithms agree; each stored alternating-state rank satisfies all messenger-choice and universal pursuer-response obligations. Thus $I = n(n - 1) - m$ gives 21, 32, 48, 62, 84 respectively. For concreteness, the seven-vertex witness is the following graph, whose guarantees require at most four messenger moves:

Vertex	Outgoing neighbors
0	1, 3, 6
1	0, 4, 6
2	0, 5, 6
3	1, 2, 5
4	0, 2, 3
5	1, 3, 4
6	2, 4, 5

Seven-vertex upper bound. If a graph has at most thirteen arcs, the arc-only bounds give $I \leq 20$; it suffices here to use monotonicity of M_E (pad by disjoint one-way arcs) and $M_E(13) \leq 20$. A graph with at least twenty-one arcs has at most $42 - 21 = 21$ missing pairs. It remains to exclude $I \geq 22$ for $14 \leq m \leq 20$.

We describe the finite reduction and every rejection rule. Write $O_s = N^+(s)$, $P_t = N^-(t)$ and $C_c = \{c\} \cup N^+(c)$. A counted pair (s, t) necessarily satisfies

1. $s \neq t$, $t \notin O_s$, $|O_s| \geq 2$ and $|P_t| \geq 2$;
2. $O_s \not\subseteq C_c$ for every $c \neq s$;
3. $P_t \not\subseteq C_c$ for every $c \neq s$.

The degree conditions are the one-neighbor camping obstructions. For condition 2, a pursuer waiting at c captures the messenger after its first departure from s . For condition 3, it waits at c and captures the messenger upon the first entry into a predecessor of t , before the next messenger move can deliver. Indefinite waiting by the messenger also fails to deliver. These arguments respect receipt priority because an indirect source is not itself a predecessor of its recipient.

Let $U(G)$ be the number of pairs satisfying these three conditions. Then $I(G) \leq U(G)$. A source contributes at most four pairs, so $I \geq 22$ requires at least six sources of outdegree at least two. Every graph can be relabeled in nondecreasing outdegree order, allowing only the first row to have degree below two. We enumerate every loopless outgoing row mask in that order, with total degree between fourteen and twenty. This includes every possible graph, with harmless repetitions. At a partial assignment, the following bounds allow safe pruning:

- A completed row of degree $d \geq 2$ contributes at most $6 - d$, or zero if a known completed row already violates condition 2. Each remaining row contributes at most $6 - \max(2, d_0)$, where d_0 is the smallest outdegree still allowed by the ordering.
- A target currently of indegree d contributes at most $6 - \max(2, d)$. Its contribution is zero if the remaining rows cannot raise its indegree to two.
- The total arc count must remain extendible to the interval $[14, 20]$.

No omitted edge can repair a rejected completed source: the covering closed neighborhood only grows. The incoming-degree bounds likewise use upper estimates on all possible completions. At every surviving complete graph we count $U(G)$ directly. The exhaustive enumeration has 109,941,961 recursion nodes and 4,299,624 complete leaves before an optional indegree tie normalization. None has $U(G) \geq 22$. A separate implementation uses boolean adjacency, descending outdegree order, and no indegree tie normalization; it completes 123,280,819 nodes and the same 4,299,624 leaves, again with no survivor. This proves $I \leq 21$ in the remaining arc range. The lower witness proves equality.

Eight-vertex upper bound. At most fifteen arcs give $I \leq M_E(15) \leq 30$; at least twenty-four arcs leave at most thirty-two missing pairs. To exclude $I \geq 33$, it therefore suffices to examine $16 \leq m \leq 23$. Each source contributes at most five pairs, so at least seven sources must have outdegree at least two. The same camping conditions and partial bounds used above apply with eight vertices.

For this slightly larger search we also remove labeling redundancy. Fix a nondecreasing outdegree sequence. After the outgoing rows numbered less than r have been chosen, partition the unassigned vertices into cells with equal prescribed outdegree and identical adjacency from all completed rows. Vertices in such a cell may be permuted without changing any completed row or degree requirement. Therefore, when choosing row r , its neighbors among the future vertices of each cell may be required to form a prefix of that cell. This loses no graph: permute within each cell to put the neighbors first, fixing row r and all previous vertices, and continue recursively. The new cells are obtained by splitting the old cells, so their contiguity is preserved. The analogous suffix rule and reversed degree order are equally complete.

The primary search completes all 178 degree sequences, with 946,676,955 recursion nodes and 22,465,453 complete leaves. Only 373 leaves satisfy the necessary pair bound $U(G) \geq 33$. Exact least-attractor evaluation rejects all of them; the largest actual I among these survivors is seventeen. Thus no graph has $I \geq 33$, proving $M_V(8) = 32$. The ordinary canonicalization and every pruning rule were independently audited; as an additional small positive check, every labeled loopless four-vertex graph was canonicalized in both prefix and suffix conventions and its relabeling and prefix conditions checked. Independent complete suffix enumeration and the explicit alternating-state game provide a second finite verification: its completed intervals $[0, 160)$ and $[160, 178)$ cover all degree profiles. They visit 1,179,975,770 nodes and 63,717,034 complete leaves, with 1,115 static survivors, all rejected. The differing counts reflect the opposite degree order and suffix convention, which choose a different redundant collection of representatives. Both implementations give the same zero-survivor conclusion.

Nonattainment at orders eight through eleven. By the vertex equality criterion, an attaining graph would be universally winning and have indegree and outdegree exactly two everywhere. Any such directed graph is the union of two arc-disjoint permutation factors. Indeed, its bipartite tail/head incidence graph is two-regular, so alternating the edges around each even cycle gives the two perfect matchings. Both permutations are fixed-point-free. Up to simultaneous relabeling, the first permutation may be put in canonical cycle form for each partition of n having no part one; enumerate every second permutation avoiding both the diagonal and the first permutation's arcs.

The source and predecessor camping conditions above reject a branch as soon as a completed two-element neighborhood is covered by a known closed outgoing neighborhood. For the predecessor condition, at least $n - 3 \geq 2$ sources are indirect, so a covering vertex can be avoided as the initial source. Every remaining complete graph is evaluated by the finite least-attractor algorithm. The exhaustive counts are:

n	Cycle partitions	Recursion nodes	Locally admissible complete graphs	Universally winning graphs
7	4	142	0	0
8	7	1,446	0	0
9	8	10,996	535	0
10	12	231,676	35,610	0
11	14	2,177,946	120,302	0

An independent implementation reverses the permutation assignment order. Its local tests forbid directed transitive triangles, two distinct length-two walks with the same ordered endpoints, and duplicate completed outgoing pairs; these are precisely the same camping obstructions in adjacency-matrix form. It then uses an explicit alternating-state AND/OR queue attractor instead of the primary synchronous bitmask iteration. Both complete enumerations give exactly the displayed counts. As a positive control, a known universal twelve-vertex graph passes every incremental pruning test and the game check of the primary enumeration. Hence equality $M_V(n) = n(n-3)$ is impossible at these orders, which lowers the integer upper bound by one. $\square$

The seven-vertex example also gives the first noncomplete universally winning graph. At orders at most three, the previously established zero maximum excludes any missing guaranteed pair. At orders four through six, the density bound for a noncomplete universal graph would require $m \leq n(n-3)$ and hence $I \geq 2n$, contradicting the exact maxima 2, 4, 7.

Verification boundary. These are finite computer-assisted proofs, with ordinary completeness and pruning arguments; they are not presently Lean formalizations. The negative seven-vertex result needs no game solver after the static camping reduction. The lower witnesses have saved rank certificates checked directly. Every remaining degree-two graph and every eight-vertex static survivor is evaluated by the exact game algorithms in the independent exhaustive runs; full per-graph ranks for those negative enumerations are not stored. The exact values at orders nine through eleven remain open; absence of a two-regular universal graph does not determine their extrema.

Primary sources are `src/small_exceptional_n7_exact.cpp` and `src/small_exceptional_regular.cpp`; independent sources are `src/small_exceptional_n7_independent.cpp` and `src/small_exceptional_independent.cpp`. The eight-vertex sources are `src/small_exceptional_canonical.cpp` and `src/small_exceptional_canonical_independent.cpp`; the independent ordinary audit is `research/small-eight-canonical-independent-audit.md`. The reproduction receipt is `research/small-exceptional-verification.json` and the research record is `research/small-exceptional-progress.md`.

21 The combined joint maximum

The sparse constructions can be placed inside a larger network by adding isolated vertices. This observation joins the arc-budget extrema to the two dense regimes and gives a single eventual formula.

Corollary 47. Let $n \geq 1800$ and $24 \leq m \leq n(n-1)$. Then the following cases determine $M(n, m)$ exactly:

$$M(n, m) = \begin{cases} k(k-3), & m = 2k \leq 2n, \\ k(k-3) + 2, & m = 2k + 1 < 2n, \\ n(n-1) - m, & 2n \leq m \leq n(n-3), \\ 2n - r - \lceil r/2 \rceil - 1, & m = n(n-3) + r, \quad 1 \leq r \leq n, \\ 0, & n(n-2) < m \leq n(n-1). \end{cases}$$

The two formulas at $m = 2n$ agree. For even m , the first formula also holds for every $24 \leq m \leq 2n$ on every host order $n \geq m/2$.

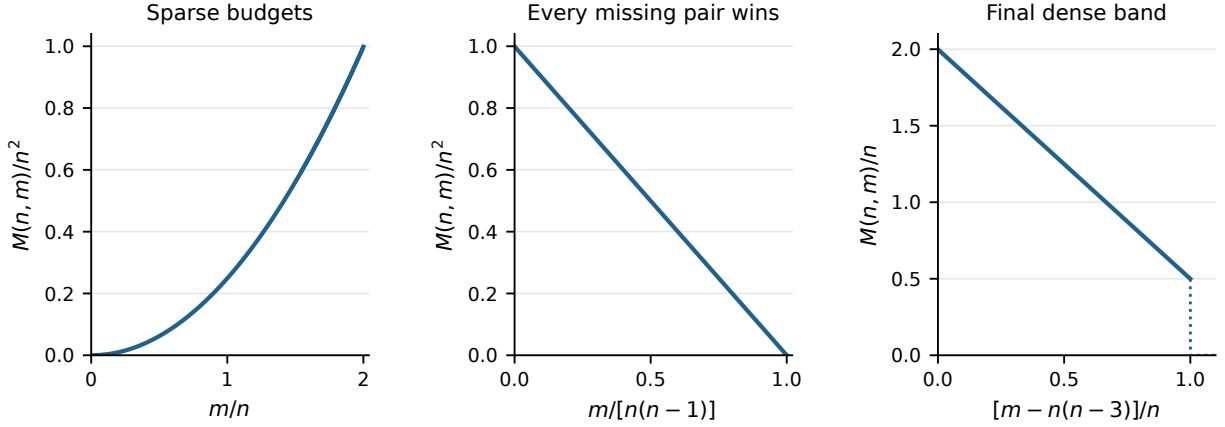


Figure 7: The three exact regimes at $n = 1800$: sparse growth, the main interval, and the final dense band. Separate axis scales show the two narrow endpoint regimes. The dotted continuation in the last panel leads to zero above $n(n-2)$.

Proof. For even $m = 2k \geq 24$, Theorem 40 constructs an attaining k -vertex graph. If $k \leq n$, add $n - k$ isolated vertices. This does not change I : pairs between components have no delivery path; a pursuer starting at an added isolate stays there and cannot obstruct an existing delivery strategy. The arc-only upper bound supplies the matching upper bound for the joint problem.

For odd $m = 2k + 1 \geq 25$, Corollary 45.1 constructs an attaining graph on $k + 1 = (m + 1)/2$ vertices. When $m < 2n$, its support has at most n vertices. Add isolates and apply the odd arc-only upper bound. This gives $k(k-3) + 2$.

Theorem 44 supplies the third case. Corollary 33 supplies the fourth, since its threshold 75 is far below the present hypothesis. Finally Theorem 13 gives zero above $n(n-2)$. The ranges cover every stated integer budget. $\square$

Theorem 53 additionally gives $M(n, 23) = 83$ for every $n \geq 12$.

This is a complete joint classification in the displayed eventual range, with deterministic attaining constructions. It does not assert the same onset for smaller host orders, or fill the remaining finite odd and very small even arc budgets. In particular, the assertion that the full extremal problem is settled for every finite pair (n, m) would be stronger than what has been proved here.

22 A quadratic dense interval from three safe intermediaries

The sparse missing graph need not retain large girth while its density increases. Three designated outgoing missing arcs per vertex suffice, provided certain differences between their endpoints stay protected. The resulting construction covers a quadratic interval of missing-arc counts by a direct modular rule.

Theorem 48. For every integer $n \geq 19$, put

$$K(n) = \max \left\{ 3, \left\lceil \frac{n-1}{13} \right\rceil \right\}.$$

At every integer missing-arc count $3n \leq h \leq nK(n)$, an explicit graph attains

$$M(n, n(n-1) - h) = h.$$

Every source–recipient pair in the construction is guaranteed within two messenger moves. Arbitrary intermediate subsets of the specified optional missing arcs are permitted.

Proof. Work in $\mathbb{Z}/n\mathbb{Z}$, and put

$$S = \{1, 3, 7\}, \quad D = \{\pm 2, \pm 4, \pm 6\}, \quad E = \{\pm 3, \pm 5, \pm 9\}.$$

We shall construct a set A containing S , not containing zero, with the following properties:

1. Each difference in D has exactly its original ordered representation between two members of S , and no other ordered representation between members of A .
2. No difference between two members of A belongs to E .

For $n \geq 19$, these properties hold for $A = S$: its six ordered nonzero differences are precisely the six distinct members of D , and none belongs to E .

Whenever $|A| = k$, adjoin any residue outside

$$\{0\} \cup A \cup (A + D) \cup (A + E).$$

The exclusion of $A + D$ preserves all protected representations, and the exclusion of $A + E$ preserves the second property; both difference sets are symmetric. At most $13k + 1$ residues are excluded. If $k < \lceil (n-1)/13 \rceil$, this quantity is strictly less than n , so the process continues. Starting with three members, it produces $|A| = K(n)$; choosing the smallest permissible representative makes the rule deterministic.

Let F be any loopless digraph satisfying

$$\{(x, x+d) : x \in \mathbb{Z}/n\mathbb{Z}, d \in S\} \subseteq E(F) \subseteq \{(x, x+d) : x \in \mathbb{Z}/n\mathbb{Z}, d \in A\}.$$

Let G be its directed complement. We prove all missing pairs of G have safe two-move intermediaries, using only the three designated missing neighbors of each initial pursuer position.

Fix a missing pair (s, t) and initial pursuer position $c \neq s$. Translate so $s = 0$, and write $a = t - s \in A$ and $z = c - s \neq 0$. The three candidates are the members of $z + S$. A candidate is suitable if it lies outside

$$A \cup (a - A).$$

Indeed, this ensures both messenger arcs are present in G , while the candidate is a missing outgoing neighbor of c and therefore cannot be reached on the pursuer's response.

At most one candidate belongs to A . Otherwise, for distinct $u, v \in S$, the difference of $z+u, z+v \in A$ is the protected difference $u-v$. Its unique representation forces $z+u = u$ and $z+v = v$, contrary to $z \neq 0$.

At most two candidates belong to $a-A$. If the candidates with steps u, v belong there, the two elements $a-z-u, a-z-v$ of A have protected difference $v-u$. Its unique representation is (v, u) , so

$$a-z = u+v.$$

If the third candidate, with step w , also belonged to $a-A$, the same argument applied to u, w would give $a-z = u+w$, a contradiction.

If all three candidates were blocked, the remaining candidate $z+w$ would have to belong to A , while the other two are in $a-A$. Setting $b = z+w \in A$, the displayed equality gives

$$b-a = w-u-v.$$

As w runs over 1, 3, 7 and u, v are the other two steps, this difference is respectively $-9, -5, 3$. Each belongs to E , contradicting the second property of A . Hence a suitable candidate exists.

The suitable candidate differs from c because $0 \notin S$. It differs from both endpoints: $a \in A$ and $0 \in a-A$. Thus the messenger first moves to it, survives every response, and then reaches t ; receipt precedes any subsequent interception. This argument uses only the envelope containments, so no translation symmetry of the chosen intermediate F is needed.

The compulsory missing graph has $3n$ arcs and the full envelope has $nK(n)$ arcs. Adding any chosen number of optional missing arcs gives each integer h in this interval, with all h missing pairs guaranteed. The missing-pair upper bound proves equality. $\square$

The interval extends to at least $n(n-1)/13$ missing arcs. Together with the incidence bridge of Theorem 50, it supplies every missing budget from $2n$ through $nK(n)$ for every $n \geq 166$. The earlier high-girth growth bound in Theorem 20 also gives an overlap, but has a larger sufficient order.

The executable construction and bounded checks are in `src/dense_protected_envelope.py`; their exact finite scope is recorded in `research/dense-protected-envelope-checks.json`. The theorem itself has an ordinary proof and does not depend on those pursuit-game tests. No new Lean coverage is asserted.

23 Exact arc maxima from eight through eleven

The six-vertex table supplies useful lower bounds, but does not itself settle an arc budget when the number of vertices is unrestricted. At eleven arcs the distinction is essential: seven vertices attain eight guarantees, exceeding the six-vertex maximum of seven.

Theorem 49 (with finite exhaustive certificates). Over all finite vertex sets,

$$(M_E(8), M_E(9), M_E(10), M_E(11)) = (4, 5, 6, 8).$$

Proof of the support reduction. Indirect guarantees add over weak components. Every component with a positive contribution has at least four arcs. At these budgets there can be at most two positive components. If there are two, each has at most seven arcs; Propositions 23 and 24 give a combined maximum of at most three. This cannot improve any of the displayed lower bounds.

There is therefore only one relevant positive component. We argue in increasing order of the budget $m = 8, 9, 10, 11$. If this component has fewer than m arcs, the already established smaller-budget maximum bounds its contribution by the displayed value at m . If it uses all m arcs and has at least m vertices, the tree-or-unicyclic argument in Proposition 24 gives at most one indirect guarantee. That argument only needs a connected underlying multigraph with at most as many edges as vertices, so applies unchanged at the present budgets.

In the remaining case the component has at most $m - 1$ vertices. Pad it with isolates to exactly $m - 1$ vertices; its arc count and indirect count do not change. Hence each upper bound reduces to the following one finite counterexample test:

Arc budget m	Host order	Excluded threshold for I
8	7	5
9	8	6
10	9	7
11	10	9

Complete finite upper checks. Both enumerators list all prescribed outdegree profiles with total m in sorted order and then all loopless outgoing rows. As in the canonical reduction of Theorem 46, future vertices with equal prescribed degree and equal incoming pattern from completed rows can be permuted while leaving every completed row unchanged. Choosing the new row's neighbors as a prefix of each such contiguous cell retains a representative of every graph. The second implementation instead uses descending degrees and suffixes. Both reduction arguments fix the current vertex and all completed vertices.

The source and predecessor camping conditions of Theorem 46 supply necessary local rejections. A completed degree-two source that points to a prescribed sink also contributes zero, by the earlier sink obstruction. Otherwise a source of degree $d \geq 2$ contributes at most $n - 1 - d$. A target of current indegree d contributes at most $n - 1 - \max(2, d)$, or zero if its indegree cannot reach two.

One additional numerical bound makes these sparse checks short. Let Q contain all sources not yet excluded, and put $q = |Q|$. Every member of Q has prescribed outdegree at least two. If the eventual number of indegree-at-least-two targets is r , the same degree-incidence count as (1) gives

$$I \leq qr - \max(0, q + r - n) - \max(0, 2q + 2r - m).$$

To bound r at a partial assignment, let R be the number of arcs still to be assigned. A target currently of indegree d needs at least $\max(0, 2 - d)$ of these arcs to reach indegree two. Discard targets with too few remaining possible predecessors, and select as many of the others as possible using the smallest such costs first, with total cost at most R . The resulting $r_{\max}$ is an upper bound: each remaining arc enters only one target. Maximizing the displayed bound over $0 \leq r \leq r_{\max}$ is therefore a safe partial rejection. No sufficiency is assumed for these degree or camping tests.

Every remaining complete graph is evaluated by the original finite reachability game. The primary implementation uses synchronous messenger-position masks; the independent implementation uses

an explicit alternating-state queue attractor. Both start with terminal receipt states and compute the least winning region, including waiting and all pursuer responses. They reject every candidate at every one of the four thresholds. The exact enumeration counts are supplied in the reproduction receipt; both runs report complete enumeration rather than termination at their time limit. This proves the four upper bounds.

Lower witnesses. The witnesses at eight, nine and ten arcs are the graphs in the six-vertex table. The eleven-arc witness has seven vertices and the following outgoing neighborhoods:

Vertex	Outgoing neighbors
0	1
1	0
2	3
3	4, 5
4	0, 2
5	3, 6
6	1, 2

Its guaranteed indirect pairs are

$$(3, 0), (3, 1), (3, 2), (4, 1), (5, 0), (5, 1), (5, 2), (6, 0).$$

Both independent game solvers check each of the four lower witnesses, and the saved explicit ranks satisfy every legal-move obligation. They attain the upper bounds and complete the proof. $\square$

The complete reproduction command is `python3 src/small_arcs_resume.py`. Its two exhaustive sources are `src/small_arcs_sparse_primary.cpp` and `src/small_arcs_sparse_independent.cpp`; the receipt, source hashes, complete enumeration summaries and all four lower rank certificates are in `research/small-arcs-resume-verification.json`. The final full reproduction took about eighteen seconds. These are finite computer-assisted proofs with an ordinary all-order support reduction; they have not yet been formalized in Lean.

24 A short dense bridge from incidence graphs

The linear gap between the sparsest missing graph and the protected dense envelope can be filled by combining three finite graph types. This avoids the large degree-ball constant in Theorem 20.

Theorem 50. For every $n \geq 166$ and every integer $2n \leq h \leq 3n$, there is an explicit graph with n vertices and exactly h missing arcs in which every source–recipient pair is guaranteed within two messenger moves. Consequently

$$M(n, n(n-1) - h) = h.$$

Proof. First construct a bipartite graph J_p for each of the primes $p = 5, 7$. One part consists of the one-dimensional subspaces of $\mathbb{F}_p^3$ and the other of its two-dimensional subspaces; join them when one is contained in the other. Equivalently, represent a point by a nonzero vector v up to nonzero scaling, a line by a nonzero linear form a up to scaling, and test $a \cdot v = 0$. Normalizing the first nonzero coordinate to one makes this construction entirely explicit.

Each part has $p^2 + p + 1$ members. Each line contains $p + 1$ points, and each point lies on $p + 1$ lines, by counting the one-dimensional subspaces of a two-dimensional vector space. Two distinct points span a unique two-dimensional subspace. Thus two points cannot share two line neighbors, excluding four-cycles. Bipartiteness excludes odd cycles, so J_p has girth at least six. In particular, J_5 has 62 vertices and degree six, while J_7 has 114 vertices and degree eight.

Every regular bipartite graph is a disjoint union of perfect matchings: in a d -regular graph the $d|X|$ edges from a set X enter its neighbor set, whose incident capacity is $d|N(X)|$, so Hall's condition holds; remove a perfect matching and repeat. Select four matchings in each J_p . Their union is a spanning four-regular graph. Orient its Euler tours, giving every vertex two incoming and two outgoing arcs. Orient each remaining edge arbitrarily. Taking any subset of those additional arcs preserves minimum outgoing degree two and girth at least six in the underlying graph.

Now write, uniquely,

$$n = 114 + 62k + r, \quad k \geq 0, \quad 52 \leq r \leq 113.$$

This is possible at every $n \geq 166$ by division with remainder. Take one copy of J_7 , k copies of J_5 , and the four-regular girth-six graph on r vertices provided by Theorem 19. Orient the last graph by Euler tours as well. The selected four-regular spanning subgraphs initially give exactly $2n$ arcs. Including every available extra arc gives

$$456 + 186k + 2r = 3n + (114 - r) \geq 3n + 1.$$

Therefore adding the extra arcs one at a time attains every integer arc count h from $2n$ through $3n$ in an oriented graph F of girth at least six and minimum outgoing degree two. Lemma 18 applied to its directed complement proves the assertion. $\square$

The only finite certificate used for arbitrary order is the already verified range $52 \leq r \leq 113$ in Theorem 19. The two incidence graphs have the direct algebraic construction above. The executable generator, matching decomposition, structural checks, and boundary order tests are in `src/dense_incidence_bridge.py`; the receipt is `research/dense-incidence-bridge-checks.json`. No pursuit-game search is needed for this proof, and no new Lean coverage is asserted.

Together with Theorem 48, this covers every missing budget from $2n$ through $n\lceil(n-1)/13\rceil$ at every $n \geq 166$. The complete main-interval threshold in Theorem 44 is consequently limited by its sparse construction, rather than by a dense graph-growth bound.

25 Sharper conditioning at an exact arc budget

The exact-budget argument improves when we use concentration around the binomial mode instead of bounding it by the reciprocal of the entire support size.

Proposition 51. Put $L = n(n-1)$ and $p = m/L$, where $0 < m < L$. If

$$(n-3)p^2(1-p) > 4\log n + \log(8/3),$$

there is a deterministic algorithm, polynomial in n , constructing an n -vertex graph with exactly m arcs in which every pair is guaranteed within two messenger moves.

Proof. Let T be the number of arcs in the independent directed random graph of density p . Then T has binomial distribution, mean m , and variance $Lp(1-p) \leq L/4$. Chebyshev's inequality gives

$$\Pr(|T - m| < \sqrt{L}) \geq \frac{3}{4}.$$

Since $\sqrt{n(n-1)} < n - 1/2$, this interval contains at most $2n - 1$ integers. Moreover m is a mode of the binomial distribution: $\lfloor (L+1)p \rfloor = \lfloor m + m/L \rfloor = m$. Each point in the interval therefore has probability at most $\Pr(T = m)$, so

$$\Pr(T = m) \geq \frac{3}{4(2n-1)} \geq \frac{3}{8n}.$$

Let Z count the bad source-recipient-pursuer triples of Theorem 43. The disjoint-intermediary calculation of Theorem 25 gives

$$\mathbb{E}Z \leq n^3 \exp(-(n-3)p^2(1-p)) < \frac{3}{8n}.$$

Consequently

$$\mathbb{E}[Z \mid T = m] \leq \frac{\mathbb{E}Z}{\Pr(T = m)} < 1.$$

Conditioning on $T = m$ is uniform over the exactly m -arc graphs. The completion-count algorithm in Theorem 43 now applies without change: its initial integer counts satisfy $B < C$, and choosing a branch with $B_j < C_j$ preserves that strict inequality until it produces a graph with zero bad triples. Its polynomial running-time proof is unchanged. $\square$

25.1 An overlap calculation for a shorter sparse bridge

Theorem 52 supplies every sparse budget $2n \leq m \leq 3n$ at every $n \geq 1075$. The following calculation therefore establishes the remaining overlap used by Theorem 44.

By Theorem 38, the explicit sparse range then extends to $n(\lceil n/7 \rceil - 1)$. Theorems 48 and 50 cover the dense end through at least $n(n-1)/13$ missing arcs. Any budget left between these ranges has density

$$\frac{n-7}{7(n-1)} < p < \frac{12}{13}.$$

For $n > 1001$ its lower bound exceeds $71/500$. Taking the minimum of $p^2(1-p)$ at the two endpoints gives

$$p^2(1-p) \geq \min \left\{ \frac{2162589}{125000000}, \frac{144}{2197} \right\} > \frac{173}{10000}.$$

At $n = 1800$,

$$\frac{173(n-3)}{10000} = 31.0881 > 31 > 4 \log 1800 + \log(8/3).$$

Here $\log(8/3) < 1$ follows from $e > 8/3$. For the other logarithm, the finite rational inequality

$$\exp(15/2) > \sum_{j=0}^{20} \frac{(15/2)^j}{j!} > 1800$$

shows that $\log 1800 < 15/2$. The last sum equals $1124776672490111045689/622120446547984384$, so the numerical comparison involves no approximation. The difference $173(n-3)/10000 - 4 \log n - \log(8/3)$ increases for $n \geq 1800$. Thus Proposition 51 covers every intervening budget for all such n .

The mode bound and this overlap have independent ordinary proof checks in `research/exact-budget-mode-bound-independent-audit.md`. No additional Lean verification or large random-graph experiment is asserted.

26 Packing the sparse construction's deleted vertices

The local strategies already used in Theorem 40 allow the deleted vertices to be much closer together than in Theorem 41. A projection between finite arenas makes this improvement compatible with the optional arcs.

Theorem 52 (with finite local certificates). For every integer $n \geq 1075$ and every integer $2n \leq m \leq 3n$, there is an explicit universally winning digraph with n vertices and m arcs. Thus

$$M(n, m) = n(n - 1) - m.$$

Every delivery takes at most $9\lceil n/6 \rceil + 26$ messenger moves.

Proof. Write $k = \lceil n/6 \rceil \geq 180$ and $r = 6k - n \in \{0, \dots, 5\}$. Begin with the six-type graph from Theorem 40 on $(\mathbb{Z}/k\mathbb{Z}) \times \{0, \dots, 5\}$. Its first permutation follows the Hamilton cycle

$$0_i \rightarrow 1_i \rightarrow 2_i \rightarrow 3_i \rightarrow 4_i \rightarrow 5_i \rightarrow 0_{i+1},$$

and its second permutation has arcs

$$0_i \rightarrow 5_i, \quad 1_i \rightarrow 4_{i+1}, \quad 2_i \rightarrow 1_{i+1}, \quad 3_i \rightarrow 0_{i-1}, \quad 4_i \rightarrow 2_{i+1}, \quad 5_i \rightarrow 3_{i-1}.$$

Delete 4_h in columns

$$S = \{0, 3, \dots, 3(r-1)\},$$

with $S = \emptyset$ if $r = 0$, and reconnect each permutation through each removed vertex. The affected paths become

$$3_h \rightarrow 5_h, \quad 1_{h-1} \rightarrow 2_{h+1}.$$

The cyclic gaps between deletions are at least three. The result has exactly n vertices, two incoming and two outgoing arcs at every vertex, and no loops or repeated arcs.

Use the eight optional arc types $(i, j) \rightarrow (i + d, t)$ from Theorem 41:

$$(j, d, t) \in \{(1, -3, 5), (3, -3, 5), (4, -3, 0), (4, 3, 1), (2, 3, 0), (3, 3, 0), (3, -3, 0), (5, -2, 0)\}.$$

Again permit an optional arc only when its source column has cyclic distance greater than 16 from every deletion. These are distinct nonloop arcs outside the base graph. For $r > 0$ the forbidden columns lie in the cyclic image of

$$[-16, 3(r-1) + 16],$$

an interval of $3r + 30 \leq 45$ integers. Consequently there are at least

$$8(k - 45) = 8k - 360 \geq 6k \geq n$$

optional arcs. With no deletions there are $8k \geq n$. Select any $m - 2n$ of them.

We next verify the local strategies in the presence of several nearby deletions. Represent messenger columns $[-4, 4]$ and pursuer columns $[-8, 8]$. Messenger exits are disallowed. Pursuer exits enter an outside state Ω , from which the pursuer may wait or enter any existing vertex in columns $\pm 6, \pm 7, \pm 8$. A local deletion set $H \subseteq [-8, 8]$ has at most five members and consecutive gaps at least three. For the moment omit all optional arcs from this arena.

Put $H_0 = H \cap [-6, 6]$ and consider the previously certified arena of Theorem 40 with these deletions, messenger columns $[-4, 4]$, pursuer columns $[-6, 6]$, and outside re-entry into columns $\pm 5, \pm 6$. Keep every messenger position and every pursuer position in $[-6, 6]$, and map all remaining pursuer positions to the smaller arena's Ω . This map includes every legal move of the larger arena:

- The messenger moves coincide. A deletion outside $[-6, 6]$ affects no messenger tail or retained messenger destination.
- A pursuer move with both endpoints in $[-6, 6]$ is unchanged. An exterior deletion can alter a represented tail only for $h = 7$, changing $1_6 \rightarrow 4_7$ into $1_6 \rightarrow 2_8$; both heads map to Ω .
- A pursuer outside $[-6, 6]$ moves at most two columns in the base graph. Any entry therefore lands in ± 5 or ± 6 , where the smaller arena already permits arbitrary re-entry. The larger arena's own outside transitions map either to Ω or to column ± 6 , also permitted.

The map preserves collisions because every messenger lies in $[-4, 4]$. It preserves terminal receipt and the messenger-turn stopping condition for safe transfer. The old strategies therefore lift to the larger arena. They safely transfer any existing messenger in column -1 to column zero within nine moves, and deliver from column -2 to any existing recipient in column zero within 27 moves.

For a window with no deletion, instead use the fully augmented arena from Theorem 41, with the same radii and three re-entry layers. Its certificates give transfer within nine moves and delivery within 35. The messenger strategies use only base arcs; the optional arcs are included only among the pursuer's possible responses. Thus removing any subset of the optional arcs leaves these strategies valid. Every actual window fits one of these two cases. If it contains a deletion, all its represented source columns have distance at most 16 from that deletion, so none of their optional arcs is present. If it contains no deletion, the fully augmented arena includes all actual pursuer moves. A deletion just outside the window changes no represented-to-represented base arc: at the right boundary, both the original and the smoothed head lie outside; at the left boundary the affected tail lies outside.

All actual moves have column displacement at most three. Since $k \geq 180$, no move can cross the exterior gap between the represented ends and land back inside. Every exterior return lands in one of the three permitted boundary layers. Mapping an actual exterior pursuer to Ω therefore remains a valid simulation even if optional arcs occur outside the current window.

At most $k - 1$ safe transfers move the messenger to the column two steps before its recipient. Each transfer ends on a messenger turn after the pursuer response, with distinct positions. The final delivery then takes at most 35 moves. The total bound is

$$9(k - 1) + 35 = 9k + 26.$$

All missing pairs are guaranteed, giving the stated equality. $\square$

Certificate boundary and reproducibility. There are 865 possible sets H : for cardinalities zero through five their counts are 1, 17, 105, 286, 330, 126. Cropping maps them into the 189 previously verified patterns. The new generator `src/six_type_budget_multihole_certificate.py` checks every arena map, lifts the saved strategies, and directly checks each supplied winning rank and every required initial state. The 5,886 objectives cover 3,365,384 required starts; all checks passed. Its complete ranks are saved in `research/six-type-budget-multihole-certificates.json.gz`, with counts, source hashes, and construction checks in `research/six-type-budget-multihole-summary.json`. Negative rank entries make no claim about losing states or optimality; only the winning strategy implications are used. The two old certificate families have independent alternating-game verifiers described in Theorems 40 and 41. An independent implementation of the new projection, `src/six_type_independent_multihole_projection.py`, checks all 865 patterns using separately written successor formulas; its receipt is `research/six-type-independent-multihole-projection.json`. The arena projection and uniform construction above are ordinary proofs; this theorem is not presently formalized in Lean.

27 The exact 23-arc endpoint

The eleven-vertex core bound and a tighter use of the same degree obstructions close the remaining gap at 23 arcs.

Theorem 53 (with finite exhaustive checks).

$$\boxed{M_E(23) = 83.}$$

Moreover, $M(n, 23) = 83$ for every $n \geq 12$.

Proof. The previously verified 12-vertex witness has 83 guaranteed indirect pairs. We prove the matching upper bound for arbitrary support size. Let A and B be the vertices of outdegree and indegree at least two, respectively, and put

$$q = |A|, \quad r = |B|, \quad b = |A \cap B|, \quad e = e(A, B).$$

Only missing pairs in $A \times B$ can contribute. Thus

$$q, r \leq 11, \quad e \geq 2q + 2r - 23, \quad I \leq qr - b - e.$$

In particular $I \leq 83$ if $q, r \leq 10$, and $I \leq 82$ if either is at most nine. Assume towards a contradiction that $I \geq 84$.

We use two elementary obstructions. If an indegree-two recipient has an indegree-zero predecessor, its entire indirect column is zero: every other source must approach through its other predecessor, where the pursuer can wait. Similarly, if an outdegree-two source has an outdegree-zero successor, its indirect row is zero. The pursuer waits at the other successor; entering the sink cannot help an indirect delivery. These are separate arguments in the original game, requiring no reversal equivalence.

First suppose $(q, r) = (10, 11)$. The displayed inequality gives $I \leq 91 - b$, excluding $b \geq 8$. Total indegree outside B is at most one, so at least $9 - b$ vertices of $A \setminus B$ have indegree zero. For $1 \leq b \leq 7$, at least two such vertices emit at least four arcs. Among these arcs, at most three can avoid indegree-two recipients of B : either B contains one indegree-three vertex and there are no arcs into its complement, or every vertex of B has indegree two and at most one arc enters its complement. Consequently some indegree-two recipient has a zero column. Removing it from the eligible rectangle removes at most two direct pairs and one diagonal; using $e \geq 19$, we obtain

$$I \leq 10 \cdot 10 - (19 - 2) - (b - 1) = 84 - b \leq 83.$$

If $b = 0$, the nine indegree-zero vertices emit at least 18 arcs, at least 15 of which enter indegree-two recipients of B . They give at least eight distinct zero columns, so $I \leq 10(11 - 8) = 30$. The case $(11, 10)$ follows by the corresponding zero-row counting argument. It remains that $q = r = 11$ and $e \geq 21$.

If $b \leq 7$, at least three vertices of $A \setminus B$ have indegree zero. They emit at least six arcs, at least three of which enter indegree-two recipients. There are at least two zero columns, and hence

$$I \leq 11 \cdot 9 - (21 - 4) = 82.$$

If $b = 9$, there is at least one indegree-zero vertex in $A \setminus B$. Its two successors cannot both avoid indegree-two recipients: using an indegree-three recipient and an arc into B^c would consume two units of surplus, whereas only one is available. Removing one zero column gives

$$I \leq 11 \cdot 10 - (21 - 2) - (9 - 1) = 83.$$

If $b = 8$, there are at least two indegree-zero vertices in $A \setminus B$ and at least two outdegree-zero vertices in $B \setminus A$. The same arguments give both an indegree-two zero column and an outdegree-two zero row. Deleting both removes at most four direct pairs and two diagonals. Therefore

$$I \leq 10 \cdot 10 - (21 - 4) - (8 - 2) = 77.$$

For $b = 10$, write $A \setminus B = \{v\}$ and $B \setminus A = \{u\}$. If v has indegree zero or u has outdegree zero, the preceding one-column or one-row argument gives $I \leq 82$. Otherwise both exceptional degrees are one. They exhaust the respective degree allowances outside B and A , so all other degrees equal two and every remaining vertex is isolated. After isolates are removed, there are 12 vertices, with distinct exceptional vertices u, v . Either $u \rightarrow v$ is absent or it is present.

For $b = 11$, put $C = A = B$. At most one arc starts outside C and at most one ends outside C . If all 23 arcs lie in C , the eleven-vertex core lemma gives $I \leq 81$. An arc with both endpoints outside C leaves a 22-arc two-regular core and also gives $I \leq 81$: the messenger cannot leave the core, and the pursuer can remain in it.

Suppose exactly one crossing arc exists, say $C \rightarrow x$. The vertex x is a sink, and exactly one vertex of C has outdegree three. If that vertex emits the crossing arc, removing it leaves a two-regular core. Entering the sink cannot aid delivery to C , so the core bound again applies. If an outdegree-two vertex emits it, that vertex has a zero row. The eligible rectangle has 88 missing pairs, at least nine of which lie in this row, giving $I \leq 79$. The case $x \rightarrow C$ has the same conclusion by the zero-column argument: an indegree-three recipient leaves a two-regular core, whereas an indegree-two recipient has an inaccessible predecessor.

Finally, if both crossing arcs exist, they are $C \rightarrow x$ and $y \rightarrow C$, and every vertex of C has both degrees two. If $x \neq y$, the outside sink gives a zero row containing nine missing pairs, while the rectangle contains 89; hence $I \leq 80$. If $x = y$, the nonisolated graph has 12 vertices, with one common vertex of indegree and outdegree one and all other degrees two.

Thus every possible counterexample belongs to one of three families: distinct degree-one exceptions with their connecting arc absent; distinct exceptions with that arc present; or one common exception. Their numbers of eligible missing pairs are 90, 89, and 89.

We exhaust these families using their two-permutation descriptions. For the first, add the missing exceptional arc $u \rightarrow v$; for the second add another copy of the existing arc; for the third add a dummy loop at the common exceptional vertex. The resulting bipartite incidence multigraph is two-regular and decomposes into two perfect matchings. This follows directly by alternating the edges of each even cycle, including a doubled-edge cycle. Regard the matchings as permutations P, Q , placing the added arc in P .

Label the exceptional source 0 and, when distinct, the exceptional recipient 1. Up to relabeling, enumerate the cycle of P containing the added arc and the unordered cycle lengths of its remaining vertices. In the common-exception case, P fixes 0. Every other cycle has length at least two. Then enumerate every admissible permutation Q , allowing only the prescribed doubled exceptional arc in the second family, and delete the added P arc before any game calculation. This covers every graph in the three families.

Partial assignments may be discarded when a stationary pursuer covers the completed outgoing neighbors of a contributing source, or the completed predecessors of a contributing recipient. A source obstruction destroys at least eight eligible pairs. A recipient obstruction destroys at least

seven, after omitting a possible source coinciding with the pursuer. Since there are at most 90 eligible pairs, all discarded graphs satisfy $I \leq 83$. Testing coverage with a partially assigned pursuer neighborhood is also safe: that neighborhood can only grow.

Every completed graph surviving these obstructions is evaluated by the exact least reachability fixed point. Evaluation may stop at a leaf once its remaining possible score falls below 84. The three finished enumerations give the following counts; none has $I \geq 84$.

Exceptional configuration	Permutation cases	Partial nodes	Completed graphs surviving camping
Distinct, arc absent	42	164,276,910	16,902,091
Distinct, arc present	42	13,448,216	1,125,588
Common vertex	14	49,814,092	3,558,672

This excludes every proposed counterexample and proves $I \leq 83$. The verified witness gives equality. Adding isolates preserves its guaranteed indirect pairs, proving the fixed-order assertion for every $n \geq 12$. $\square$

Reproducibility and proof boundary. The new source `src/odd_endpoint_near23_threshold.cpp` is a separately saved, parameterized copy of the earlier enumerator; it accepts cutoffs at least 84. The coverage, game calculation, and pruning are unchanged. The complete receipts are `research/odd-endpoint-threshold84-absent.json`, `research/odd-endpoint-threshold84-present.json`, and `research/odd-endpoint-threshold84-same.json`. Their cut-off, completion flags, exact counts, source hash, and dependencies are consolidated in `research/odd-endpoint-threshold84-summary.json`. The unchanged dependency on the eleven-vertex enumeration is audited in `research/odd-endpoint-eleven-enumeration-audit.md`. The full strengthened degree reduction, outside-core cases, and cutoff 84 pruning are independently reviewed in `research/odd-endpoint-threshold84-ordinary-audit.md`. A separate software audit of the coverage, pruning, and evaluation is `research/odd-endpoint-threshold84-independent-audit.md`. It also compares 37 leaf-game and early-exit cases across all three families with an explicit alternating-state solver, including full rank and losing-closure checks; the receipt is `research/odd-endpoint-threshold84-independent-tests.json`. The lower-bound certificate is `research/odd-endpoint-m23-lower-certificate.json.gz`, checked by two independently implemented game solvers and every saved rank obligation. The exhaustive upper-bound computation is reviewed at the source and mathematical level; it is not a separately implemented second enumeration or a Lean proof.

28 Exact computation and formal verification

For a fixed recipient t , let $X_j(r)$ be the set of initial pursuer positions from which delivery can be forced in at most j messenger moves. Start with $X_0(t) = V$ and $X_0(r) = \emptyset$ otherwise. Set

$$S_j(v) = \{c : C(c) \subseteq X_j(v)\} \quad (v \neq t), \quad S_j(t) = V,$$

and update

$$X_{j+1}(r) = \left(\bigcup_{v \in C(r)} S_j(v) \right) \setminus \{r\} \quad (r \neq t), \quad X_{j+1}(t) = V.$$

The recurrence includes waits and excludes all nonterminal collisions. Its least fixed point enforces eventual delivery. A separate implementation builds the explicit alternating graph of states (r, c, turn) and computes its reachability attractor. This is standard finite reachability-game machinery (Berwanger 2009), rather than a new general algorithm for graph games.

The certificate checker independently verifies decreasing ranks on the selected messenger moves and on every pursuer response. Outside the winning region it verifies closure under every messenger move and at least one pursuer response. Thus a negative answer comes with an avoidance strategy, not simply failure of a search. Standard adjacency-list attractor analysis gives an all-recipient time bound $O(n^2m + n^3)$; that bound describes the standard implementation, not a claimed running-time bound for the simpler synchronous reference solver.

28.1 Certified finite extrema

Complete enumeration gives

n	1	2	3	4	5	6
$M_V(n)$	0	0	0	2	4	7

For $n \leq 5$ every labeled digraph was enumerated. For $n = 6$, sorting the vertex labels by degree reduces the search while retaining at least one representative of every isomorphism class; valid degree bounds prune only candidates unable to improve the current maximum. This certifies extrema, not labeled distributions. Every attaining graph is independently checked by the explicit alternating solver. The complete joint tables, enumeration code, and receipts accompany the paper.

Universal two-regular witnesses attain Theorem 5 for every $12 \leq n \leq 24$. Consequently Theorem 6 is attained at $m = 24, 26, \dots, 48$. Deletion witnesses attain Theorem 7 at $m = 37, 39, \dots, 47$. Theorem 46 proves $M_V(7) = 21$, $M_V(8) = 32$, and bounds the still-open orders 9 through 11.

One clean 12-vertex example is a Cayley digraph of A_4 , with right multiplication by two suitable 3-cycles. Its actual graph and all-recipient finite-rank certificate are included. The Lean theorem checks the explicit 12-vertex graph directly, so correctness does not depend on trusting the group-label interpretation or the search that discovered it.

28.2 Lean coverage

The formal artifact uses Lean 4.33.1 and its standard library. The game implementation checks terminal receipt before collision, permits waiting, and quantifies over arbitrary legal history-dependent pursuer choices. Finite winning trees are proved to generate actual delivering plays; rank certificates generate those winning trees. Safe branching and predecessor camping are connected to the same semantics. A fully kernel-checked 12-vertex certificate proves universal delivery and computes $I = 108$.

The dense-counting module additionally proves the staircase arithmetic from its displayed cardinality inequalities, the attaining count identity, and complementary-block gluing as actual two-move

delivery against every legal history-dependent pursuer. The counting inequalities themselves are established by the ordinary graph argument, rather than silently asserted as formal theorems. Existence of the blocks is also a separate premise of the formal gluing result.

These are meaningful statements about the physical game. The general graph-to-counting arguments, arithmetic existence lemma, infinite constructions, probabilistic existence results, and exhaustive-search completeness arguments remain ordinary proofs. The downloadable coverage map lists exact theorem names and source hashes; it does not equate a numerical certificate or assumed interface with a full general formalization.

29 Further consequences and open problems

For the random loopless digraph with each arc independently present with a fixed probability $0 < p < 1$, all distinct pairs are guaranteed with probability tending to one. For fixed distinct s, t and $c \neq s$, each candidate $u \notin \{s, t, c\}$ independently supplies the arcs $s \rightarrow u, u \rightarrow t$ and the absence of $c \rightarrow u$ with probability $p^2(1 - p)$. A union bound gives failure probability at most

$$n^3(1 - p^2(1 - p))^{n-3}.$$

Thus

$$\mathbb{E}[I(G)] = (1 - p)n(n - 1) - o(1).$$

The error is absolute $o(1)$ because the probability bound is exponential. Theorem 25 supplies a separate conditioning argument for an exact prescribed number of arcs; the expectation alone does not do so.

Angel Raychev’s conjecture that every sufficiently large order attains the missing-pair bound at every $2n \leq m \leq n(n - 3)$ is now proved by Theorems 38–44. The sparse endpoint constructions also settle the vertex maximum at every $n \geq 12$, and the even and odd arc maxima at all sufficiently large budgets. Corollary 47 assembles the eventual joint classification.

The remaining central questions are the exact vertex maxima at orders nine through eleven, the arc budgets twelve through twenty-two, the joint maximum at moderate orders outside the stated intervals, and classification of all maximizing graphs. Improving the sufficient 1800 threshold would clarify how early the full interval begins. A finite construction that works at one circumference still does not prove a uniform family: the local games and projection arguments are essential to the results above.

A correct future proof must retain enough information about both players. Simple distance-to-recipient potentials fail: some states at a fixed small distance from the recipient require detours whose length grows with the graph. Nor does safe indefinite motion imply delivery. The finite witnesses and failed conjectures are preserved to make those distinctions reproducible.

30 Construction appendix: choosing the circulant parameter

Proof of Lemma 12. If $n \equiv 0 \pmod{4}$, choose $h = n/2 - 1$; any common divisor with n divides two and is odd. If $n \equiv 2 \pmod{4}$, choose $h = n/2 - 2$; the same argument uses four. If $n \equiv 3 \pmod{4}$, use $h = (n - 1)/2$.

It remains to handle $n \equiv 1 \pmod{4}$. Write $n - 1 = 2^e d$ with d odd and $e \geq 2$. If $d \geq 7$, take $h = d$: it divides $n - 1$ and is at most $(n - 1)/4$. If $d \in \{1, 3, 5\}$ and $3 \nmid n$, use $h = (n - 3)/2$.

If $d = 1$ and $3 \mid n$, take $h = (n - 7)/2$. The number $2^e + 1$ is not divisible by seven, since powers of two modulo seven are 1, 2, 4. The case $d = 3$ and $3 \mid n$ is impossible. Finally, if $d = 5$ and $3 \mid n$, then e is even. Use $h = (n - 11)/2$. Divisibility of $5 \cdot 2^e + 1$ by eleven would imply $2^e \equiv 2 \pmod{11}$, whereas even powers have residues 1, 3, 4, 5, 9.

All choices are odd and smaller than $n/2$. Their smallest possible lower estimate is $(n - 11)/2 \geq 7$, valid for $n \geq 25$. $\square$

31 Provenance and assistance

Alexandra Ignatova’s original problem and layered construction, developed under Angel Raychev’s mentorship in 2023 and written in the retained February 2024 manuscript, are credited throughout. The upper bound $I(G) \leq n(n - 3)$ and the investigation of two-in/two-out-regular graphs also belong to Ignatova and Raychev’s original work; the present account does not claim that bound or research direction as a new contribution. The September 2026 work repairs the balancing and strategy arguments, corrects the original count, and develops the new extremal bounds, circulant constructions, dense families, computational certificates, and formalization. Astra 6, operating through the Codex harness, assisted with these deductions, counterexample searches, proof writing, programming, and Lean development. The current document is a research checkpoint; it has not been submitted to arXiv and does not assign an unresolved final human author line.

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